CoE shell model vs. Migkas cluster anisotropy — every step (v3)

This notebook reproduces the v3 term-by-term sheet (`thinshell_termmatch_equations_table_v3.pdf`, Sections 1–4) and the Migkas $\langle H \rangle$ anchor, one step at a time. Each step has a short explanation and then the code that does it.

Rules used throughout

- Gravity only, $\Lambda = 0$, no dark matter. All parameters are free, with no age, boundedness or ΛCDM limits.
- Chris's A: $a_f = |\Delta\vec{V}|/\mathrm{separation}$, used linearly. It is never squared and never turned into D_L^{-2}.
- Symbols: D = CoE–Earth distance; R = the 14.3 Gpc vision sphere; shell edges $R_{\rm in}$, $R_{\rm out}$. Lengths are in Gpc.
- $H_0 = 70$ **is not an input.** It appears in only two labelled places: (i) Section 3's Table 1 masses, which were originally computed with Earth's rate set to 70, and are shown unrescaled with a label; (ii) the unit convention of the LTB integration, which is then rescaled to the Migkas-anchored $\langle H \rangle$.

Sources. Taylor, G. I. 1950, Proc. R. Soc. Lond. A 201, 159; Sedov, L. I. 1959, *Similarity and Dimensional Methods in Mechanics*; Migkas, K. et al. 2020, A&A 636, A15 (arXiv:2004.03305); Migkas, K. et al. 2021, A&A 649, A151 (arXiv:2103.13904); Planck Collaboration 2020, A&A 641, A6 and A1 (Planck 2018 VI and I); Riess, A. G. et al. 2022, ApJL 934, L7 (SH0ES); Watkins, R. et al. 2023, MNRAS 524, 1885 (CosmicFlows-4 bulk flow).

The heavy integrators are Chris's code, used unchanged: `ptshell.py` / `prof_core.py` (Newtonian) and `relfree.py` / `rm_core.py` (LTB). Everything else is written out in the cells.

```
# Step 0 - paths, Chris's integrators (unchanged), plotting
import sys, os, json, csv, numpy as np
import matplotlib; matplotlib.use("Agg"); import matplotlib.pyplot as plt
%matplotlib inline
BUB = "/workspace/cosmology/papers/spreadsheet_model/bubble"
THS = "/workspace/cosmology/papers/A_equals_fourier/thinshell"
sys.path.insert(0, BUB); sys.path.insert(0, BUB + "/gr_ltb/refit_mass"); os.chdir(BUB)
import ptshell as S, prof_core as P                # Newtonian shell + core integrator (Chris's code)
np.set_printoptions(precision=4, suppress=True)
TH1 = np.arange(0, 360, 1.0)                       # Galactic longitude grid, 1 deg (the code's 360 directions)
RV = 14.30048287260589                             # R, vision sphere [Gpc]
G = 6.674e-11; c = 299792.458; MPC = 3.0857e19; GPCm = 3.0857e25; MSUN = 1.989e30
def rhoc(H_kms_Mpc):                               # critical density 3H^2/(8 pi G) [kg/m^3]
    h = H_kms_Mpc / MPC; return 3 * h * h / (8 * np.pi * G)
print("R =", round(RV, 2), "Gpc;", len(TH1), "directions")
```

```
R = 14.3 Gpc; 360 directions
```

Step 1 — Model setup (Section 1)

Flow. Sedov–Taylor self-similar expansion (Taylor 1950; Sedov 1959). Every point moves away from the Centre of Expansion (CoE) as

$$\vec{V} = k\,\vec{r} \qquad (\vec{r} = \text{position from the CoE}).$$

Vector subtraction. An observer O (Earth) sees a source P move at

$$\vec{V}_P - \vec{V}_O = k(\vec{r}_P - \vec{r}_O) = k\,\vec{s},$$

so with no mass present, $|\Delta\vec{V}|/s = k$ in every direction and the sky is isotropic. The anisotropy comes only from gravity.

Mass. A point-like core M_c at the CoE, plus a **uniform spherical shell** of ordinary matter with density $\rho_s = \Omega_s \rho_c$ between $R_{\rm in}$ and $R_{\rm out}$ (both centred on the CoE and fixed). Earth sits at distance D from the CoE, and all sources are on the $R = 14.3$ Gpc sphere around Earth, inside the shell. Each fit keeps its own shell rule: Newtonian Ω_s free uses $R_{\rm in} = D - R$, $R_{\rm out} = D + R$; Newtonian $\Omega_s = 0.049$ uses $D - 1.028R$, $D + 1.475R$; LTB uses $D - 1.040R$, $D + 1.171R$.

```
# Step 1a - with no mass, V = k r gives |dV|/s = k in every direction (vector subtraction)
k = 1.0; D = 3.0; rO = np.array([D, 0.0])                    # Earth at distance D from the CoE (units arbitrary)
th = np.radians(TH1); rP = rO + RV / 14.3 * np.c_[np.cos(th), np.sin(th)]
dV = k * rP - k * rO                                         # V_P - V_O
af = np.linalg.norm(dV, axis=1) / np.linalg.norm(rP - rO, axis=1)
print("pure Sedov-Taylor: af/k min, max =", af.min(), af.max())
```

```
pure Sedov-Taylor: af/k min, max = 1.0 1.0
```

```
# Step 1b - the three fitted models (parameters from the fit files; all free, nothing imposed)
T1 = json.load(open(BUB + "/universe_thin.json"))            # Table 1 of universe_bestfit_thinshell.pdf
hug = json.load(open(BUB + "/thin_b_free_hug.json"))         # Newtonian, Omega_s free
omb = json.load(open(BUB + "/universe_newt_omb.json"))["Rs14.3|sine"]   # Newtonian, Omega_s = 0.049
REL = json.load(open(BUB + "/universe_rel_mcfree.json"))     # relativistic LTB, all free
NEWT = {
 "free": dict(DR=hug["DR"], Os=hug["Os"], mcc=hug["mcc"], rin=hug["rin"], rout=hug["rout"], Lc=0.005, mc=0.0, Ob=0.0, l0=hug["l0"], amp=hug["amp"],
phi=hug["phi"]),
 "omb":  dict(DR=omb["D"] / RV, Os=0.049, mcc=omb["mcc"], rin=omb["R_in"] / omb["D"], rout=omb["R_out"] / omb["D"], Lc=0.005, mc=0.0, Ob=0.0,
l0=omb["l0"], amp=omb["amp"], phi=omb["phi"]),
}
for kk, a in NEWT.items():
    D = a["DR"] * RV
    print(f"{kk:5s} D = {D:6.2f} Gpc  D/R = {a['DR']:.3f}  R_in = {a['rin']*D:6.1f}  R_out = {a['rout']*D:6.1f} Gpc  "
          f"Omega_s = {a['Os']:.3f}  Mc/Mcrit(D) = {a['mcc']:.5f}")
print(f"LTB   D = {REL['D']:6.2f} Gpc  D/R = {REL['D']/RV:.3f}  R_in = {REL['R_in']:6.1f}  R_out = {REL['R_out']:6.1f} Gpc  "
      f"Omega_s = 0.049  Mc/Mcrit(D) = {REL['mcc']:.4f}  (k_in {REL['kin']:.3f}, k_out {REL['kout']:.3f})")
```

```
free  D =  93.72 Gpc  D/R = 6.554  R_in =   79.4  R_out =  108.0 Gpc  Omega_s = 1.012  Mc/Mcrit(D) = 0.00265
omb   D =  49.17 Gpc  D/R = 3.439  R_in =   34.5  R_out =   70.3 Gpc  Omega_s = 0.049  Mc/Mcrit(D) = 0.00171
LTB   D =  32.94 Gpc  D/R = 2.303  R_in =   18.1  R_out =   49.7 Gpc  Omega_s = 0.049  Mc/Mcrit(D) = 0.0835  (k_in 1.040, k_out 1.171)
```

Step 2 — Chris's A, exactly as his code computes it

The code integrates the flow from $\tau_i = 0.01$ to today (time unit t_0, length unit D), with Earth and 360 sources on the R sphere (one per degree of Galactic longitude, in the Galactic plane). For each source,

$$a_f(\ell) = \frac{|\vec{V}_P - \vec{V}_O|}{s_{\rm nominal}}, \qquad A(\ell) = \frac{a_f(\ell)}{\langle a_f \rangle_{\rm sky}}.$$

This is compared **linearly** with Migkas's $A/A_{\rm all}$: it is not squared, and not converted to D_L^{-2} or H^2. In code units $1/t_0 = k$, so $\langle a_f \rangle / k$ is dimensionless. The code also returns Earth's radial rate H_E/k (`Ht`) and speed v_E (units D/t_0).

```
# Step 2 - run Chris's integrator (inlined from prof_core.run) and form af = |dV| / separation
def run_newt(a, taui=0.01, nsteps=200, iters=3):
    _, _, m1 = S.build(a)                                  # core + uniform shell enclosed-mass table (ptshell.build)
    fac = max(1.5, 1.02 * a["rout"] / (1 + 1 / a["DR"]))
    q, rr = P.geometry(TH1, a["DR"]); Rb = P.rb_of(a["DR"], fac); Ht = 0.95
    _, MT = P.tables("ptshell", a, Rb)
    for _ in range(iters):                                  # self-consistent Earth rate H_E t0
        mu = m1 * Ht**2 / 2
        X, V, VR = P.integrate(q, mu, P.RG, MT, 0.0, Rb, taui, nsteps)
        Ht = np.hypot(*V[0]) / np.hypot(*X[0])
    dV = V[1:] - V[0]                                       # V_P - V_O
    af = np.hypot(dV[:, 0], dV[:, 1]) / rr                  # / nominal separation (R in units of D)
    return dict(af=af, Ht=Ht, vE=np.hypot(*V[0]), rE=np.hypot(*X[0]), mu=mu, MT=MT, Rb=Rb)
RUN = {kk: run_newt(a) for kk, a in NEWT.items()}
for kk, r in RUN.items():
    A = r["af"] / r["af"].mean()
    print(f"{kk:5s} <af>/k = {r['af'].mean():.4f}  A min/max = {A.min():.3f}/{A.max():.3f}  H_E/k = {r['Ht']:.4f}  v_E = {r['vE']:.4f} D/t0")
```

```
free  <af>/k = 0.9757  A min/max = 0.921/1.200  H_E/k = 0.9902  v_E = 0.8990 D/t0
omb   <af>/k = 1.0119  A min/max = 0.924/1.124  H_E/k = 0.9942  v_E = 0.9353 D/t0
```

Step 3 — Migkas data: $A/A_{\rm all}$ vs. ℓ and the Fourier form A_F

The points are the digitised Migkas et al. 2020 Fig. 6 profile, $A/A_{\rm all}$ against Galactic longitude (71 points). A_F is its 3-harmonic Fourier fit:

$$A_F(\ell) = 0.9965 + 0.0296\cos\ell + 0.0434\sin\ell + 0.0636\cos 2\ell + 0.0366\sin 2\ell - 0.0299\cos 3\ell - 0.0114\sin 3\ell.$$

```
# Step 3 - load the Migkas profile and define A_F
mig = list(csv.DictReader(open("/workspace/cosmology/migkas/fig6_digitized.csv")))
ML = np.array([float(r["l_deg"]) for r in mig]); MA = np.array([float(r["A_over_Aall"]) for r in mig]); IMIG = ML.astype(int)
AFc = {0: (0.9965, 0.0), 1: (0.0296, 0.0434), 2: (0.0636, 0.0366), 3: (-0.0299, -0.0114)}      # (cos, sin) coefficients
def AFfour(l):
    l = np.radians(l); return sum(cs * np.cos(m * l) + sn * np.sin(m * l) for m, (cs, sn) in AFc.items())
def r2(y, m): return 1 - np.sum((y - m) ** 2) / np.sum((y - y.mean()) ** 2)
print(len(ML), "points; l range", ML.min(), "-", ML.max(), "; R2(A_F vs points) =", round(r2(MA, AFfour(ML)), 3))
```

```
71 points; l range 5.0 - 355.0 ; R2(A_F vs points) = 0.915
```

Step 4 — The diagnostic sine term and the headline R^2

The geometry alone gives only $\cos m\psi$ terms about the CoE axis ($\psi = \ell - \ell_0$). A single rotation/tilt factor absorbs the sine parts:

$$A(\ell) = a_f(\ell - \ell_0)\,\big[1 - \mathrm{Amp}\,\sin(\psi + \varphi)\big].$$

It is a diagnostic factor, not a velocity. The headline number is R^2 of this full curve against the 71 Migkas points. Each fit's ℓ_0, Amp and φ are its Table 1 values. The LTB curve comes from the relativistic ray-tracing code (heavy cell below, about 1 s).

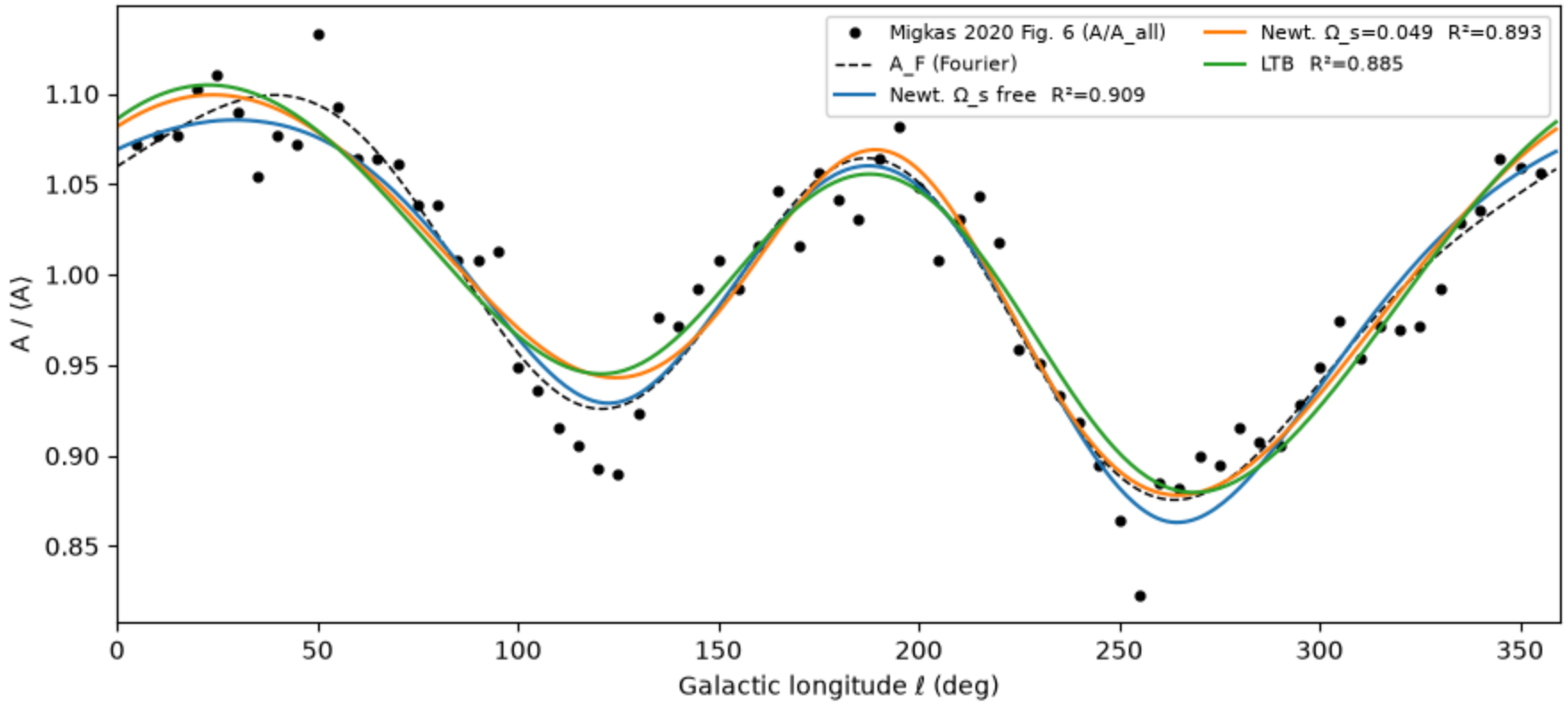
```
# Step 4a - Newtonian: shift to l0, apply the sine factor, R^2 vs Migkas
def shifted(af, l0): return np.interp((TH1 - l0) % 360, np.append(TH1, 360), np.append(af, af[0]))
def A_from(af, th, amp, phi): return (1 - amp * np.sin(np.radians(th) + phi)) * af
CURVE, GEO = {}, {}
for kk, a in NEWT.items():
    g = RUN[kk]["af"] / RUN[kk]["af"].mean(); GEO[kk] = shifted(g, a["l0"])
    CURVE[kk] = A_from(GEO[kk], TH1 - a["l0"], a["amp"], a["phi"])
    print(f"{kk:5s} l0 = {a['l0']:.1f}  Amp = {a['amp']:.4f}  phi = {a['phi']:.3f}   R2 vs Migkas = {r2(MA, CURVE[kk][IMIG]):.3f} "
          f"(geometry only {r2(MA, GEO[kk][IMIG]):.3f})   R2 vs A_F = {r2(AFfour(TH1), CURVE[kk]):.3f}")
```

```
free  l0 = 12.8  Amp = -0.1240  phi = 1.264   R2 vs Migkas = 0.909 (geometry only -0.734)   R2 vs A_F = 0.987
omb   l0 = 13.0  Amp = -0.0622  phi = 0.929   R2 vs Migkas = 0.893 (geometry only 0.497)   R2 vs A_F = 0.977
```

```
# Step 4b - HEAVY CELL (relativistic LTB): Chris's LTB solver + ray tracing at the Table-1 parameters (~1 s)
# Unit note: the LTB code works in units where the 3D-mean local expansion rate is 70 km/s/Mpc (its internal convention);
# only dimensionless ratios are used from it here, and Section 4 rescales to the Migkas-anchored <H>.
import relfree as RF, rm_core as RC
ltb = RF.ev(REL["Mc_kg"], REL["D"], REL["kin"], REL["kout"], full=True)
Ad = np.array(ltb["Ad"])
def ltb_curve(Ad, l0, amp=0.0, phi=0.0):                    # inlined rm_core.curve: 0.5-deg ray table -> 1-deg grid, normalise, sine
    idx = (((TH1 - l0 - 180.0) % 360) / 0.5).astype(int); A = Ad[idx]; A = A / A.mean()
    return A_from(A, TH1 - l0, amp, phi) if amp else A
GEO["rel"] = ltb_curve(Ad, ltb["l0_s"]); CURVE["rel"] = ltb_curve(Ad, ltb["l0_s"], ltb["amp"], ltb["phi"])
json.dump(dict(ltb={k: v for k, v in ltb.items() if k != "Ad"}, A=CURVE["rel"].tolist()),
open("/workspace/cosmology/papers/A_equals_fourier/steps_v3/ltb_cache.json", "w"), default=float)
print(f"LTB   l0 = {ltb['l0_s']:.1f}  Amp = {ltb['amp']:.4f}  phi = {ltb['phi']:.3f}   R2 vs Migkas = {r2(MA, CURVE['rel'][IMIG]):.3f} "
      f"(geometry only {r2(MA, GEO['rel'][IMIG]):.3f})   R2 vs A_F = {r2(AFfour(TH1), CURVE['rel']):.3f}")
print(f"      H_perp = {ltb['H_perp']:.2f}, H_r = {ltb['H_r']:.2f} (fit units), age = {ltb['age_Gyr']:.2f} Gyr (fit units)")
```

```
LTB   l0 = 13.5  Amp = -0.0558  phi = 0.848   R2 vs Migkas = 0.885 (geometry only 0.569)   R2 vs A_F = 0.963
      H_perp = 65.92, H_r = 78.15 (fit units), age = 12.98 Gyr (fit units)
```

```python
# Step 4c - figure: the three full curves vs the Migkas points
fig, ax = plt.subplots(figsize=(9, 4.2))
ax.plot(ML, MA, "ko", ms=3.5, label="Migkas 2020 Fig. 6 (A/A_all)")
ax.plot(TH1, AFfour(TH1), "k--", lw=1, label="A_F (Fourier)")
for kk, col, lab in (("free", "C0", "Newt. Ω_s free"), ("omb", "C1", "Newt. Ω_s=0.049"), ("rel", "C2", "LTB")):
    ax.plot(TH1, CURVE[kk], col, label=f"{lab}  R²={r2(MA, CURVE[kk][IMIG]):.3f}")
ax.set_xlabel("Galactic longitude ℓ (deg)"); ax.set_ylabel("A / ⟨A⟩"); ax.set_xlim(0, 360); ax.legend(fontsize=8, ncol=2); plt.tight_layout();
plt.show()
```



Step 5 — Section 2: term-by-term solution against A_F

To first order in $\varepsilon = R/D$, with $h(q) = V/(kq)$ the radial rate profile, $\beta = h'/h$ and $\gamma = h''/h$ at Earth:

$$a_f^2 \propto 1 + B\mu^2 + \varepsilon(P\mu + Q\mu^3), \quad B = 2\beta + \beta^2,\ P = \beta(3+\beta),\ Q = (1+\beta)(\gamma+\beta) - 2\beta.$$

Taking the square root and projecting onto harmonics gives

$$\frac{a_f}{\langle a_f\rangle} = 1 + a_1\cos\psi + a_2\cos 2\psi + a_3\cos 3\psi, \quad a_2 = \frac{B/4 - B^2/16}{m_0},\ m_0 = 1 + \tfrac{B}{4} - \tfrac{3B^2}{64},$$

$$a_1 = \frac{\varepsilon}{2m_0}\Big[P + \tfrac{3}{4}\big(Q - \tfrac{PB}{2}\big) - \tfrac{5QB}{16}\Big], \qquad a_3 = \frac{\varepsilon}{2m_0}\Big[\tfrac{1}{4}\big(Q - \tfrac{PB}{2}\big) - \tfrac{5QB}{32}\Big].$$

The sine factor adds $C_1 = a_1 - \mathrm{Amp}\sin\varphi(1 + a_2/2)$, $S_1 = -\mathrm{Amp}\cos\varphi(1 - a_2/2)$, $C_3 = a_3 - \frac{\mathrm{Amp}\,a_2}{2}\sin\varphi$, $S_3 = -\frac{\mathrm{Amp}\,a_2}{2}\cos\varphi$. Solve order: 2ℓ phase → ℓ_0; 2ℓ amplitude → β; C_1, S_1, C_3 → Amp, φ, ε (so $D = R/\varepsilon$); S_3 is then a check. Point-core shape: $\gamma = -4\beta$. Mass: $\beta \simeq 3K/(1-K)$, $K = \frac{1}{2}\frac{M_c}{M_{\mathrm{crit}}(D)}(\frac{1}{\tau_i} - 1)$.

```
# Step 5 - sequential (hand) solve with the series, then exact least squares on C_m, S_m
from scipy.optimize import brentq, fsolve, least_squares
TH = np.radians(TH1)
def harm(y, m): return (y.mean(), 0.0) if m == 0 else (2 * np.mean(y * np.cos(m * TH)), 2 * np.mean(y * np.sin(m * TH)))
def rot(cs, sn, m, l0):                 # coefficients about l -> about psi = l - l0
    t = np.radians(m * l0); return cs * np.cos(t) + sn * np.sin(t), -cs * np.sin(t) + sn * np.cos(t)
def series(beta, gam, eps):
    B = 2 * beta + beta**2; Pp = beta * (3 + beta); Q = (1 + beta) * (gam + beta) - 2 * beta
    m0 = 1 + B / 4 - 3 * B**2 / 64; m2 = B / 4 - B**2 / 16
    m1 = eps / 2 * (Pp + 0.75 * (Q - Pp * B / 2) - Q * B / 2 * 10 / 16); m3 = eps / 2 * (0.25 * (Q - Pp * B / 2) - Q * B / 2 * 5 / 16)
    return dict(a1=m1 / m0, a2=m2 / m0, a3=m3 / m0, Q=Q)
C2, S2 = AFc[2]; l0 = (np.degrees(np.arctan2(S2, C2)) / 2) % 180                    # 2-l phase -> axis
Dt = {m: rot(*AFc[m], m, l0) for m in range(4)}; C0 = Dt[0][0]; (C1, S1), (C2r, _), (C3, S3) = Dt[1], Dt[2], Dt[3]
gb = -4.0                                                                           # point core: gamma = -4 beta
beta = brentq(lambda b: series(b, gb * b, 0)["a2"] - C2r / C0, 1e-4, 1.0)
s1 = series(beta, gb * beta, 1.0); a2, g1, g3 = s1["a2"], s1["a1"], s1["a3"]
def F(x):
    amp, phi, eps = x
    return [C0 * (eps * g1 - amp * np.sin(phi) * (1 + a2 / 2)) - C1, -C0 * amp * np.cos(phi) * (1 - a2 / 2) - S1, C0 * (eps * g3 - amp * a2 / 2 *
np.sin(phi)) - C3]
amp, phi, eps = fsolve(F, [-0.06, 0.9, 0.3])
S3_pred = -C0 * amp * a2 / 2 * np.cos(phi); N = C0 / (1 - amp * eps * g1 * np.sin(phi) / 2)
K = beta / (3 + beta); mcc_lin = 2 * K / (1 / 0.01 - 1)
print(f"2l phase  : l0 = {l0:.2f} deg      2l amplitude: C2 = {np.hypot(*AFc[2]):.4f} -> beta = {beta:.3f}")
print(f"1l, 3l cos: Amp = {amp:.3f}, phi = {phi % (2*np.pi):.2f}, eps = {eps:.3f} -> D = R/eps = {RV/eps:.0f} Gpc, eps*Q = {eps*s1['Q']:.3f}")
print(f"3l sin    : predicted S3 = {S3_pred:.4f} vs data {S3:.4f} (not produced)     m=0: N = {N:.4f}")
print(f"mass      : K = {K:.4f} -> Mc/Mcrit(D) = {mcc_lin:.4f} (linear)")
# exact least squares: exact radial field with h(q) = 1 + beta (q-1) + (gamma/2)(q-1)^2, gamma = -4 beta
def A_radial(hfun, eps):
    x = 1 + eps * np.cos(TH); y = eps * np.sin(TH); qs = np.hypot(x, y); hs = hfun(qs); h1 = hfun(np.array([1.0]))[0]
    af = np.hypot(hs * x - h1, hs * y) / eps; return af / af.mean()
def model(p):
    l0_, b, e, am, ph = p; g = A_radial(lambda q: 1 + b * (q - 1) + 0.5 * gb * b * (q - 1)**2, e); return A_from(shifted(g, l0_), TH1 - l0_, am, ph)
def res(p):
    A = model(p); out = []
    for m in range(4):
        a_, b_ = harm(A, m); cs, sn = AFc[m]; out += [a_ - cs, b_ - sn] if m else [a_ - cs]
```

```
    return np.array(out)
ls = least_squares(res, [15, 0.16, 0.3, -0.05, 0.9], bounds=([-30, 0, 0.02, -0.3, -10], [60, 1, 0.95, 0.3, 10])).x
print(f"exact LSQ : l0 = {ls[0]:.1f}, beta = {ls[1]:.3f}, Amp = {ls[3]:.3f}, phi = {ls[4]:.2f}, D = {RV/ls[2]:.0f} Gpc")
```

```
2l phase  : l0 = 14.96 deg      2l amplitude: C2 = 0.0734 -> beta = 0.161
1l, 3l cos: Amp = -0.068, phi = 1.02, eps = 0.348 -> D = R/eps = 41 Gpc, eps*Q = -0.308
3l sin    : predicted S3 = 0.0013 vs data 0.0130 (not produced)     m=0: N = 0.9971
mass      : K = 0.0511 -> Mc/Mcrit(D) = 0.0010 (linear)
exact LSQ : l0 = 12.8, beta = 0.210, Amp = -0.072, phi = 1.03, D = 47 Gpc
```

Step 6 — Section 3: input parameters, term-by-term solve vs. Table 1

For each Newtonian fit, the unknowns $M_c/M_{\rm crit}(D)$, D, ℓ_0, Amp and φ are solved from C_m, S_m ($m = 1$–3) of A_F by least squares. Each solve uses the code's **exact** $h(q)$ from the integrator and keeps that fit's shell rule. The Table 1 columns are the full point fits; for LTB only the Table 1 column is shown.

Label: the masses in this section were computed with Earth's rate set to $H_E = 70$ km s^{-1} Mpc^{-1}, as in the original sheet ($\rho_s = \Omega_s \rho_c(70)$, $M_{\rm crit} = \frac{4\pi}{3}\rho_c(70) D^3$). They are shown unrescaled; Section 4 (Step 8) derives the masses from the anchored $\langle H \rangle$.

```
# Step 6a - term-by-term solve (exact h(q)) for the two Newtonian cases  [masses at H_E = 70: LABELLED, original-sheet convention]
RHOC70 = rhoc(70.0)                                          # <-- only for the Section 3 mass columns (original convention)
def af_code(D, mcc, Os, kin, kout):
    a = dict(DR=D / RV, Os=Os, mcc=mcc, rin=1 - kin * RV / D, rout=1 + kout * RV / D, Lc=0.005, mc=0.0, Ob=0.0)
    if not (0.01 < a["rin"] < 1): return None
    af = run_newt(a)["af"]; return af / af.mean() if np.all(np.isfinite(af)) else None
def hres(A):
    out = []
    for m in (1, 2, 3):
        a_, b_ = harm(A, m); cs, sn = AFc[m]; out += [a_ - cs, b_ - sn]
    return np.array(out)
def masses(D, mcc, Os, kin, kout, rho_c):
    Dm = D * GPCm; Mc = mcc * 4 * np.pi / 3 * rho_c * Dm**3; rhos = Os * rho_c; Rin, Rout = D - kin * RV, D + kout * RV
    Msh = lambda Rr: 4 * np.pi / 3 * rhos * ((min(max(Rr, Rin), Rout) * GPCm)**3 - (Rin * GPCm)**3)
    return dict(Mc=Mc, Mtot=Mc + Msh(D), Mshtot=Msh(Rout), Mvis=4 * np.pi / 3 * rhos * (RV * GPCm)**3, R_in=Rin, R_out=Rout)
def tbt(Os, kin, kout, x0):
    def res6(x):
        lm, lD, l0_, am, ph = x; A = af_code(10**lD, 10**lm, Os, kin, kout)
        return np.ones(6) if A is None else hres(A_from(shifted(A, l0_), TH1 - l0_, am, ph))
    x = least_squares(res6, x0, diff_step=1e-3, x_scale=[0.3, 0.1, 3, 0.03, 0.2]).x
    lm, lD, l0_, am, ph = x; D = 10**lD; A = af_code(D, 10**lm, Os, kin, kout); A = A_from(shifted(A, l0_), TH1 - l0_, am, ph)
    if am > 0: am, ph = -am, ph + np.pi
    return dict(D=D, mcc=10**lm, Os=Os, l0=l0_ % 360, amp=am, phi=ph % (2 * np.pi), R2=r2(MA, A[IMIG]), R2F=r2(AFfour(TH1), A), **masses(D, 10**lm,
Os, kin, kout, RHOC70))
TBT = {"free": tbt(T1["free"]["Os"], 1.0, 1.0, [np.log10(T1["free"]["mcc"]), np.log10(T1["free"]["D"]), 13, -0.12, 1.26]),
       "omb":  tbt(0.049, omb["kin"], omb["kout"], [np.log10(omb["mcc"]), np.log10(omb["D"]), 13, -0.06, 0.93])}
for kk, v in TBT.items(): print(kk, {q: float("%.4g" % v[q]) for q in ("D", "mcc", "l0", "amp", "phi", "R2", "R2F", "Mc", "Mtot")})
```

```
free {'D': 92.58, 'mcc': 0.00265, 'l0': 13.19, 'amp': -0.1246, 'phi': 1.267, 'R2': 0.9086, 'R2F': 0.9868, 'Mc': 2.382e+54, 'Mtot': 3.623e+56}
omb {'D': 37.03, 'mcc': 0.001637, 'l0': 12.75, 'amp': -0.06372, 'phi': 0.9413, 'R2': 0.8808, 'R2F': 0.9698, 'Mc': 9.415e+52, 'Mtot': 2.294e+54}
```

```
# Step 6b - Section 3 table (Table-1 columns from the fit files; masses at H_E = 70 as originally computed - NOT rescaled)
def col_t1(t):
    return dict(D=t["D"], R_in=t["Rin"], R_out=t["Rout"], Mc=t["Mc"], mcc=t["mcc"], Mtot=t["Mtot"], Mshtot=t["Mshtot"], Mvis=t["Mvis"],
                Os=t["Os"], l0=t["l0"], amp=t["amp"], phi=t["phi"], R2=t["R2"], R2F=t["R2F"])
COLS = [("Newt Ω_s free, term-by-term", TBT["free"]), ("Newt Ω_s free, Table 1", col_t1(T1["free"])),
        ("Newt Ω_s=0.049, term-by-term", TBT["omb"]), ("Newt Ω_s=0.049, Table 1", col_t1(T1["newt_omb"])), ("LTB, Table 1", col_t1(T1["rel"]))]
rows = [("D/R", lambda v: f"{v['D']/RV:.3f}"), ("D [Gpc]", lambda v: f"{v['D']:.1f}"), ("R_in [Gpc]", lambda v: f"{v['R_in']:.1f}"),
        ("R_out [Gpc]", lambda v: f"{v['R_out']:.1f}"), ("Mc [kg] (H_E=70)", lambda v: f"{v['Mc']:.2e}"), ("Mc [M_sun] (H_E=70)", lambda v: f"
{v['Mc']/MSUN:.2e}"),
        ("Mc/Mcrit(D)", lambda v: f"{v['mcc']:.3g}"), ("M(<D) total [kg] (H_E=70)", lambda v: f"{v['Mtot']:.2e}"),
        ("M_shell total [kg] (H_E=70)", lambda v: f"{v['Mshtot']:.2e}"), ("M within R [kg] (H_E=70)", lambda v: f"{v['Mvis']:.2e}"),
        ("Ω_s", lambda v: f"{v['Os']:.3f}"), ("ℓ0 [deg]", lambda v: f"{v['l0']:.1f}"), ("Amp", lambda v: f"{v['amp']:.4f}"),
        ("φ [rad]", lambda v: f"{v['phi']:.3f}"), ("R² vs Migkas", lambda v: f"{v['R2']:.3f}"), ("R² vs A_F", lambda v: f"{v['R2F']:.3f}")]
from IPython.display import Markdown, display
tab = "| Quantity | " + " | ".join(n for n, _ in COLS) + " |\n|" + "---|" * (len(COLS) + 1) + "\n"
tab += "\n".join("| " + lab + " | " + " | ".join(f(v) for _, v in COLS) + " |" for lab, f in rows)
display(Markdown(tab))
print("\nThe relativistic term-by-term column (D not fixed; R2 0.824/0.899) is left out, as in the v3 sheet; it can be restored.")
```

Quantity	Newt Ω_s free, term-by-term	Newt Ω_s free, Table 1	Newt Ω_s=0.049, term-by-term	Newt Ω_s=0.049, Table 1	LTB, Table 1
D/R	6.474	6.554	2.589	3.439	2.303
D [Gpc]	92.6	93.7	37.0	49.2	32.9
R_in [Gpc]	78.3	79.4	22.3	34.5	18.1
R_out [Gpc]	106.9	108.0	58.1	70.3	49.7
Mc [kg] (H_E=70)	2.38e+54	2.47e+54	9.41e+52	2.31e+53	3.38e+54
Mc [M_sun] (H_E=70)	1.20e+24	1.24e+24	4.73e+22	1.16e+23	1.70e+24
Mc/Mcrit(D)	0.00265	0.00265	0.00164	0.00171	0.0835
M(<D) total [kg] (H_E=70)	3.62e+56	3.72e+56	2.29e+54	4.56e+54	5.04e+54
M_shell total [kg] (H_E=70)	8.50e+56	8.71e+56	1.03e+55	1.70e+55	6.48e+54
M within R [kg] (H_E=70)	3.35e+54	3.35e+54	1.62e+53	1.62e+53	1.62e+53
Ω_s	1.012	1.012	0.049	0.049	0.049
ℓ0 [deg]	13.2	12.8	12.8	13.0	13.5
Amp	-0.1246	-0.1240	-0.0637	-0.0622	-0.0558
ϕ [rad]	1.267	1.264	0.941	0.929	0.848
R^2 vs Migkas	0.909	0.909	0.881	0.893	0.885
R^2 vs A_F	0.987	0.987	0.970	0.977	0.963

```
The relativistic term-by-term column (D not fixed; R2 0.824/0.899) is left out, as in the v3 sheet; it can be restored.
```

Step 7 — Anchoring $\langle H \rangle$ on Migkas's own H_0 values

The fits fix only the **shape** $A(\ell)/\langle A \rangle$. The absolute sky-mean rate comes from Migkas et al.'s own H_0 numbers by fitting

$$H(\ell) = \langle H \rangle \, \frac{A(\ell)}{\langle A \rangle}$$

by weighted least squares, $\langle H \rangle = \sum_i w_i H_i m_i / \sum_i w_i m_i^2$ with $w_i = 1/\sigma_i^2$. Here m_i is the model at the region's cone centre, or the model mean outside the cone for "rest of sky" (75° cones for LX–T and the joint result, 60° for YSZ–T, 90° for "opposite hemisphere"). Only ℓ is used because the model is planar. The values below were checked against the arXiv full texts (2004.03305 §7, §8.1–8.2; 2103.13904 §6.1–6.4, App. B). The **representative value** is the 2021 all-combined joint result.

Caveat: Migkas's absolute H_0 values rest on their fiducial calibration ($H_0 = 70$ assumed when calibrating the scaling relations; M21 §3), so the anchor inherits that calibration. The H0 maps are published only as figures, not tables, so they are not used.

```
# Step 7 - all 11 Migkas pairs -> <H> for each fit
CH = [  # (label, [(H0, sigma, kind, l, cone)])   kind: 'pt' = cone centre, 'out' = outside cone / opposite hemisphere
 ("M20 our sample LX-T, max/min",           [(75.17, 1.81, "pt", 17, 0), (66.20, 1.72, "pt", 274, 0)]),
 ("M20 ACC LX-T, max/min",                  [(78.76, 4.15, "pt", 77, 0), (58.12, 2.68, "pt", 317, 0)]),
 ("M20 XCS-DR1 LX-T, max/min",              [(77.91, 2.20, "pt", 31, 0), (63.56, 2.32, "pt", 281, 0)]),
 ("M20 joint (3 samples), max/min",         [(76.64, 1.41, "pt", 34, 0), (65.20, 1.48, "pt", 303, 0)]),
 ("M21 our LX-T, region/rest",              [(66.0, 1.7, "pt", 274, 0), (72.1, 1.4, "out", 274, 75)]),
 ("M21 ours+ACC LX-T, region/rest",         [(66.2, 1.6, "pt", 284, 0), (72.7, 1.5, "out", 284, 75)]),
 ("M21 our YSZ-T, region/rest",             [(62.3, 2.2, "pt", 268, 0), (72.1, 0.9, "out", 268, 60)]),
 ("M21 LBCG-T, region/opp. hemisphere",     [(66.2, 2.5, "pt", 257, 0), (72.6, 2.3, "out", 257, 90)]),
 ("M21 ACC LX-T, region/rest",              [(61.3, 4.2, "pt", 318, 0), (72.2, 2.4, "out", 318, 75)]),
 ("M21 ACC YSZ-T, region/opp. hemisphere",  [(60.6, 3.6, "pt", 311, 0), (73.4, 1.9, "out", 311, 90)]),
 ("M21 ALL COMBINED (joint), region/rest",  [(66.5, 1.0, "pt", 273, 0), (72.8, 0.6, "out", 273, 75)]),
]
SHAPE = {kk: CURVE[kk] / CURVE[kk].mean() for kk in ("free", "omb", "rel")}
def m_of(s, kind, l, th):
    if kind == "pt": return s[int(round(l)) % 360]
    dl = np.abs((TH1 - l + 180) % 360 - 180); return s[dl > th].mean()
HM = {kk: [] for kk in SHAPE}
print("choice".ljust(42) + "Newt free    Newt 0.049   LTB")
for lab, pts in CH:
    line = lab.ljust(42)
    for kk, s in SHAPE.items():
        m = np.array([m_of(s, kd, l, th) for _, _, kd, l, th in pts]); H = np.array([p[0] for p in pts]); wt = 1 / np.array([p[1] for p in pts])**2
        Hm = (wt * H * m).sum() / (wt * m * m).sum(); err = 1 / np.sqrt((wt * m * m).sum()); HM[kk].append((Hm, err)); line += f"{Hm:6.2f}±{err:.2f}
"
    print(line)
ANCH = {kk: dict(H=v[-1][0], err=v[-1][1], mean=np.mean([x[0] for x in v]), sd=np.std([x[0] for x in v], ddof=1)) for kk, v in HM.items()}
```

```
print("\nrepresentative (M21 joint):", {kk: f"{a['H']:.1f}±{a['err']:.1f}" for kk, a in ANCH.items()})
print("mean ± s.d. of all 11     :", {kk: f"{a['mean']:.1f}±{a['sd']:.1f}" for kk, a in ANCH.items()})
```

```
choice                                  Newt free    Newt 0.049   LTB
M20 our sample LX-T, max/min             71.84±1.27   71.16±1.26   71.01±1.26
M20 ACC LX-T, max/min                    64.03±2.24   64.73±2.26   65.11±2.28
M20 XCS-DR1 LX-T, max/min                71.46±1.60   71.05±1.59   71.00±1.59
M20 joint (3 samples), max/min           69.55±0.99   69.61±0.99   69.68±1.00
M21 our LX-T, region/rest                71.78±1.11   71.53±1.11   71.58±1.11
M21 ours+ACC LX-T, region/rest           71.94±1.13   71.86±1.13   72.04±1.13
M21 our YSZ-T, region/rest               70.07±0.82   69.94±0.82   69.96±0.82
M21 LBCG-T, region/opp. hemisphere       72.72±1.77   72.40±1.76   72.13±1.75
M21 ACC LX-T, region/rest                69.06±2.07   69.32±2.08   69.44±2.08
M21 ACC YSZ-T, region/opp. hemisphere    70.24±1.67   70.49±1.68   70.70±1.68
M21 ALL COMBINED (joint), region/rest    71.79±0.52   71.61±0.52   71.63±0.52

representative (M21 joint): {'free': '71.8±0.5', 'omb': '71.6±0.5', 'rel': '71.6±0.5'}
mean ± s.d. of all 11     : {'free': '70.4±2.4', 'omb': '70.3±2.1', 'rel': '70.4±2.0'}
```

Step 8 — Section 4: everything that follows from $\langle H \rangle$

Newtonian. $\langle H \rangle$ is the sky mean of the full curve $a_f[1 - \mathrm{Amp}\sin(\psi + \varphi)]$ in code units ($1/t_0 = k$), so $\langle H \rangle / k$ is a pure number and

$$k = \frac{\langle H \rangle}{\langle H \rangle / k}, \quad t_0 = k^{-1}, \quad H_E = (H_E/k)\,k, \quad \rho_s = \Omega_s \rho_c(H_E), \quad M_{\mathrm{crit}} = \tfrac{4\pi}{3}\rho_c(H_E) D^3,$$

$$V_O = v_E\, k\, D, \qquad \text{bulk flow} = \left|\tfrac{\epsilon_b^2}{10}(h'' + 4h')\right| V_O,\ \epsilon_b = r_b/D,\ r_b = 150\,h^{-1}\,\mathrm{Mpc} \text{ with } h = \langle H \rangle / 100.$$

LTB. $\langle H \rangle$ is the in-plane mean of the local rate, $(H_\perp + H_r)/2$. From the LTB run, $\langle H \rangle t_0 = 0.956$ (dimensionless). Masses are rescaled from the fit's unit convention (3D-mean local rate 70) by $(\langle H \rangle / 72.04)^2$. This rescaling is approximate in GR, and the bulk flow is not computed.

Accepted values: Planck 2018 VI ($H_0 = 67.36 \pm 0.54$, $\Omega_b h^2 = 0.02237$, $\Omega_m = 0.3153$, age 13.797 Gyr); SH0ES (73.04 ± 1.04); Planck 2018 I (CMB dipole 369.82 ± 0.11 km/s; Local Group 620 ± 15 km/s); Watkins 2023 (bulk flow 395 ± 29 km/s at $150\,h^{-1}$ Mpc).

```
# Step 8a - accepted values (derived masses use Planck's own H0 = 67.36, not 70)
Pl = dict(H0=67.36, Obh2=0.02237, Om=0.3153, age=13.797); VOL = 4 * np.pi / 3 * (RV * GPCm)**3
Ob = Pl["Obh2"] / (Pl["H0"] / 100)**2; rcP = rhoc(Pl["H0"])
ACC = dict(Ob=Ob, rho_b=Ob * rcP, rhoc=rcP, Mb=Ob * rcP * VOL, Mm=Pl["Om"] * rcP * VOL, vcmb=369.82, vlg=620.0, bulk=395.0)
print({k: float("%.4g" % v) for k, v in ACC.items()})
```

```
{'Ob': 0.0493, 'rho_b': 4.202e-28, 'rhoc': 8.523e-27, 'Mb': 1.512e+53, 'Mm': 9.672e+53, 'vcmb': 369.8, 'vlg': 620.0, 'bulk': 395.0}
```

```
# Step 8b - model columns from the anchored <H> (no H0 = 70 input for the Newtonian columns)
def radial_h(a, r, qs):                                  # h(q) = V(q)/q along the CoE-Earth axis, code units
    q = np.vstack([[1.0, 0.0], np.c_[qs, np.zeros_like(qs)]])
    X, V, VR = P.integrate(q, r["mu"], P.RG, r["MT"], 0.0, r["Rb"], 0.01, 200); return V[1:, 0] / qs
```

```
SEC4 = {}
for kk, a in NEWT.items():
    r = RUN[kk]
    Hk = A_from(shifted(r["af"], a["l0"]), TH1 - a["l0"], a["amp"], a["phi"]).mean()   # <H>/k: sky mean of af x sine factor, code units (1/t0 = k)
    H = ANCH[kk]["H"]; k = H / Hk; HE = r["Ht"] * k; D = a["DR"] * RV
    kin, kout = (1 - a["rin"]) * a["DR"], (a["rout"] - 1) * a["DR"]           # shell edges in units of R: R_in = D - kin R, R_out = D + kout R
    ms = masses(D, a["mcc"], a["Os"], kin, kout, rhoc(HE))                    # rho_c at Earth's rate H_E (the code's Omega reference)
    qs = np.linspace(0.95, 1.05, 41); h = radial_h(a, r, qs); cf = np.polyfit(qs - 1, h, 6); h1 = np.polyval(cf, 0)
    hp = np.polyval(np.polyder(cf), 0) / h1; hpp = np.polyval(np.polyder(cf, 2), 0) / h1
    VO = r["vE"] * k * D * 1000; eb = 0.150 / (H / 100) / D
    SEC4[kk] = dict(H=H, Hk=Hk, k=k, t0=977.8 / k, rho_s=a["Os"] * rhoc(HE), Mvis=ms["Mvis"], Mc=ms["Mc"], Mtot=ms["Mtot"], Os=a["Os"], Os_H=a["Os"] *
(HE / H)**2,
                    rhoc=rhoc(H), VO=VO, bulk=abs(eb**2 / 10 * (hpp + 4 * hp)) * VO)
# LTB: dimensionless <H> t0 from the run; masses from the fit's unit convention (3D local mean = 70) rescaled by (<H>/<H>_fit)^2
Hloc_fit = (ltb["H_perp"] + ltb["H_r"]) / 2; Ht0 = Hloc_fit * ltb["age_Gyr"] / 977.8; H = ANCH["rel"]["H"]; g = H / Hloc_fit; t3 = T1["rel"]
SEC4["rel"] = dict(H=H, Hk=Ht0, k=H / Ht0, t0=977.8 * Ht0 / H, rho_s=0.049 * rhoc(70.0) * g**2, Mvis=t3["Mvis"] * g**2, Mc=t3["Mc"] * g**2,
Mtot=t3["Mtot"] * g**2,
                   Os=0.049, Os_H=0.049 * (70.0 / Hloc_fit)**2,   # 70 = LTB fit's unit convention (3D local mean), labelled
                   rhoc=rhoc(H), VO=t3["vE"] * c * g, bulk=None)
print(f"LTB: <H>_fit = {Hloc_fit:.2f} (fit units), <H> t0 = {Ht0:.3f}, rescale g = {g:.4f}")
```

```
LTB: <H>_fit = 72.04 (fit units), <H> t0 = 0.956, rescale g = 0.9943
```

```
# Step 8c - Section 4 table
F4 = [("(H) [km/s/Mpc]", "67.36±0.54 (CMB); 73.04±1.04 (local)", lambda v: f"{v['H']:.1f}  (= {v['Hk']:.3f} k)" if v is not SEC4['rel'] else f"
{v['H']:.1f}  (= {v['Hk']:.3f}/t0)"),
      ("k (LTB: 1/t0) [km/s/Mpc]", "-", lambda v: f"{v['k']:.1f}"),
      ("Age / k^-1 = t0 [Gyr]", "13.797±0.023", lambda v: f"{v['t0']:.2f}"),
      ("Ordinary mass within R [kg]", f"{ACC['Mb']:.2e}", lambda v: f"{v['Mvis']:.2e}"),
      ("Total matter within R [kg]", f"{ACC['Mm']:.2e}", lambda v: "same (no DM)"),
      ("rho_b [kg/m^3]", f"{ACC['rho_b']:.2e}", lambda v: f"{v['rho_s']:.2e}"),
      ("Omega_b (= Omega_m in model)", f"{ACC['Ob']:.4f}; Om 0.3153", lambda v: f"{v['Os']:.3f} ({v['Os_H']:.3f} vs ⟨H⟩)"),
      ("rho_c at ⟨H⟩ [kg/m^3]", f"{ACC['rhoc']:.2e} (67.36)", lambda v: f"{v['rhoc']:.2e}"),
      ("Earth speed vs CMB V_O", "369.82±0.11 km/s", lambda v: f"{v['VO']:.2e} km/s = {v['VO']/c:.1f}c"),
      ("V_O / CMB dipole", "1", lambda v: f"{v['VO']/369.82:,.0f}"),
      ("Local Group vs CMB", "620±15 km/s", lambda v: "not modelled"),
      ("Bulk flow, 150/h Mpc [km/s]", "395±29", lambda v: "not computed" if v["bulk"] is None else f"{v['bulk']:.2g}"),
      ("Core mass Mc [kg] (x M_b)", "-", lambda v: f"{v['Mc']:.2e} ({v['Mc']/ACC['Mb']:.1f}x)"),
      ("Mass inside D [kg] (x M_b)", "-", lambda v: f"{v['Mtot']:.2e} ({v['Mtot']/ACC['Mb']:,.0f}x)")]
from IPython.display import Markdown, display
tab = "| Quantity | Accepted | Newt Ω_s free (0.909) | Newt Ω_s=0.049 (0.893) | LTB (0.885) |\n|---|---|---|---|---|\n"
tab += "\n".join("| " + lab + " | " + accv + " | " + " | ".join(f(SEC4[kk]) for kk in ("free", "omb", "rel")) + " |" for lab, accv, f in F4)
display(Markdown(tab))
json.dump(dict(anchor=ANCH, sec4=SEC4, tbt=TBT, R2={kk: r2(MA, CURVE[kk][IMIG]) for kk in CURVE}),
open("/workspace/cosmology/papers/A_equals_fourier/steps_v3/notebook_results.json", "w"), indent=1, default=float)
```

Quantity	Accepted	Newt Ω_s free (0.909)	Newt Ω_s=0.049 (0.893)	LTB (0.885)
⟨H⟩ [km/s/Mpc]	67.36±0.54 (CMB); 73.04±1.04 (local)	71.8 (= 0.971 k)	71.6 (= 1.012 k)	71.6 (= 0.956/t0)
k (LTB: 1/t0) [km/s/Mpc]	-	74.0	70.8	74.9
Age / k^-1 = t0 [Gyr]	13.797±0.023	13.22	13.81	13.05
Ordinary mass within R [kg]	1.51e+53	3.67e+54	1.64e+53	1.60e+53
Total matter within R [kg]	9.67e+53	same (no DM)	same (no DM)	same (no DM)
rho_b [kg/m^3]	4.20e-28	1.02e-26	4.56e-28	4.46e-28
Omega_b (= Omega_m in model)	0.0493; Om 0.3153	1.012 (1.053 vs ⟨H⟩)	0.049 (0.047 vs ⟨H⟩)	0.049 (0.046 vs ⟨H⟩)
rho_c at ⟨H⟩ [kg/m^3]	8.52e-27 (67.36)	9.68e-27	9.63e-27	9.64e-27
Earth speed vs CMB V_O	369.82±0.11 km/s	6.23e+06 km/s = 20.8c	3.26e+06 km/s = 10.9c	2.29e+06 km/s = 7.6c
V_O / CMB dipole	1	16,848	8,804	6,199
Local Group vs CMB	620±15 km/s	not modelled	not modelled	not modelled
Bulk flow, 150/h Mpc [km/s]	395±29	5.7	0.32	not computed
Core mass Mc [kg] (x M_b)	-	2.71e+54 (17.9x)	2.33e+53 (1.5x)	3.34e+54 (22.1x)
Mass inside D [kg] (x M_b)	-	4.07e+56 (2,692x)	4.61e+54 (30x)	4.98e+54 (33x)

Summary

- Headline fits (full curve with sine, vs 71 Migkas points): R^2 = 0.909 (Newtonian, Ω_s free), 0.893 (Newtonian, $\Omega_s = 0.049$), 0.885 (LTB).
- Migkas-anchored sky-mean rate (2021 joint result): $\langle H \rangle \approx 71.8/71.6/71.6$ km s^{-1} Mpc^{-1}; the mean over all 11 Migkas pairs is $\approx 70.4 \pm 2$.
- With that anchor, $t_0 = k^{-1} \approx 13.2/13.8/13.05$ Gyr. Earth's speed about the CoE is $V_O \approx 21c/11c/7.6c$, thousands of times the CMB dipole speed. This is the model's main tension.
- The absolute anchor inherits Migkas's fiducial calibration.

Part B — Loops and maps

Part B redraws every loop and map figure made earlier in the project, using **the latest script for each figure, run verbatim**. The only edit to each script is where it writes its output: the images go to `steps_v3/figs/` so the approved originals on disk are not overwritten. That edit is

listed in each step. After each script runs, the new image is shown next to the approved reference image and compared pixel by pixel.

Shared geometry. With a pure CoE flow, A is axisymmetric about the CoE axis, so any value read at angle θ from the anti-CoE direction holds on the whole small circle (cone) of half-angle θ about that direction. The model $A(\ell)$ after the 2ℓ step has two maxima ($\theta \approx 30^\circ$, 185°) and two minima ($\theta \approx 115^\circ$, 265°). Each one becomes a loop: a transparent band of $\pm 9^\circ$ (18° end to end, 5% of 360°) about the centre θ, plus one solid mid-line. The axis can be tilted to latitude b_0 and rolled about itself. Points on a loop are $\hat{n} = \cos\theta\,\hat{a} + \sin\theta(\cos\psi\,\hat{e}_1 + \sin\psi\,\hat{e}_2)$, with $\hat{a}$ the anti-CoE unit vector and $\hat{e}_{1,2}$ perpendicular to it. They are converted to Galactic (ℓ, b) and drawn on a Mollweide projection (equal-area, Snyder 1987, *Map Projections — A Working Manual*, USGS PP 1395) with ℓ increasing to the left, as in Migkas Fig. 8.

Data. Migkas et al. 2020 (A&A 636, A15; arXiv:2004.03305): Table 2 directions of the A extrema, and the Fig. 8 $\theta = 75^\circ$ $A/A_{\rm all}$ map (background image loaded from `/workspace/cosmology/map_compare/FULL_migkas_theta75.png`). Markers: MAX at $(17^\circ, -9^\circ)$ and $(185^\circ, 0^\circ)$; MIN at $(115^\circ, 0^\circ)$ and $(274^\circ, -22^\circ)$.

```
# Part B setup - run each figure script verbatim in its own namespace; outputs go to steps_v3/figs
import os, sys, shlex, matplotlib, numpy as np
from pathlib import Path
from PIL import Image
from IPython.core.magic import register_cell_magic
from IPython.display import display
FIGS = Path("/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs"); FIGS.mkdir(exist_ok=True)
for d in ("journal_rows", "stack_rows"): (FIGS / d).mkdir(exist_ok=True)
FIGDIR = "/workspace/cosmology/figures"
if FIGDIR not in sys.path: sys.path.insert(0, FIGDIR)

@register_cell_magic
def runscript(line, cell):
    """%%runscript <script name> [--cwd DIR] [--argv ARGS...]: exec the cell as that script (__name__ == '__main__'),
    __file__ placed in steps_v3/figs (so OUT_DIR = figs), matplotlib defaults, then restore cwd/argv/rc/modules."""
    tok = shlex.split(line); name = tok[0]; cwd = FIGS; argv = []
    if "--cwd" in tok: cwd = Path(tok[tok.index("--cwd") + 1])
    if "--argv" in tok: argv = tok[tok.index("--argv") + 1:]
    old = (os.getcwd(), sys.argv[:], matplotlib.rcParams.copy(), dict(sys.modules), sys.path[:])
    import matplotlib.pyplot as plt
    try:
        os.chdir(cwd); sys.argv = [name] + argv; matplotlib.rcdefaults(); matplotlib.use("Agg")
        g = {"__name__": "__main__", "__file__": str(FIGS / name)}
        exec(compile(cell, str(FIGS / name), "exec"), g)
    finally:
        plt.close("all"); os.chdir(old[0]); sys.argv = old[1]; matplotlib.rcParams.update(old[2])
        for k in list(sys.modules):
            if k not in old[3] and (k.startswith("coe_") or k.startswith("make_")): del sys.modules[k]
        for k, v in old[3].items():
            if k.startswith("coe_") or k.startswith("make_"): sys.modules[k] = v
        sys.path[:] = old[4]
        if FIGDIR not in sys.path: sys.path.insert(0, FIGDIR)

MATCH = []
```

```
def compare(new, ref, label=None):
    """approved reference (left) | notebook re-run (right), side by side; prints sizes and pixel differences."""
    from IPython.display import Image as IImage
    a = Image.open(ref).convert("RGB"); b = Image.open(new).convert("RGB")
    same = a.size == b.size; bb = b if same else b.resize(a.size, Image.LANCZOS)
    d = np.abs(np.asarray(a, float) - np.asarray(bb, float)); mad = d.mean(); frac = (d.max(axis=2) > 40).mean()
    verdict = "identical" if d.max() == 0 else ("match (tiny rendering differences)" if frac < 0.005 else "DIFFERS")
    MATCH.append(dict(figure=label or Path(new).name, reference=str(ref), size_ref=a.size, size_new=b.size, mean_abs_diff=round(mad, 3),
frac_px_diff=round(frac, 5), verdict=verdict))
    print(f"{Path(new).name}: new {b.size} vs ref {a.size} | mean |diff| = {mad:.3f}/255, pixels differing >40: {100*frac:.3f}% -> {verdict}")
    print("left: approved reference " + str(ref).replace("/workspace/cosmology/", "") + "   |   right: notebook re-run steps_v3/figs/" +
Path(new).name)
    W = 1100; h = round(a.size[1] * W / a.size[0]); pair = Image.new("RGB", (2 * W + 24, h), "white")
    pair.paste(a.resize((W, h), Image.LANCZOS), (0, 0)); pair.paste(b.resize((W, h), Image.LANCZOS), (W + 24, 0))
    out = FIGS / ("cmp_" + Path(new).stem + ".jpg"); pair.save(out, quality=85); display(IImage(filename=str(out)))
```

Step 9 — Mollweide maps of Migkas Table 2 A extrema, and Fig. 8 with the published markers

What it shows. (i) A clean 2D Galactic Mollweide map of the Migkas et al. 2020 Table 2 A maximum and minimum directions for every sample and cone size, with no CoE loops and no model curves. (ii) The same published MAX/MIN markers placed on the real Migkas Fig. 8 ($\theta = 75^\circ$) $A/A_{\rm all}$ map. **Geometry.** Mollweide: $x = \frac{2\sqrt{2}}{\pi}\lambda\cos\vartheta$, $y = \sqrt{2}\sin\vartheta$, with $2\vartheta + \sin 2\vartheta = \pi \sin b$ and $\lambda = -\mathrm{wrap}(\ell)$, so $\ell = 0$ is at the centre and ℓ increases to the left. The Fig. 8 image is registered by its oval: centre and semi-axes $(c_x, c_y, R_x, R_y) = (863.0, 763.25, 693.99, 392.54)$ px.

Source script: `/workspace/cosmology/figures/make_migkas_A_extrema_maps.py` (last modified 2026-09-25 20:22 CT), run verbatim. *Output-path edit:* none. The script's `OUT_DIR` follows `__file__`, which is set to `steps_v3/figs/`. *Reference image:* `figures/migkas_A_extrema_table2_mollweide.png`, `figures/migkas_fig8_theta75_A_extrema_only.png`.

```
%%runscript make_migkas_A_extrema_maps.py
#!/usr/bin/env python3
"""
Clean 2D Galactic Mollweide maps of Migkas et al. 2020 Table 2 A extrema.

Deliverables (this directory):
  migkas_A_extrema_table2_mollweide.png / .pdf
  migkas_fig8_theta75_A_extrema_only.png
  migkas_A_extrema_table2.json

Convention (match Migkas Fig. 8 / existing overlay):
  lon_disp = -wrap(ℓ)  →  ℓ=0 at center, ℓ increases LEFT
  (ℓ=90 left of center, ℓ=270 right of center)

No CoE rings, no model curves — published Table 2 directions only.
"""
```

```
from __future__ import annotations

import json
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
from matplotlib.lines import Line2D
from matplotlib.patches import Ellipse, FancyBboxPatch
from PIL import Image

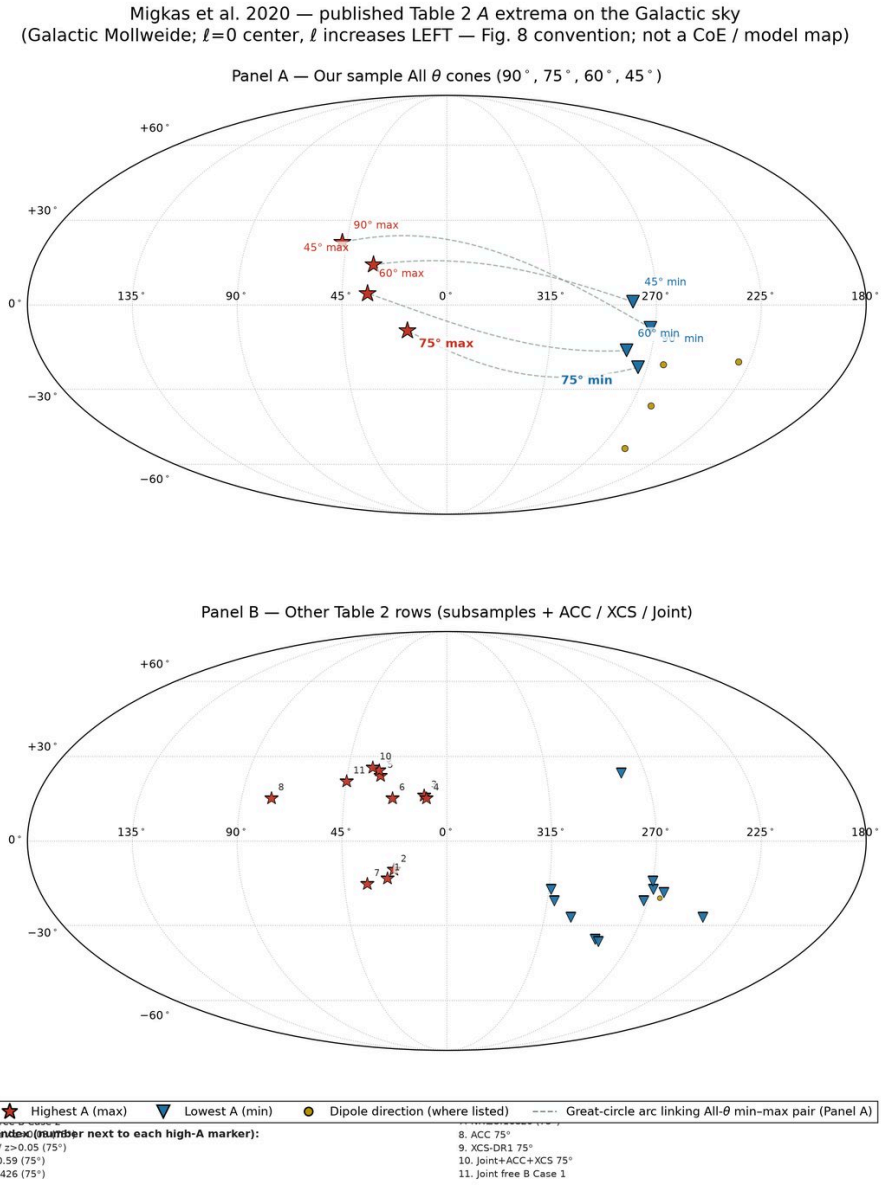
OUT_DIR = Path(__file__).resolve().parent
DPI = 220

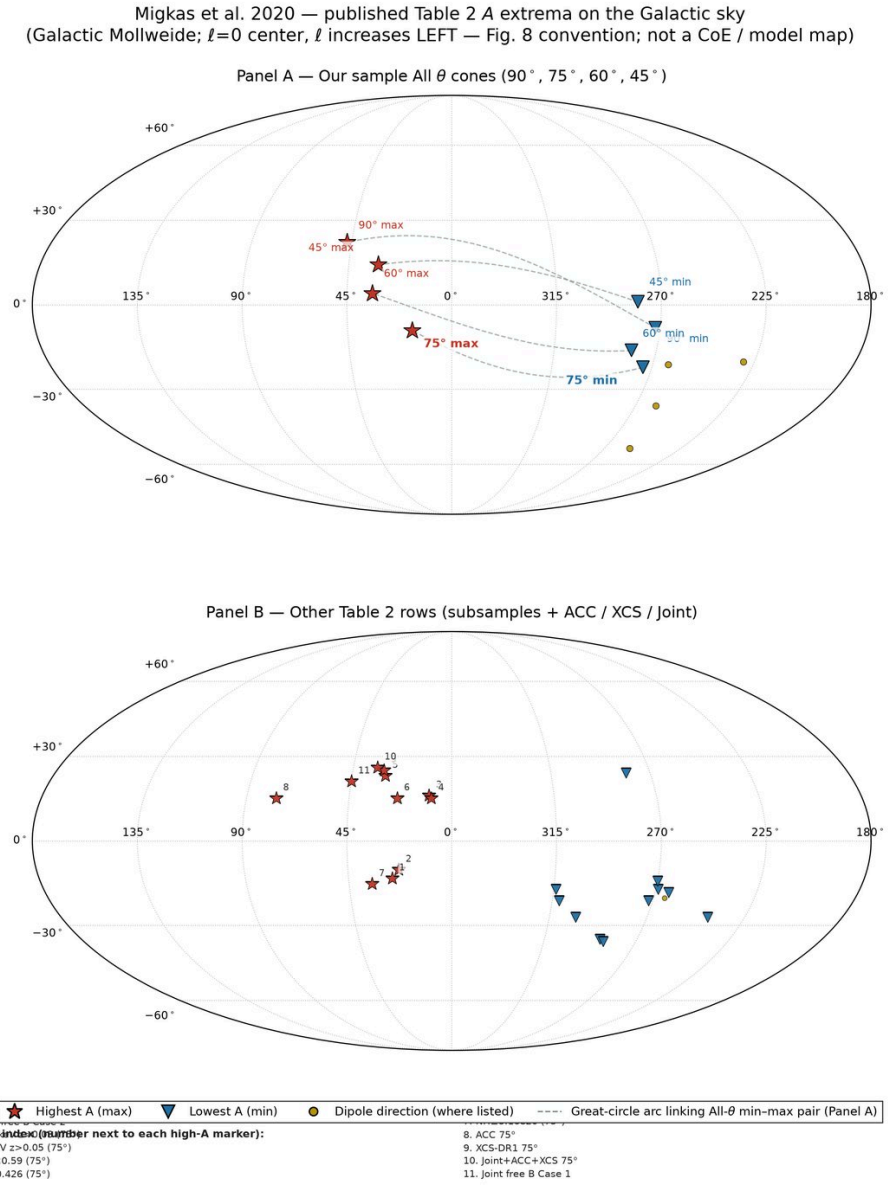
# ... 740 more lines of the verbatim script: see coe_steps_v3.ipynb / .html / .md (or coe_steps_v3_fullcode.pdf)
```

```
wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/migkas_A_extrema_table2_mollweide.png
wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/migkas_A_extrema_table2_mollweide.pdf
Fig8 source=FULL_migkas_theta75.png size=1975x1353; max px=(798, 812) RGB=[237.  166.3   0. ]; min px=(1179, 881) RGB=[ 96.6   0.  205.4]; dipole px=
(1221, 875); warm_check=OK
wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/migkas_fig8_theta75_A_extrema_only.png
wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/migkas_A_extrema_table2.json
--- done ---
/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/migkas_A_extrema_table2_mollweide.png
/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/migkas_A_extrema_table2_mollweide.pdf
/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/migkas_fig8_theta75_A_extrema_only.png
/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/migkas_A_extrema_table2.json
```

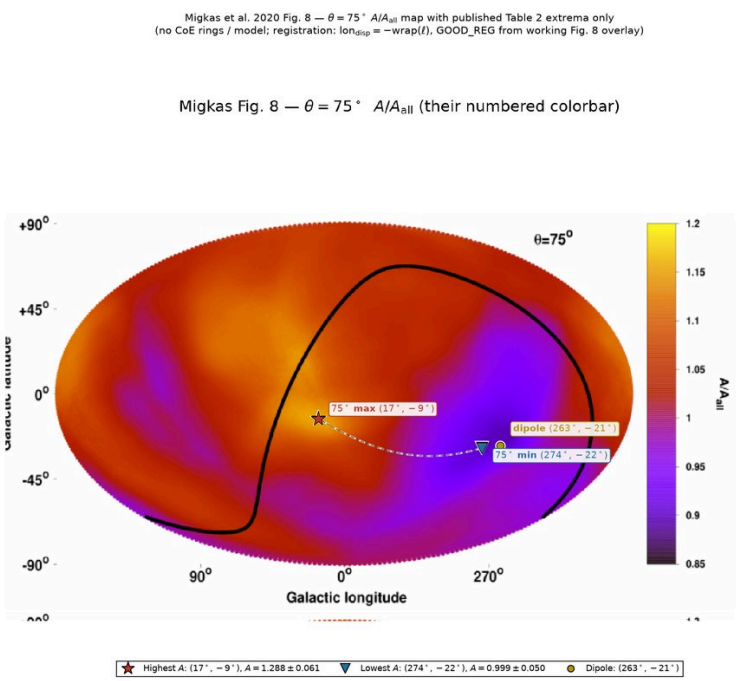
```
compare(FIGS / 'migkas_A_extrema_table2_mollweide.png', '/workspace/cosmology/figures/migkas_A_extrema_table2_mollweide.png')
compare(FIGS / 'migkas_fig8_theta75_A_extrema_only.png', '/workspace/cosmology/figures/migkas_fig8_theta75_A_extrema_only.png')
```

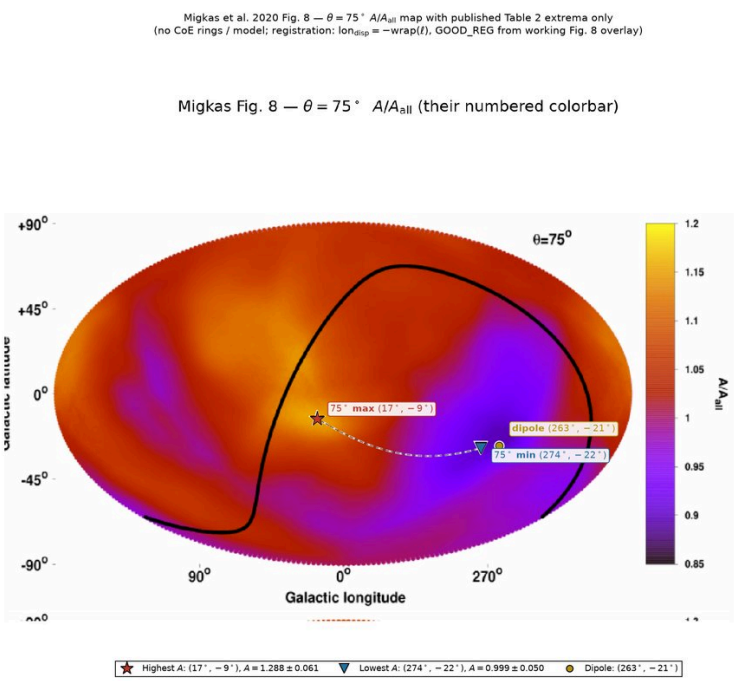
```
migkas_A_extrema_table2_mollweide.png: new (2079, 2612) vs ref (2079, 2612) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference figures/migkas_A_extrema_table2_mollweide.png   |   right: notebook re-run steps_v3/figs/migkas_A_extrema_table2_mollweide.png
```





migkas_fig8_theta75_A_extrema_only.png: new (1975, 1486) vs ref (1975, 1486) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference figures/migkas_fig8_theta75_A_extrema_only.png | right: notebook re-run steps_v3/figs/migkas_fig8_theta75_A_extrema_only.png





Step 10 — First three-panel loop figure (axis tilted $+30^\circ$)

What it shows. Panel A: the flat model-frame angle circle with the four extrema bands (two reds for MAX, two blues for MIN) and Galactic ℓ ticks. Panel B: the four loops on a Galactic Mollweide map with the Migkas markers. Panel C: the same loops on a 3D sphere. In this first version the CoE axis is tilted $+30^\circ$: anti-CoE at $(0^\circ, +30^\circ)$, CoE at $(180^\circ, -30^\circ)$. **Maths.** Loop centres $\theta = 30^\circ, 185^\circ$ (MAX) and $115^\circ, 265^\circ$ (MIN) from the anti-CoE direction. Bands $[\theta - 9^\circ, \theta + 9^\circ]$. A small circle of radius θ about $\hat{a}$ is traced by $\psi \in [0, 2\pi)$.

Source script: `/workspace/cosmology/figures/make_coe_rings_migkas_pin_axis.py` (last modified 2026-09-25 20:22 CT), run verbatim. *Output-path edit:* none. The script's `OUT_DIR` follows `__file__`, which is set to `steps_v3/figs/`. *Reference image:* `figures/coe_rings_migkas_pin_axis.png`.

```
%%runscript make_coe_rings_migkas_pin_axis.py
#!/usr/bin/env python3
"""
CoE rings + Migkas A(ell) extrema loops in the Galactic frame.

Concept:
  Pure CoE axisymmetry → constant A on small-circle rings about x = CoE↔anti-CoE.
  Scan A vs angle from +x (anti-CoE) around 360°. After-2ℓ A(ℓ) has TWO maxima
```

```
  and TWO minima (from best_fit_2ell_multiplier.npz A_after):
    MAX at θ = 30° and 185° from +x,
    MIN at θ = 115° and 265° from +x.

  Exactly FOUR loops — each ONE transparent highlight band, 18° end-to-end
  (= ±9° = 5% of 360°) about the center θ:
    MAX: 30±9 → [21°, 39°] and 185±9 → [176°, 194°]
    MIN: 115±9 → [106°, 124°] and 265±9 → [256°, 274°]
  Drawing: ONE smooth transparent filled ribbon between θ−9° and θ+9°
  + ONE solid mid-stroke at the exact center θ. No multi-stroke wash,
  no dense linspace hairball, no dual strong boundaries.
  Four distinct colors (two reds + two blues). No orange, no 283° companion.

  Galactic orientation: anti-CoE (+x) at ℓ=0°, b=+30°; CoE (−x) at ℓ=180°, b=−30°
  (axis tilted +30° so features near ℓ=0 move north on Mollweide; Migkas markers fixed).
  Sky markers (ℓ, b):
    Maxes: map extrema (17°, −9°) for ~30° max; strip (185°, 0°) for second max.
    Mins:  strip (115°, 0°) for ~115° min; map extrema (274°, −22°) for ~265° min.

Outputs:
  coe_rings_migkas_pin_axis.png
  coe_rings_migkas_pin_axis.json

# ... 810 more lines of the verbatim script: see coe_steps_v3.ipynb / .html / .md (or coe_steps_v3_fullcode.pdf)
```

```
{
  "png": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_migkas_pin_axis.png",
  "script": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/make_coe_rings_migkas_pin_axis.py",
  "migkas_max_lb_deg": [
    [
      17.0,
      -9.0
    ],
    [
      185.0,
      0.0
    ]
  ],
  "migkas_min_lb_deg": [
    [
      115.0,
      0.0
    ],
    [
      274.0,
      -22.0
    ]
  ],
  "strip_theta_max_deg": [
    30.0,
    185.0
  ],
  "strip_theta_min_deg": [
    115.0,
    265.0
  ],
  "band_half_width_deg": 9.0,
  "band_full_width_deg": 18.0,
  "band_total_width_deg": 18.0,
  "band_width_note": "5% of 360\u00b0 = 18\u00b0 end-to-end = \u00b19\u00b0 about each MAX/MIN center",
  "max_bands_deg": [
    [
      21.0,
      39.0
    ],
    [
      176.0,
      194.0
    ]
  ],
  "min_bands_deg": [
    [
      106.0,
      124.0
    ],
    [
      256.0,
      274.0
    ]
  ],
  "band_intervals": {
    "MAX_30": [
```

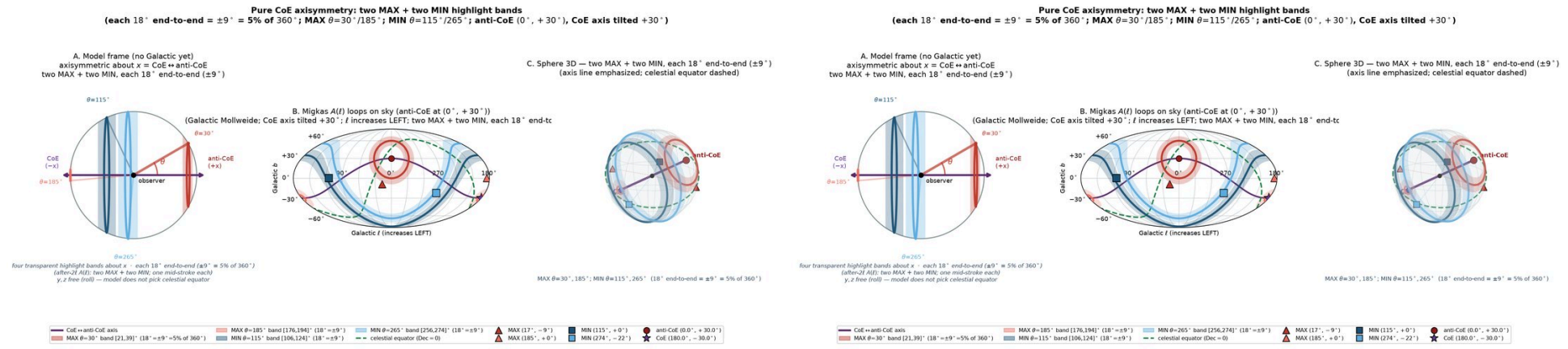
```
      21.0,
      39.0
    ],
    "MAX_185": [
      176.0,
      194.0
    ],
    "MIN_115": [
      106.0,
      124.0
    ],
    "MIN_265": [
      256.0,
      274.0
    ]
  },
  "scan_stop_source": "best_fit_2ell_multiplier.npz A_after extrema (two max, two min); each drawn as one transparent 18\u00b0 end-to-end highlight
(\u00b19\u00b0 = 5% of 360\u00b0) + one solid mid-stroke",
  "theta_max_from_plus_x_deg": [
    30.0,
    185.0
  ],
  "theta_min_from_plus_x_deg": [
    115.0,
    265.0
  ],
  "mu_max_bands": [
    [
      0.93358043,
      0.77714596
    ],
    [
      -0.99756405,
      -0.97029573
    ]
  ],
  "mu_min_bands": [
    [
      -0.27563736,
      -0.5591929
    ],
    [
      -0.2419219,
      0.06975647
    ]
  ],
  "CoE_lb_deg": [
    180.0,
    -30.0
  ],
  "anti_CoE_lb_deg": [
    0.0,
    30.0
  ],
  "D_hat_galactic_xyz": [
    0.8660254037844387,
    0.0,
```

```
    0.49999999999999994
  ],
  "angular_sep_migkas_max0_min1_deg": 98.476384,
  "axis_tilt_b_deg": 30,
  "axis_tilt_reason": "Chris 2026-09-22 \u2014 markers fixed, rings rotate up 30\u00b0",
  "orientation": "Galactic orientation: anti-CoE at (ell=0\u00b0, b=+30\u00b0), CoE at (ell=180\u00b0, b=\u221230\u00b0); +D = anti-CoE = +x; axis
tilted +30\u00b0 (Chris 2026-09-22 \u2014 markers fixed, rings rotate up 30\u00b0)",
  "drawing": "Exactly 4 features. Each = ONE smooth transparent highlight band between \u03b8\u22129\u00b0 and \u03b8+9\u00b0 (18\u00b0 end-to-end = 5%
of 360\u00b0) + ONE solid mid-stroke at the exact center \u03b8. No multi-stroke wash / hairball. Four distinct colors (two reds + two blues); no
orange; no 283\u00b0 companion. Panel A: clean vertical strip clipped to unit circle; theta rays from observer to each loop (emphasized at MAX
30\u00b0). Mollweide seam-split honestly. Sphere: single Poly3D strip (no n_wash stack).",
  "note": "Stops are after-2\u2113 A(\u2113) extrema from best_fit_2ell_multiplier.npz A_after: two MAX at \u03b8=30\u00b0/185\u00b0 and two MIN at
\u03b8=115\u00b0/265\u00b0, each drawn as one 18\u00b0 end-to-end highlight (\u00b19\u00b0 = 5% of 360\u00b0). Rings use \u03bc=cos(\u03b8); sky
markers are map extrema (17\u00b0,\u22129\u00b0) and (274\u00b0,\u221222\u00b0) plus strip stand-ins (185\u00b0,0\u00b0) and (115\u00b0,0\u00b0).",
  "plot_convention": "Migkas Galactic Mollweide: lon_disp=-ell_wrapped (ell increases LEFT)",
  "legend_summary": "two MAX + two MIN highlight bands, each 18\u00b0 end-to-end (\u00b19\u00b0 = 5% of 360\u00b0); four distinct colors",
  "colors": {
    "MAX_30": "#c0392b",
    "MAX_185": "#ec7063",
    "MIN_115": "#1a5276",
    "MIN_265": "#5dade2",
    "no_orange": true
  }
}

Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_migkas_pin_axis.png
Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_migkas_pin_axis.json
```


```
compare(FIGS / 'coe_rings_migkas_pin_axis.png', '/workspace/cosmology/figures/coe_rings_migkas_pin_axis.png')
```

```
coe_rings_migkas_pin_axis.png: new (2283, 1027) vs ref (2283, 1027) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference figures/coe_rings_migkas_pin_axis.png   |   right: notebook re-run steps_v3/figs/coe_rings_migkas_pin_axis.png
```



Step 11 — The Step 10 loops overlaid on Migkas Fig. 8

What it shows. The four MAX/MIN bands of Step 10 (axis $+30^\circ$) drawn on the real Migkas Fig. 8 $\theta = 75^\circ$ map, in high-contrast colours against magma (cyan, hot pink, lime, sky blue), with the published MAX triangles and MIN squares. Drawing is clipped to the coloured oval.

Source script: `/workspace/cosmology/figures/make_migkas_fig8_coe_rings_overlay.py` (last modified 2026-09-25 20:22 CT), run verbatim. *Output-path edit:* none. The script's `OUT_DIR` follows `__file__`, which is set to `steps_v3/figs/`. *Reference image:* `figures/migkas_fig8_theta75_with_coe_rings.png`.

```
%%runscript make_migkas_fig8_coe_rings_overlay.py
#!/usr/bin/env python3
"""
Superimpose CoE MAX/MIN highlight bands + Migkas markers onto the REAL Migkas
Fig. 8 theta=75 A/A_all Mollweide map.

Matches make_coe_rings_migkas_pin_axis.py (three-panel Panel B) geometry:
  Exactly FOUR features — each ONE transparent highlight band, 18° end-to-end
  (= ±9° = 5% of 360°) about the center θ + ONE solid mid-stroke:
    MAX 30°  → [21°, 39°]   color #00e5ff (bright cyan)
    MAX 185° → [176°, 194°] color #ff2d95 (magenta/hot pink)
    MIN 115° → [106°, 124°] color #b8f200 (lime)
    MIN 265° → [256°, 274°] color #4fc3f7 (sky blue)
  Drawing: clean PolyCollection fill + one mid-stroke (optional thin white
  underlay on mid only). NO black rim, NO thick white halo stacks, NO dual
  boundary-edge strokes. No 283° companion. No Dec=0 overlay.
  MAX triangles: (17°, −9°) and (185°, 0°)
  MIN squares:   (115°, 0°) and (274°, −22°)
  CoE axis tilted +30°: anti-CoE (0°,+30°), CoE (180°,−30°) (purple).

Registration (unchanged from working Fig. 8):
  GOOD_REG = (cx, cy, Rx, Ry) = (863.0, 763.25, 693.9927, 392.542425)
  Source: FULL_migkas_theta75.png
  CoE axis tilted +30°; clip/inset so overlays stay inside colored oval
  lon_disp = -wrap(ℓ)

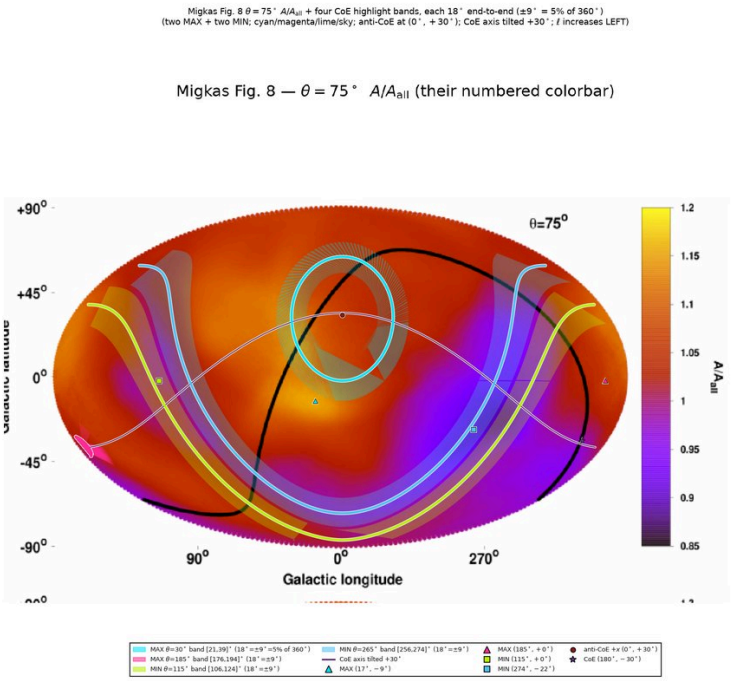
Outputs:
  migkas_fig8_theta75_with_coe_rings.png
  migkas_fig8_theta75_with_coe_rings.json
"""

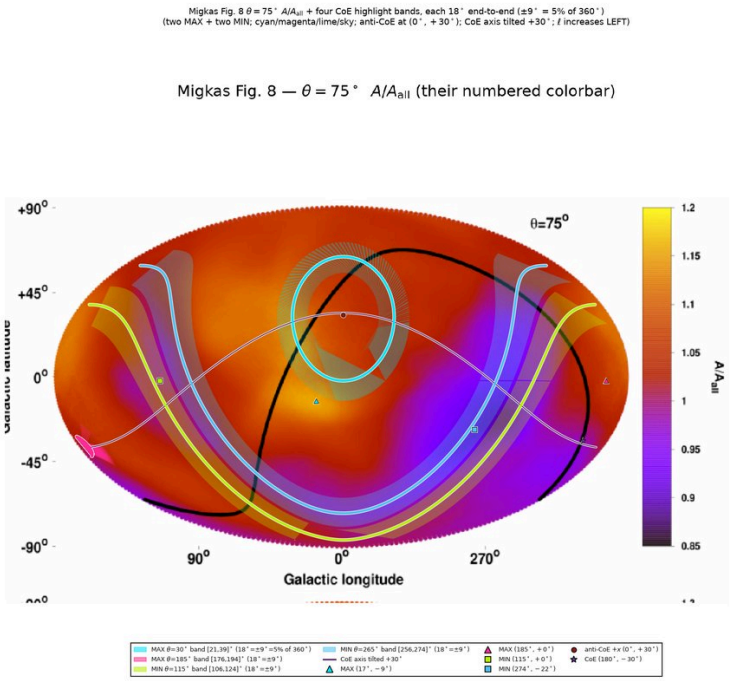
# ... 1032 more lines of the verbatim script: see coe_steps_v3.ipynb / .html / .md (or coe_steps_v3_fullcode.pdf)
```

```
source: /workspace/cosmology/map_compare/FULL_migkas_theta75.png
oval seed: {'cx': 856.0, 'cy': 754.0, 'Rx': 677.0, 'Ry': 398.5, 'x0': 179, 'x1': 1533, 'y0': 356, 'y1': 1152}
restored reg: {'cx': 863.0, 'cy': 763.25, 'Rx': 693.9927, 'Ry': 392.542425, 'ceq_median_px': 1.0}
validation OK — max0 RGB [237.0, 166.33333333333334, 0.0] min1 RGB [96.61728395061728, 0.0, 205.39506172839506]
wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/migkas_fig8_theta75_with_coe_rings.png
wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/migkas_fig8_theta75_with_coe_rings.json
```

```
compare(FIGS / 'migkas_fig8_theta75_with_coe_rings.png', '/workspace/cosmology/figures/migkas_fig8_theta75_with_coe_rings.png')
```

```
migkas_fig8_theta75_with_coe_rings.png: new (1975, 1515) vs ref (1975, 1515) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference figures/migkas_fig8_theta75_with_coe_rings.png   |   right: notebook re-run
steps_v3/figs/migkas_fig8_theta75_with_coe_rings.png
```





Step 12 — Half-loops: CoE axis $+15^\circ$, and the $+45^\circ$ CCW roll pair

What it shows. The same three panels, but with the axis tilted $+15^\circ$ (anti-CoE $(0^\circ, +15^\circ)$) and each loop kept only on its θ-ray side, an open 180° arc. First run: roll 0. Second run: roll 45° counter-clockwise about the CoE axis (looking along $+D$). **Maths.** Half-cut: keep ring points with $(\hat{n} \cdot \hat{Y})\ \sin\theta_{\rm mid} \geq 0$ (top half when $\sin\theta > 0$, bottom when $\sin\theta < 0$). A roll by ρ rotates $\hat{e}_{1,2}$ about $\hat{a}$: $\hat{e}_1' = \cos\rho\,\hat{e}_1 + \sin\rho\,\hat{e}_2$.

Source script: `/workspace/cosmology/figures/make_coe_rings_half_loops.py` (last modified 2026-09-25 20:22 CT), run verbatim. *Output-path edit:* none. The script's `OUT_DIR` follows `__file__`, which is set to `steps_v3/figs/`. *Reference image:* `figures/coe_rings_half_loops_tilt15.png`, `figures/coe_rings_half_loops_tilt15_rot45ccw.png`.

```
%%runscript make_coe_rings_half_loops.py --argv --roll 0
#!/usr/bin/env python3
"""
CoE rings — theta-ray-side half-loops only (Chris 2026-09-23 whiteout, CORRECTED).

Flags (CLI or module-level defaults):
  AXIS_TILT_B_DEG = 15   -> anti-CoE (0 deg, +15 deg), CoE (180 deg, -15 deg)
```

```
  ROLL_ABOUT_COE_DEG = 0 or 45  -> CCW about +D looking along +D
  HALF_LOOPS = True       -> open 180 deg arcs only (never full loops)

Half-cut (theta-ray side — NOT outer limb):
  foreshortened ex=mu+0.045*rho*cos(phi), ey=rho*sin(phi);
  keep ey*sin(theta_mid) >= 0  (TOP if sin theta>0, BOTTOM if sin theta<0).
  3D: keep (n·Y_hat)*sin(theta_mid) >= 0. Discard sign(mu)*cos(phi)>=0.

Outputs depend on roll:
  roll=0  -> coe_rings_half_loops_tilt15.{png,pdf,json}
  roll=45 -> coe_rings_half_loops_tilt15_rot45ccw.{png,pdf,json}
"""

from __future__ import annotations

import argparse
import json
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
from matplotlib.lines import Line2D
from matplotlib.collections import PolyCollection

# ... 629 more lines of the verbatim script: see coe_steps_v3.ipynb / .html / .md (or coe_steps_v3_fullcode.pdf)
```

```
{
  "png": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_half_loops_tilt15.png",
  "pdf": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_half_loops_tilt15.pdf",
  "script": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/make_coe_rings_half_loops.py",
  "geometry_module": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_half_loop_geometry.py",
  "HALF_LOOPS": true,
  "half_loop_rule": "\u03b8-ray side: keep (n\u00b7\u0176)*sin(\u03b8_mid)>=0 (Panel A: ey*sin(\u03b8_mid)>=0; TOP if sin\u03b8>0, BOTTOM if
sin\u03b8<0)",
  "axis_tilt_b_deg": 15.0,
  "roll_about_coe_deg": 0.0,
  "migkas_max_lb_deg": [
    [
      17.0,
      -9.0
    ],
    [
      185.0,
      0.0
    ]
  ],
  "migkas_min_lb_deg": [
    [
      115.0,
      0.0
    ],
    [
      274.0,
      -22.0
    ]
  ],
  "strip_theta_max_deg": [
    30.0,
    185.0
  ],
  "strip_theta_min_deg": [
    115.0,
    265.0
  ],
  "band_half_width_deg": 9.0,
  "band_full_width_deg": 18.0,
  "max_bands_deg": [
    [
      21.0,
      39.0
    ],
    [
      176.0,
      194.0
    ]
  ],
  "min_bands_deg": [
    [
      106.0,
      124.0
    ],
    [
      256.0,
```

```
      274.0
    ]
  ],
  "CoE_lb_deg": [
    180.0,
    -15.0
  ],
  "anti_CoE_lb_deg": [
    0.0,
    15.0
  ],
  "D_hat_galactic_xyz": [
    0.9659258262890683,
    0.0,
    0.25881904510252074
  ],
  "angular_sep_migkas_max0_min1_deg": 98.476384,
  "orientation": "anti-CoE at (0\u00b0, +15\u00b0), CoE at (180\u00b0, \u221215\u00b0); theta-ray-side half-loops (n\u00b7Y)*sin(theta)>=0; roll about
CoE = 0\u00b0 CCW",
  "plot_convention": "Migkas Galactic Mollweide: lon_disp=-ell_wrapped (ell increases LEFT)",
  "colors": {
    "MAX_30": "#c0392b",
    "MAX_185": "#ec7063",
    "MIN_115": "#1a5276",
    "MIN_265": "#5dade2"
  }
}
```


```
Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_half_loops_tilt15.png
Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_half_loops_tilt15.pdf
Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_half_loops_tilt15.json
```

```
%%runscript make_coe_rings_half_loops.py_roll45 --argv --roll 45
# second run of the same script with the +45 deg CCW roll: load its definitions (everything above its __main__ block), then call main()
exec(open('/workspace/cosmology/figures/make_coe_rings_half_loops.py').read().split('if __name__ == "__main__":')[0])
main(tilt_b=15.0, roll_deg=45.0)
```

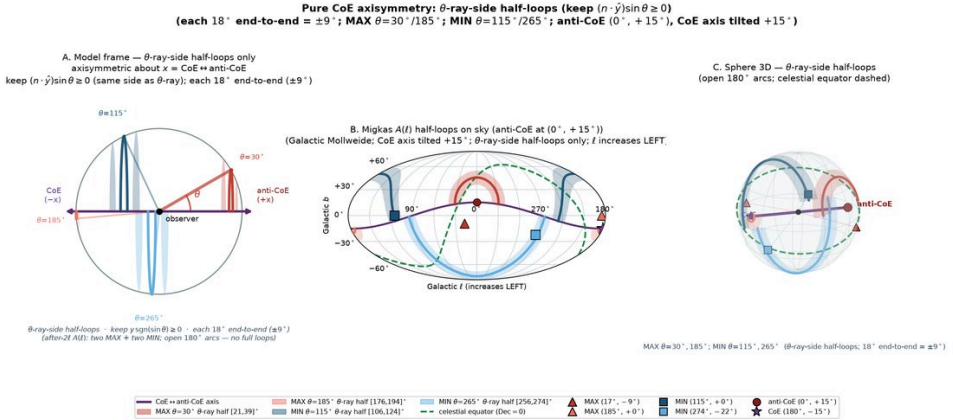
```
{
  "png": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_half_loops_tilt15_rot45ccw.png",
  "pdf": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_half_loops_tilt15_rot45ccw.pdf",
  "script": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/make_coe_rings_half_loops.py",
  "geometry_module": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_half_loop_geometry.py",
  "HALF_LOOPS": true,
  "half_loop_rule": "\u03b8-ray side: keep (n\u00b7\u0176)*sin(\u03b8_mid)>=0 (Panel A: ey*sin(\u03b8_mid)>=0; TOP if sin\u03b8>0, BOTTOM if
sin\u03b8<0)",
  "axis_tilt_b_deg": 15.0,
  "roll_about_coe_deg": 45.0,
  "migkas_max_lb_deg": [
    [
      17.0,
      -9.0
    ],
    [
      185.0,
      0.0
    ]
  ],
  "migkas_min_lb_deg": [
    [
      115.0,
      0.0
    ],
    [
      274.0,
      -22.0
    ]
  ],
  "strip_theta_max_deg": [
    30.0,
    185.0
  ],
  "strip_theta_min_deg": [
    115.0,
    265.0
  ],
  "band_half_width_deg": 9.0,
  "band_full_width_deg": 18.0,
  "max_bands_deg": [
    [
      21.0,
      39.0
    ],
    [
      176.0,
      194.0
    ]
  ],
  "min_bands_deg": [
    [
      106.0,
      124.0
    ],
    [
      256.0,
```

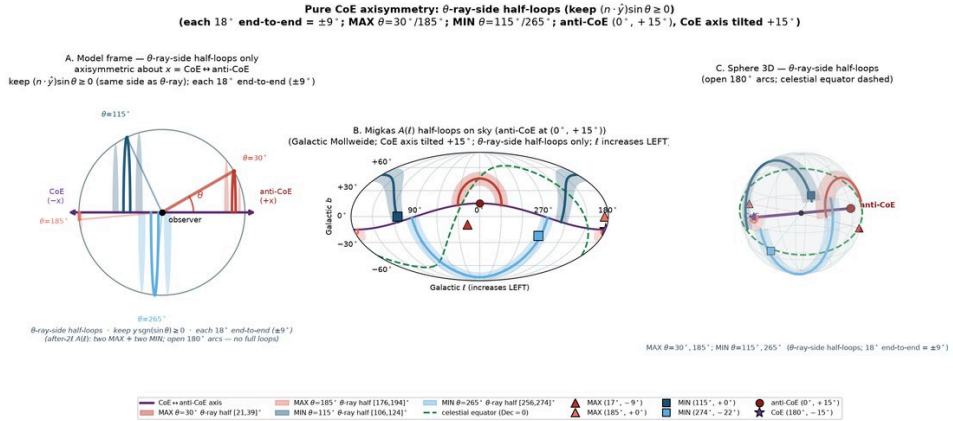
```
      274.0
    ]
  ],
  "CoE_lb_deg": [
    180.0,
    -15.0
  ],
  "anti_CoE_lb_deg": [
    0.0,
    15.0
  ],
  "D_hat_galactic_xyz": [
    0.9659258262890683,
    0.0,
    0.25881904510252074
  ],
  "angular_sep_migkas_max0_min1_deg": 98.476384,
  "orientation": "anti-CoE at (0\u00b0, +15\u00b0), CoE at (180\u00b0, \u221215\u00b0); theta-ray-side half-loops (n\u00b7Y)*sin(theta)>=0; roll about
CoE = 45\u00b0 CCW",
  "plot_convention": "Migkas Galactic Mollweide: lon_disp=-ell_wrapped (ell increases LEFT)",
  "colors": {
    "MAX_30": "#c0392b",
    "MAX_185": "#ec7063",
    "MIN_115": "#1a5276",
    "MIN_265": "#5dade2"
  }
}

Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_half_loops_tilt15_rot45ccw.png
Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_half_loops_tilt15_rot45ccw.pdf
Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_half_loops_tilt15_rot45ccw.json
```

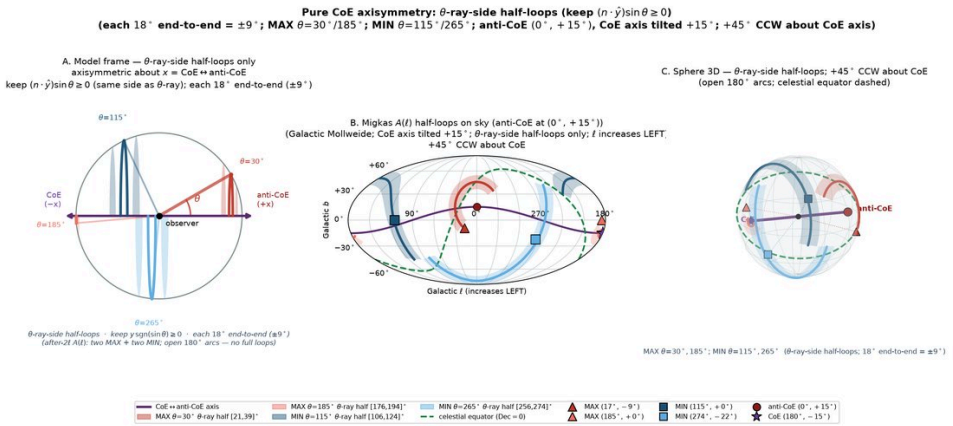

```
compare(FIGS / 'coe_rings_half_loops_tilt15.png', '/workspace/cosmology/figures/coe_rings_half_loops_tilt15.png')
compare(FIGS / 'coe_rings_half_loops_tilt15_rot45ccw.png', '/workspace/cosmology/figures/coe_rings_half_loops_tilt15_rot45ccw.png')
```

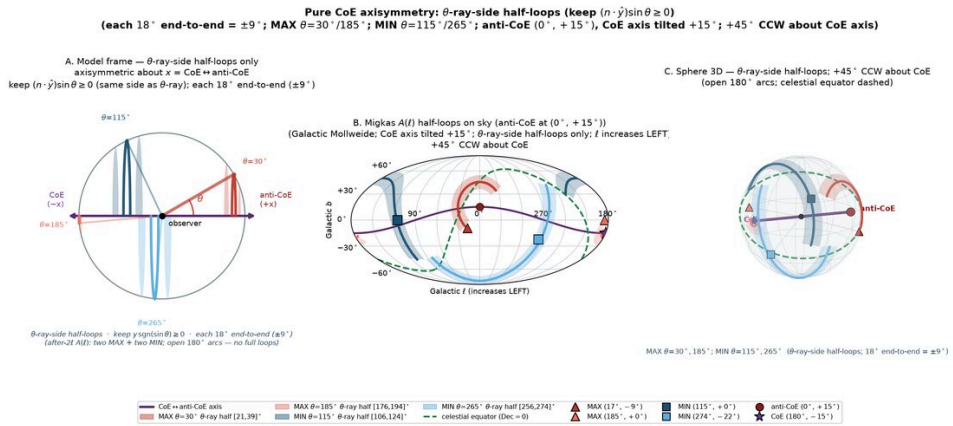
```
coe_rings_half_loops_tilt15.png: new (2272, 1027) vs ref (2272, 1027) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference figures/coe_rings_half_loops_tilt15.png   |   right: notebook re-run steps_v3/figs/coe_rings_half_loops_tilt15.png
```






```
coe_rings_half_loops_tilt15_rot45ccw.png: new (2272, 1027) vs ref (2272, 1027) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference figures/coe_rings_half_loops_tilt15_rot45ccw.png   |   right: notebook re-run
steps_v3/figs/coe_rings_half_loops_tilt15_rot45ccw.png
```





Step 13 — Half-loops on Migkas Fig. 8 (roll 0 and $+45^\circ$ CCW)

What it shows. The Step 12 half-arcs drawn on the Migkas Fig. 8 $\theta = 75^\circ$ map, with the same registration and marker positions as Step 11. The Migkas markers stay fixed; no Dec $= 0$ line is drawn.

Source script: `/workspace/cosmology/figures/make_migkas_fig8_half_loops_overlay.py` (last modified 2026-09-25 20:22 CT), run verbatim. *Output-path edit:* none. The script's `OUT_DIR` follows `__file__`, which is set to `steps_v3/figs/`. *Reference image:* `figures/migkas_fig8_half_loops_tilt15.png`, `figures/migkas_fig8_half_loops_tilt15_rot45ccw.png`.

```
%%runscript make_migkas_fig8_half_loops_overlay.py --argv --roll 0
#!/usr/bin/env python3
"""
Migkas Fig. 8 overlay — OUTER CoE half-loops (Chris 2026-09-23 whiteout).

Matches make_coe_rings_half_loops.py geometry:
  AXIS_TILT_B_DEG = 15 → anti-CoE (0°, +15°), CoE (180°, −15°)
  ROLL_ABOUT_COE_DEG = 0 or 45
  OUTER half-arcs only (sign(mu)*cos(phi-roll) >= 0)

GOOD_REG and lon_disp = -wrap(ℓ) unchanged from working Fig. 8 overlay.
Migkas markers FIXED. No Dec=0 overlay (published CEQ kept).

Outputs:
  roll=0  → migkas_fig8_half_loops_tilt15.{png,json}
  roll=45 → migkas_fig8_half_loops_tilt15_rot45ccw.{png,json}
"""

from __future__ import annotations

import argparse
import json
```

```
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
from matplotlib.collections import PolyCollection
from matplotlib.lines import Line2D
from matplotlib.patches import Ellipse, Patch
from PIL import Image

# ... 702 more lines of the verbatim script: see coe_steps_v3.ipynb / .html / .md (or coe_steps_v3_fullcode.pdf)
```

```
source: /workspace/cosmology/map_compare/FULL_migkas_theta75.png
validation OK
wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/migkas_fig8_half_loops_tilt15.png
wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/migkas_fig8_half_loops_tilt15.json
```

```
%%runscript make_migkas_fig8_half_loops_overlay.py_roll45 --argv --roll 45
# second run of the same script with the +45 deg CCW roll: load its definitions (everything above its __main__ block), then call main()
exec(open('/workspace/cosmology/figures/make_migkas_fig8_half_loops_overlay.py').read().split('if __name__ == "__main__":')[0])
main(tilt_b=15.0, roll_deg=45.0)
```

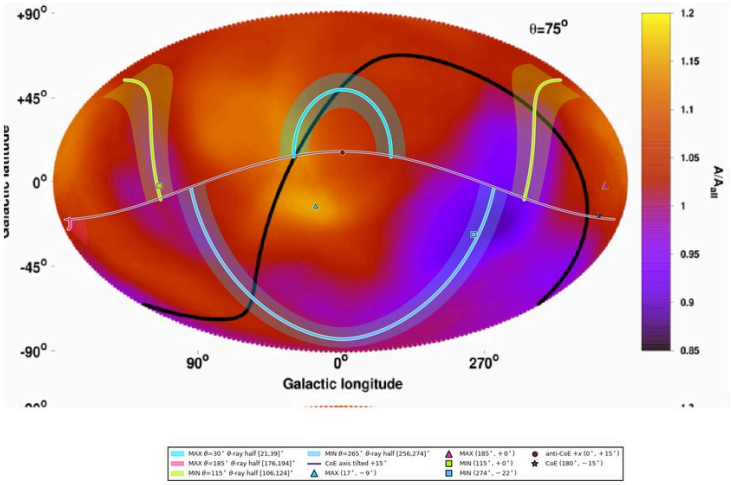
```
source: /workspace/cosmology/map_compare/FULL_migkas_theta75.png
validation OK
wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/migkas_fig8_half_loops_tilt15_rot45ccw.png
wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/migkas_fig8_half_loops_tilt15_rot45ccw.json
```

```
compare(FIGS / 'migkas_fig8_half_loops_tilt15.png', '/workspace/cosmology/figures/migkas_fig8_half_loops_tilt15.png')
compare(FIGS / 'migkas_fig8_half_loops_tilt15_rot45ccw.png', '/workspace/cosmology/figures/migkas_fig8_half_loops_tilt15_rot45ccw.png')
```

```
migkas_fig8_half_loops_tilt15.png: new (1975, 1515) vs ref (1975, 1515) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference figures/migkas_fig8_half_loops_tilt15.png   |   right: notebook re-run steps_v3/figs/migkas_fig8_half_loops_tilt15.png
```

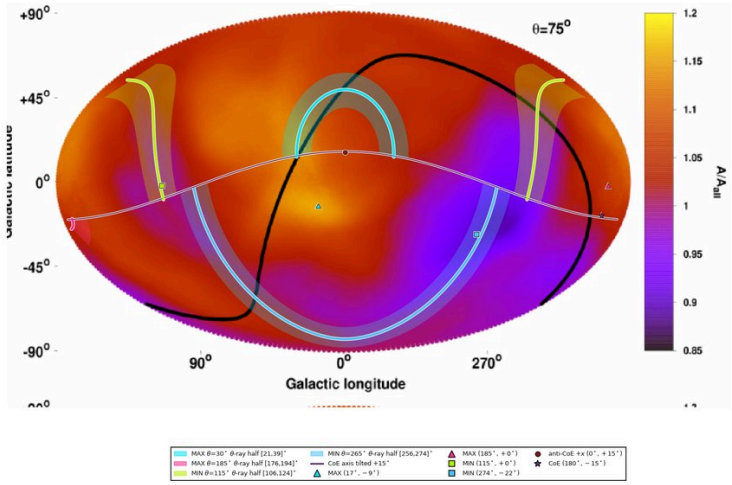
Migkas Fig. 8 $\theta = 75^\circ$ $A/A_{\rm all}$ + θ-ray-side CoE half-loops (each 18° end-to-end = ±9°)
(anti-CoE at (0°, +15°); CoE axis tilted +15°; θ-ray-side half-loops; ℓ increases LEFT)

Migkas Fig. 8 — $\theta = 75^\circ$ $A/A_{\rm all}$ (their numbered colorbar)

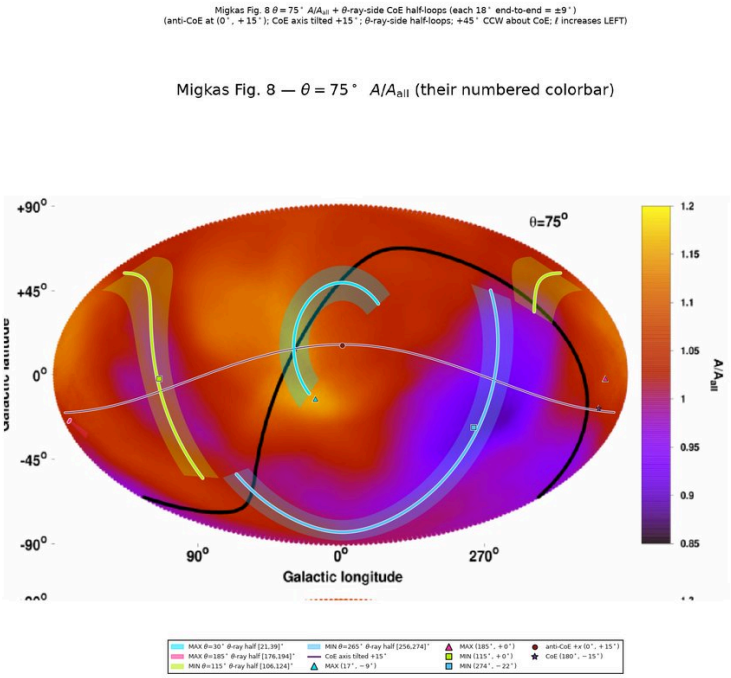


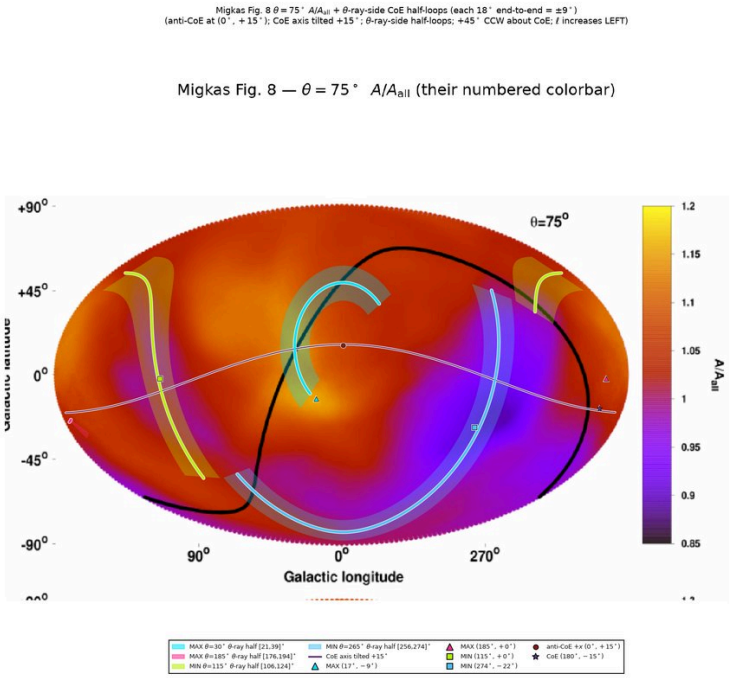
Migkas Fig. 8 $\theta = 75^\circ$ $A/A_{\rm all}$ + θ-ray-side CoE half-loops (each 18° end-to-end = ±9°)
(anti-CoE at (0°, +15°); CoE axis tilted +15°; θ-ray-side half-loops; ℓ increases LEFT)

Migkas Fig. 8 — $\theta = 75^\circ$ $A/A_{\rm all}$ (their numbered colorbar)



migkas_fig8_half_loops_tilt15_rot45ccw.png: new (1975, 1515) vs ref (1975, 1515) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference figures/migkas_fig8_half_loops_tilt15_rot45ccw.png | right: notebook re-run steps_v3/figs/migkas_fig8_half_loops_tilt15_rot45ccw.png





Step 14 — Full loops with an opacity gradient (roll 0 and $+45^\circ$ CCW)

What it shows. Full loops instead of half-loops. Each loop is drawn as short segments whose opacity fades with ring azimuth ψ from the θ-ray point: darkest there, fully transparent opposite. Panel A is always drawn at roll 0; the roll applies to Panels B and C only. **Maths.** $\alpha(\psi) = \alpha_0 \, \frac{1}{2}(1 + \cos\psi)$, where ψ is measured from the ring point furthest along $\mathrm{sign}(\sin\theta)\,\hat{Y}$ ($\hat{Y}$ rolled).

Source script: `/workspace/cosmology/figures/make_coe_rings_gradient_loops.py` (last modified 2026-09-25 20:22 CT), run verbatim. *Output-path edit:* none. The script's `OUT_DIR` follows `__file__`, which is set to `steps_v3/figs/`. *Reference image:* `figures/coe_rings_gradient_loops_tilt15.png`, `figures/coe_rings_gradient_loops_tilt15_rot45ccw.png`.

```
%%runscript make_coe_rings_gradient_loops.py --argv --roll 0
#!/usr/bin/env python3
"""
CoE rings — FULL loops with opacity gradient (Chris 2026-09-25).

Derived from make_coe_rings_half_loops.py (left unchanged). Instead of erasing
the opposite half of each ring, the full loop (±9° band fill + solid mid line)
is drawn as many short elements (LineCollection segments / PolyCollection quads)
with per-element alpha = base_alpha * (1 + cos psi)/2, psi = ring azimuth from
```

```
the θ-ray point (ring point most along sign(sinθ)·Ŷ; Ŷ rolled for B/C only).
Darkest at the θ-ray point, fully transparent at the diametrically opposite point.

Panel A is ALWAYS drawn with roll 0. Roll applies to Panels B and C only.

Outputs:
  roll=0  -> coe_rings_gradient_loops_tilt15.{png,pdf,json}
  roll=45 -> coe_rings_gradient_loops_tilt15_rot45ccw.{png,pdf,json}
"""

from __future__ import annotations

import argparse
import json
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
from matplotlib.lines import Line2D
from matplotlib.collections import LineCollection, PolyCollection
from matplotlib.patches import Circle, Patch, Polygon

# ... 664 more lines of the verbatim script: see coe_steps_v3.ipynb / .html / .md (or coe_steps_v3_fullcode.pdf)
```

```
{
  "png": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_gradient_loops_tilt15.png",
  "pdf": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_gradient_loops_tilt15.pdf",
  "script": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/make_coe_rings_gradient_loops.py",
  "geometry_module": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_half_loop_geometry.py",
  "gradient_module": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_gradient_loop_geometry.py",
  "FULL_GRADIENT_LOOPS": true,
  "gradient_rule": "full loops; alpha *= (1+cos psi)/2, psi = ring azimuth from the \u03b8-ray point (ring point most along
sign(sin\u03b8_mid)\u00b7\u0176, \u0176 rolled for B/C; A unrolled); equivalently alpha_scale = (1 + sign(sin\u03b8_mid)\u00b7sin(phi))/2",
  "legacy_half_loop_rule_equiv_alpha_scale_ge_half": "\u03b8-ray side: keep (n\u00b7\u0176)*sin(\u03b8_mid)>=0 (Panel A: ey*sin(\u03b8_mid)>=0; TOP if
sin\u03b8>0, BOTTOM if sin\u03b8<0)",
  "panel_A_roll_deg": 0.0,
  "roll_applies_to_panels": [
    "B",
    "C"
  ],
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    "fill_C": 0.25,
    "mid_line": 0.95
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    ],
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      0.0
    ]
  ],
  "migkas_min_lb_deg": [
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      0.0
    ],
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  ],
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  "max_bands_deg": [
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```

```
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      194.0
    ]
  ],
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    15.0
  ],
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    0.0,
    0.25881904510252074
  ],
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  "orientation": "anti-CoE at (0\u00b0, +15\u00b0), CoE at (180\u00b0, \u221215\u00b0); full loops, opacity gradient darkest at \u03b8 ray, fading to
12% opacity opposite; roll about CoE = 0\u00b0 CCW (panels B/C only; panel A unrolled)",
  "plot_convention": "Migkas Galactic Mollweide: lon_disp=-ell_wrapped (ell increases LEFT)",
  "colors": {
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    "MAX_185": "#ec7063",
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    "MIN_265": "#5dade2"
  }
}

Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_gradient_loops_tilt15.png
Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_gradient_loops_tilt15.pdf
Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_gradient_loops_tilt15.json
```


```
%%runscript make_coe_rings_gradient_loops.py_roll45 --argv --roll 45
# second run of the same script with the +45 deg CCW roll: load its definitions (everything above its __main__ block), then call main()
exec(open('/workspace/cosmology/figures/make_coe_rings_gradient_loops.py').read().split('if __name__ == "__main__":')[0])
main(tilt_b=15.0, roll_deg=45.0)
```

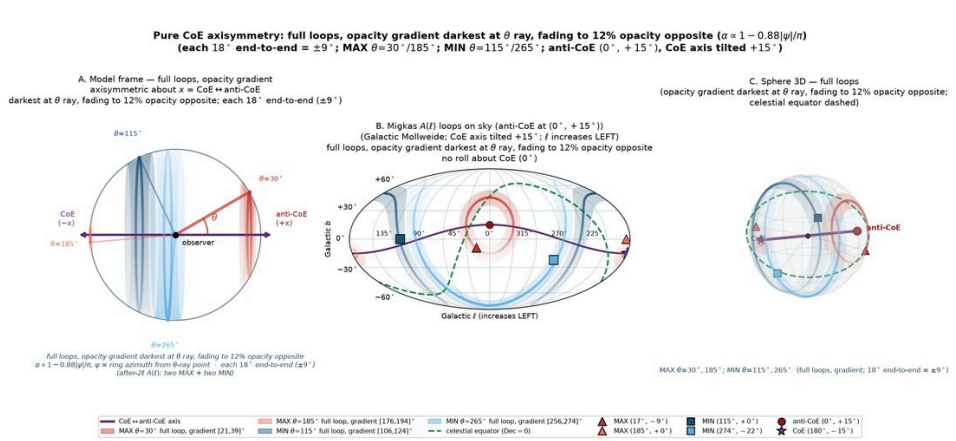
```
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  "gradient_module": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_gradient_loop_geometry.py",
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    "C"
  ],
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    "fill_B": 0.26,
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      0.0
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  ],
  "migkas_min_lb_deg": [
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      0.0
    ],
    [
      274.0,
      -22.0
    ]
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    185.0
  ],
  "strip_theta_min_deg": [
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    265.0
  ],
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```

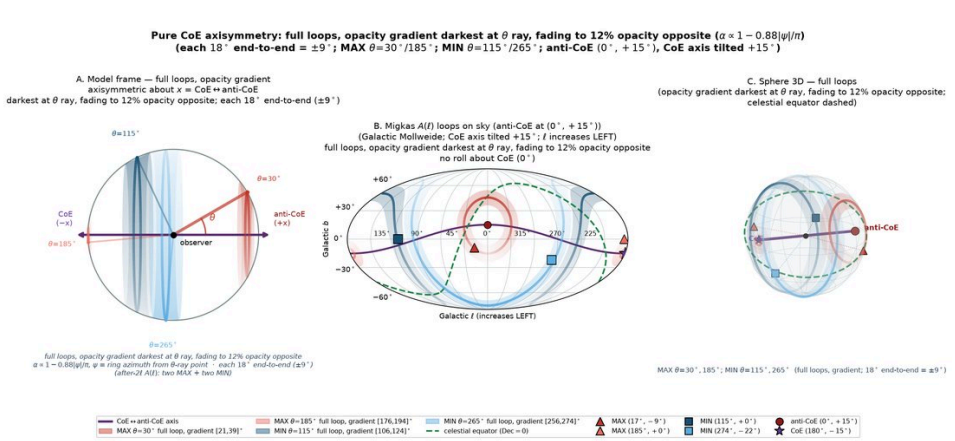
```
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    [
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      194.0
    ]
  ],
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      124.0
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    15.0
  ],
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    0.0,
    0.25881904510252074
  ],
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  "orientation": "anti-CoE at (0\u00b0, +15\u00b0), CoE at (180\u00b0, \u221215\u00b0); full loops, opacity gradient darkest at \u03b8 ray, fading to
12% opacity opposite; roll about CoE = 45\u00b0 CCW (panels B/C only; panel A unrolled)",
  "plot_convention": "Migkas Galactic Mollweide: lon_disp=-ell_wrapped (ell increases LEFT)",
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    "MAX_185": "#ec7063",
    "MIN_115": "#1a5276",
    "MIN_265": "#5dade2"
  }
}
```


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Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_gradient_loops_tilt15_rot45ccw.pdf
Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/coe_rings_gradient_loops_tilt15_rot45ccw.json
```

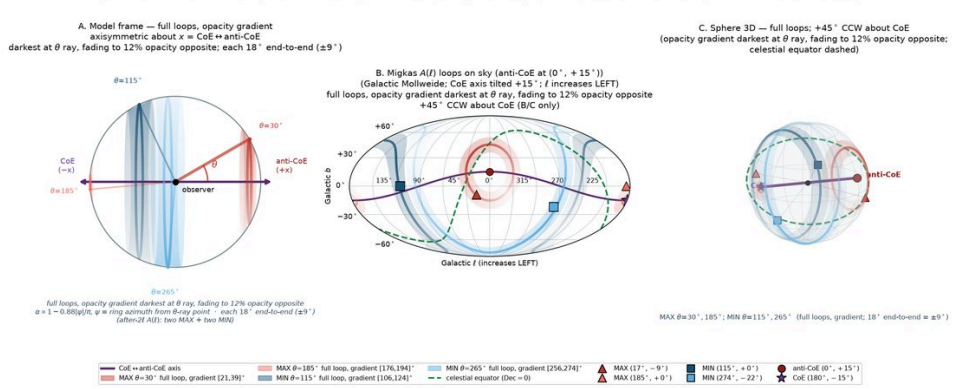
```
compare(FIGS / 'coe_rings_gradient_loops_tilt15.png', '/workspace/cosmology/figures/coe_rings_gradient_loops_tilt15.png')
compare(FIGS / 'coe_rings_gradient_loops_tilt15_rot45ccw.png', '/workspace/cosmology/figures/coe_rings_gradient_loops_tilt15_rot45ccw.png')
```

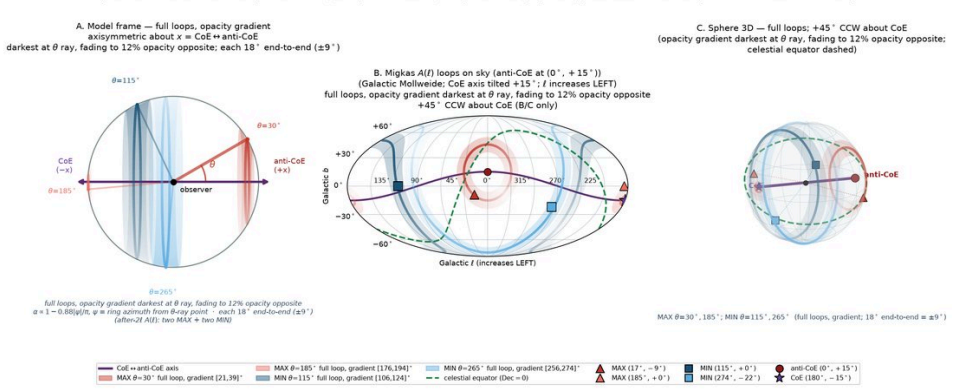
```
coe_rings_gradient_loops_tilt15.png: new (2334, 1027) vs ref (2334, 1027) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference figures/coe_rings_gradient_loops_tilt15.png   |   right: notebook re-run steps_v3/figs/coe_rings_gradient_loops_tilt15.png
```






```
coe_rings_gradient_loops_tilt15_rot45ccw.png: new (2334, 1027) vs ref (2334, 1027) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference figures/coe_rings_gradient_loops_tilt15_rot45ccw.png   |   right: notebook re-run
steps_v3/figs/coe_rings_gradient_loops_tilt15_rot45ccw.png
```





Step 15 — Gradient loops on Migkas Fig. 8 (roll 0 and $+45^\circ$ CCW)

What it shows. The Step 14 gradient loops on the Migkas Fig. 8 map, with the same registration, markers and gradient rule.

Source script: `/workspace/cosmology/figures/make_migkas_fig8_gradient_loops_overlay.py` (last modified 2026-09-25 20:22 CT), run verbatim. *Output-path edit:* none. The script's `OUT_DIR` follows `__file__`, which is set to `steps_v3/figs/`. *Reference image:* `figures/migkas_fig8_gradient_loops_tilt15.png`, `figures/migkas_fig8_gradient_loops_tilt15_rot45ccw.png`.

```
%%runscript make_migkas_fig8_gradient_loops_overlay.py --argv --roll 0
#!/usr/bin/env python3
"""
Migkas Fig. 8 overlay — FULL CoE loops with opacity gradient (Chris 2026-09-25).

Derived from make_migkas_fig8_half_loops_overlay.py (left unchanged).
```

```
Each FULL ring (±9° band fill + solid mid line with its white underlay) is drawn
with alpha multiplied by (1+cos psi)/2, psi = ring azimuth from the θ-ray point
(ring point most along sign(sin0)·Ŷ, Ŷ from ring_basis rolled by roll_deg) —
same helper as make_coe_rings_gradient_loops.py (coe_gradient_loop_geometry).
Contiguous elements sharing a quantized alpha level (64 levels) are merged into
one strip polygon / polyline (no beading, few AA seams).

GOOD_REG, lon_disp = -wrap(ℓ), oval clipping, antimeridian splitting, colours,
markers all unchanged. Published map untouched (drawn underneath only).

Outputs:
  roll=0  → migkas_fig8_gradient_loops_tilt15.{png,json}
  roll=45 → migkas_fig8_gradient_loops_tilt15_rot45ccw.{png,json}
"""

from __future__ import annotations

import argparse
import json
from pathlib import Path

import matplotlib.pyplot as plt
import numpy as np
from matplotlib.collections import LineCollection, PolyCollection

# ... 673 more lines of the verbatim script: see coe_steps_v3.ipynb / .html / .md (or coe_steps_v3_fullcode.pdf)
```

```
source: /workspace/cosmology/map_compare/FULL_migkas_theta75.png
validation OK
wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/migkas_fig8_gradient_loops_tilt15.png
wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/migkas_fig8_gradient_loops_tilt15.json
```

```
%%runscript make_migkas_fig8_gradient_loops_overlay.py_roll45 --argv --roll 45
# second run of the same script with the +45 deg CCW roll: load its definitions (everything above its __main__ block), then call main()
exec(open('/workspace/cosmology/figures/make_migkas_fig8_gradient_loops_overlay.py').read().split('if __name__ == "__main__":')[0])
main(tilt_b=15.0, roll_deg=45.0)
```

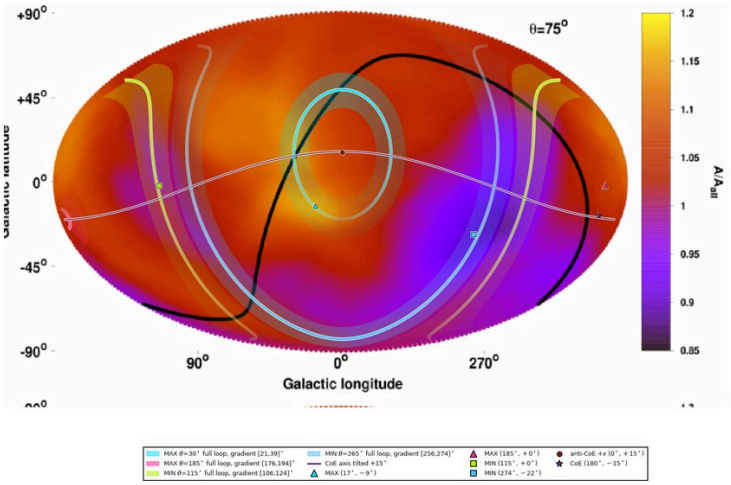
```
source: /workspace/cosmology/map_compare/FULL_migkas_theta75.png
validation OK
wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/migkas_fig8_gradient_loops_tilt15_rot45ccw.png
wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/migkas_fig8_gradient_loops_tilt15_rot45ccw.json
```

```
compare(FIGS / 'migkas_fig8_gradient_loops_tilt15.png', '/workspace/cosmology/figures/migkas_fig8_gradient_loops_tilt15.png')
compare(FIGS / 'migkas_fig8_gradient_loops_tilt15_rot45ccw.png', '/workspace/cosmology/figures/migkas_fig8_gradient_loops_tilt15_rot45ccw.png')
```

```
migkas_fig8_gradient_loops_tilt15.png: new (1975, 1515) vs ref (1975, 1515) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference figures/migkas_fig8_gradient_loops_tilt15.png   |   right: notebook re-run steps_v3/figs/migkas_fig8_gradient_loops_tilt15.png
```

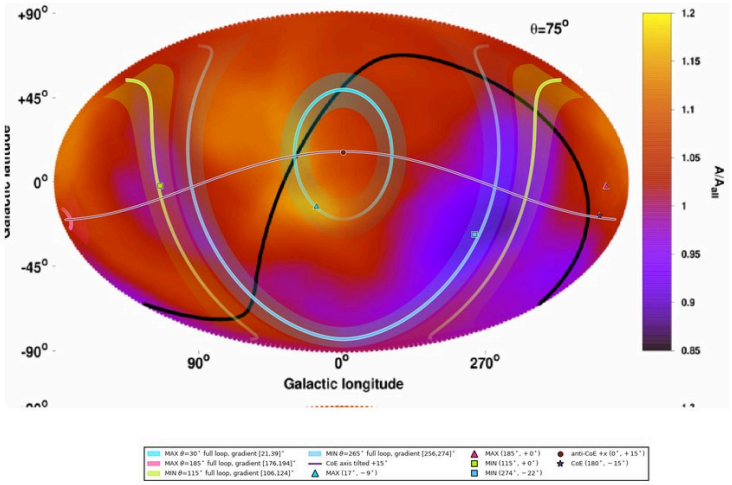
Migkas Fig. 8 $\theta = 75^\circ$ $A/A_{\rm all}$ + CoE full loops, opacity gradient darkest at θ ray, fading to 12% opacity opposite (anti-CoE at $(0^\circ, +15^\circ)$); CoE axis tilted $+15^\circ$; each 18° end-to-end = $\pm 9^\circ$; $\alpha \propto 1 - 0.88|\psi|/\pi$; ℓ increases LEFT)

Migkas Fig. 8 — $\theta = 75^\circ$ $A/A_{\rm all}$ (their numbered colorbar)

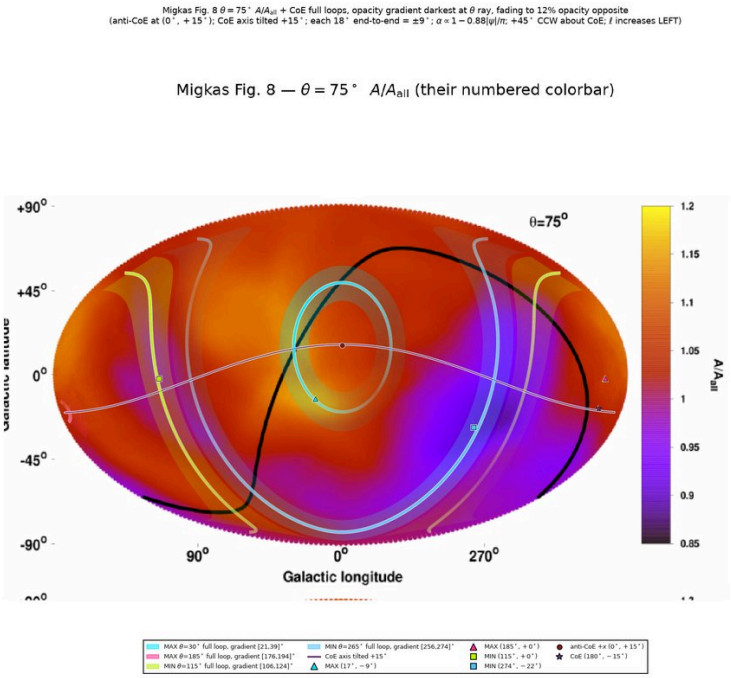


Migkas Fig. 8 $\theta = 75^\circ$ $A/A_{\rm all}$ + CoE full loops, opacity gradient darkest at θ ray, fading to 12% opacity opposite (anti-CoE at $(0^\circ, +15^\circ)$); CoE axis tilted $+15^\circ$; each 18° end-to-end = $\pm 9^\circ$; $\alpha \propto 1 - 0.88|\psi|/\pi$; ℓ increases LEFT)

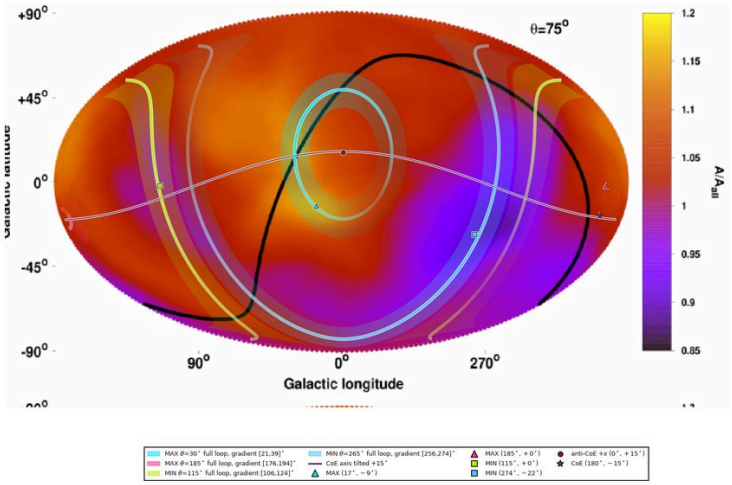
Migkas Fig. 8 — $\theta = 75^\circ$ $A/A_{\rm all}$ (their numbered colorbar)



migkas_fig8_gradient_loops_tilt15_rot45ccw.png: new (1975, 1515) vs ref (1975, 1515) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference figures/migkas_fig8_gradient_loops_tilt15_rot45ccw.png | right: notebook re-run steps_v3/figs/migkas_fig8_gradient_loops_tilt15_rot45ccw.png



Migkas Fig. 8 θ = 75° A/A_all + CoE full loops, opacity gradient darkest at θ ray, fading to 12% opacity opposite
(anti-CoE at (0°, +15°); CoE axis tilted +15°; each 18° end-to-end = ±9°; α ∝ 1 − 0.88|ψ|/π; +45° CCW about CoE; ℓ increases LEFT)
Migkas Fig. 8 — θ = 75° A/A_all (their numbered colorbar)



Step 16a — Matched three-panel: 2 MAX + 2 MIN loops, $\pm 9^\circ$ bands, axis $+15^\circ$

What it shows. The final three-panel layout: two MAX loops (two reds) and two MIN loops (two blues), each a clean transparent $\pm 9^\circ$ band with a mid-line, and no orange loop. Axis tilted $+15^\circ$. Panel A is the flat model-frame angle circle with Galactic ℓ ticks; Panel B the Mollweide map; Panel C the 3D sphere, drawn **at the same diameter as Panel A** (the canvas was widened rather than anything shrunk).

Source script: `/workspace/cosmology/figures/make_coe_rings_migkas_pin_axis_matched.py` (last modified 2026-09-26 22:54 CT), run verbatim. *Output-path edit:* `/workspace/cosmology/papers/spreadsheet_model/loop_threepanel_matched` → `/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/loop_threep`. *Reference image:* `papers/spreadsheet_model/loop_threepanel_matched.png`.

```
%%runscript make_coe_rings_migkas_pin_axis_matched.py
#!/usr/bin/env python3
"""
CoE rings + Migkas A(ell) extrema loops in the Galactic frame.

Concept:
  Pure CoE axisymmetry → constant A on small-circle rings about x = CoE↔anti-CoE.
  Scan A vs angle from +x (anti-CoE) around 360°. After-2ℓ A(ℓ) has TWO maxima
```

```
  and TWO minima (from best_fit_2ell_multiplier.npz A_after):
    MAX at θ = 30° and 185° from +x,
    MIN at θ = 115° and 265° from +x.

  Exactly FOUR loops — each ONE transparent highlight band, 18° end-to-end
  (= ±9° = 5% of 360°) about the center θ:
    MAX: 30±9 → [21°, 39°] and 185±9 → [176°, 194°]
    MIN: 115±9 → [106°, 124°] and 265±9 → [256°, 274°]
  Drawing: ONE smooth transparent filled ribbon between θ−9° and θ+9°
  + ONE solid mid-stroke at the exact center θ. No multi-stroke wash,
  no dense linspace hairball, no dual strong boundaries.
  Four distinct colors (two reds + two blues). No orange, no 283° companion.

  Galactic orientation: anti-CoE (+x) at ℓ=0°, b=+15°; CoE (−x) at ℓ=180°, b=−15°
  (axis tilted +15° so features near ℓ=0 move north on Mollweide; Migkas markers fixed).
  Sky markers (ℓ, b):
    Maxes: map extrema (17°, −9°) for ~30° max; strip (185°, 0°) for second max.
    Mins:  strip (115°, 0°) for ~115° min; map extrema (274°, −22°) for ~265° min.

Outputs:
  coe_rings_migkas_pin_axis.png
  coe_rings_migkas_pin_axis.json

# ... 837 more lines of the verbatim script: see coe_steps_v3.ipynb / .html / .md (or coe_steps_v3_fullcode.pdf)
```

```
PANEL_DIAM A=367.9px C=367.9x368.0px zoom=1.399
```

```
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      0.0
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    [
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  ],
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  ],
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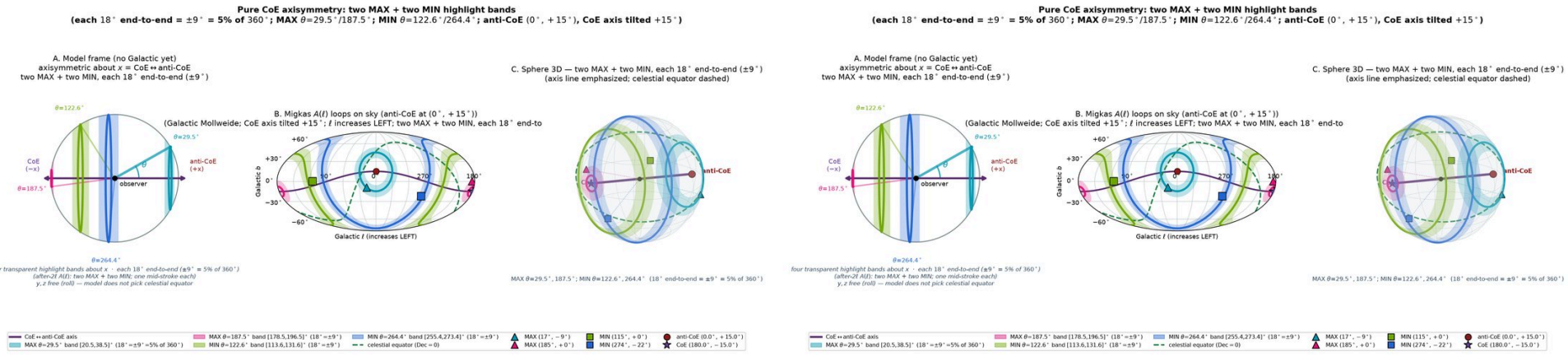
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    [
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      0.05930637
    ]
  ],
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  ],
  "anti_CoE_lb_deg": [
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    15.0
  ],
  "D_hat_galactic_xyz": [
    0.9659258262890683,
    0.0,
```

```
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  ],
  "angular_sep_migkas_max0_min1_deg": 98.476384,
  "axis_tilt_b_deg": 15,
  "axis_tilt_reason": "Chris 2026-09-22 \u2014 markers fixed, rings rotate up 30\u00b0",
  "orientation": "Galactic orientation: anti-CoE at (ell=0\u00b0, b=+15\u00b0), CoE at (ell=180\u00b0, b=\u221215\u00b0); +D = anti-CoE = +x; axis
tilted +15\u00b0 (Chris 2026-09-22 \u2014 markers fixed, rings rotate up 30\u00b0)",
  "drawing": "Exactly 4 features. Each = ONE smooth transparent highlight band between \u03b8\u22129\u00b0 and \u03b8+9\u00b0 (18\u00b0 end-to-end = 5%
of 360\u00b0) + ONE solid mid-stroke at the exact center \u03b8. No multi-stroke wash / hairball. Four distinct colors (two reds + two blues); no
orange; no 283\u00b0 companion. Panel A: clean vertical strip clipped to unit circle; theta rays from observer to each loop (emphasized at MAX
30\u00b0). Mollweide seam-split honestly. Sphere: single Poly3D strip (no n_wash stack).",
  "note": "Stops are after-2\u2113 A(\u2113) extrema from best_fit_2ell_multiplier.npz A_after: two MAX at \u03b8=29.5\u00b0/187.5\u00b0 and two MIN at
\u03b8=122.6\u00b0/264.4\u00b0, each drawn as one 18\u00b0 end-to-end highlight (\u00b19\u00b0 = 5% of 360\u00b0). Rings use \u03bc=cos(\u03b8); sky
markers are map extrema (17\u00b0,\u22129\u00b0) and (274\u00b0,\u221222\u00b0) plus strip stand-ins (185\u00b0,0\u00b0) and (115\u00b0,0\u00b0).",
  "plot_convention": "Migkas Galactic Mollweide: lon_disp=-ell_wrapped (ell increases LEFT)",
  "legend_summary": "two MAX + two MIN highlight bands, each 18\u00b0 end-to-end (\u00b19\u00b0 = 5% of 360\u00b0); four distinct colors",
  "colors": {
    "MAX_30": "#0097b2",
    "MAX_185": "#e0107a",
    "MIN_115": "#6fa800",
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  }
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Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/loop_threepanel_matched.png
Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/loop_threepanel_matched.json
```


```
compare(FIGS / 'loop_threepanel_matched.png', '/workspace/cosmology/papers/spreadsheet_model/loop_threepanel_matched.png')
```

```
loop_threepanel_matched.png: new (2262, 1027) vs ref (2262, 1027) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference papers/spreadsheet_model/loop_threepanel_matched.png   |   right: notebook re-run steps_v3/figs/loop_threepanel_matched.png
```



Step 16b — Matched three-panel with the opacity gradient

What it shows. Step 16a with each loop fading from its θ-ray point, $\alpha \propto \frac{1}{2}(1+\cos\psi)$, in Panels B and C.

Source script: `/workspace/cosmology/figures/make_coe_rings_migkas_pin_axis_matched_grad.py` (last modified 2026-09-26 22:54 CT), run verbatim. *Output-path edit:* `/workspace/cosmology/papers/spreadsheet_model/loop_threepanel_matched_` → `/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/loop_threep`. *Reference image:* `papers/spreadsheet_model/loop_threepanel_matched_grad.png`.

```
%%runscript make_coe_rings_migkas_pin_axis_matched_grad.py
#!/usr/bin/env python3
"""
CoE rings + Migkas A(ell) extrema loops in the Galactic frame.

Concept:
  Pure CoE axisymmetry → constant A on small-circle rings about x = CoE↔anti-CoE.
  Scan A vs angle from +x (anti-CoE) around 360°. After-2ℓ A(ℓ) has TWO maxima
  and TWO minima (from best_fit_2ell_multiplier.npz A_after):
    MAX at θ = 30° and 185° from +x,
    MIN at θ = 115° and 265° from +x.

  Exactly FOUR loops — each ONE transparent highlight band, 18° end-to-end
  (= ±9° = 5% of 360°) about the center θ:
    MAX: 30±9 → [21°, 39°] and 185±9 → [176°, 194°]
    MIN: 115±9 → [106°, 124°] and 265±9 → [256°, 274°]
  Drawing: ONE smooth transparent filled ribbon between θ−9° and θ+9°
  + ONE solid mid-stroke at the exact center θ. No multi-stroke wash,
  no dense linspace hairball, no dual strong boundaries.
  Four distinct colors (two reds + two blues). No orange, no 283° companion.

  Galactic orientation: anti-CoE (+x) at ℓ=0°, b=+15°; CoE (−x) at ℓ=180°, b=−15°
  (axis tilted +15° so features near ℓ=0 move north on Mollweide; Migkas markers fixed).
  Sky markers (ℓ, b):
    Maxes: map extrema (17°, −9°) for ~30° max; strip (185°, 0°) for second max.
    Mins:  strip (115°, 0°) for ~115° min; map extrema (274°, −22°) for ~265° min.

Outputs:
  coe_rings_migkas_pin_axis.png
  coe_rings_migkas_pin_axis.json

# ... 939 more lines of the verbatim script: see coe_steps_v3.ipynb / .html / .md (or coe_steps_v3_fullcode.pdf)
```

```
PANEL_DIAM A=367.9px C=367.9x368.0px zoom=1.399
```

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```

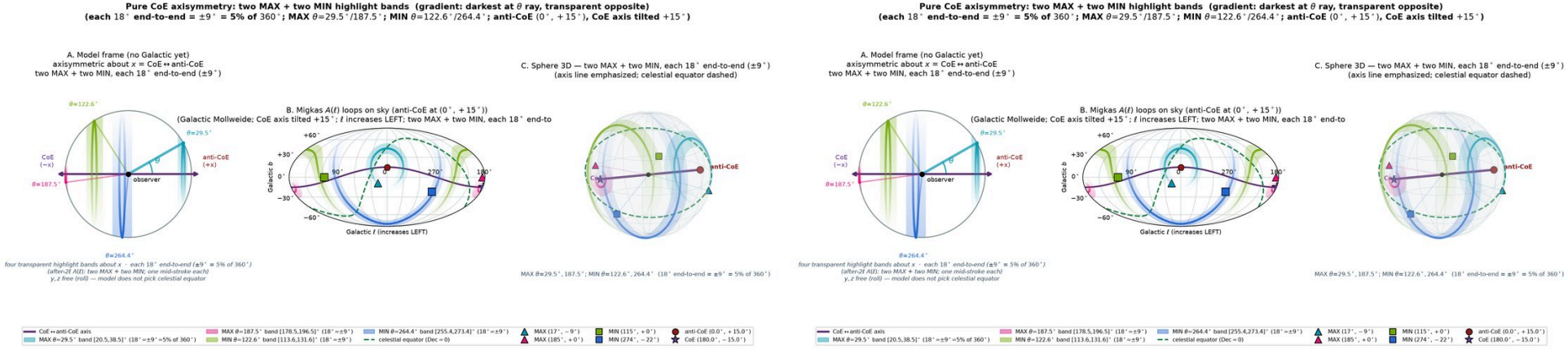
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(\u00b19\u00b0 = 5% of 360\u00b0) + one solid mid-stroke",
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  "orientation": "Galactic orientation: anti-CoE at (ell=0\u00b0, b=+15\u00b0), CoE at (ell=180\u00b0, b=\u221215\u00b0); +D = anti-CoE = +x; axis
tilted +15\u00b0 (Chris 2026-09-22 \u2014 markers fixed, rings rotate up 30\u00b0)",
  "drawing": "Exactly 4 features. Each = ONE smooth transparent highlight band between \u03b8\u22129\u00b0 and \u03b8+9\u00b0 (18\u00b0 end-to-end = 5%
of 360\u00b0) + ONE solid mid-stroke at the exact center \u03b8. No multi-stroke wash / hairball. Four distinct colors (two reds + two blues); no
orange; no 283\u00b0 companion. Panel A: clean vertical strip clipped to unit circle; theta rays from observer to each loop (emphasized at MAX
30\u00b0). Mollweide seam-split honestly. Sphere: single Poly3D strip (no n_wash stack).",
  "note": "Stops are after-2\u2113 A(\u2113) extrema from best_fit_2ell_multiplier.npz A_after: two MAX at \u03b8=29.5\u00b0/187.5\u00b0 and two MIN at
\u03b8=122.6\u00b0/264.4\u00b0, each drawn as one 18\u00b0 end-to-end highlight (\u00b19\u00b0 = 5% of 360\u00b0). Rings use \u03bc=cos(\u03b8); sky
markers are map extrema (17\u00b0,\u22129\u00b0) and (274\u00b0,\u221222\u00b0) plus strip stand-ins (185\u00b0,0\u00b0) and (115\u00b0,0\u00b0).",
  "plot_convention": "Migkas Galactic Mollweide: lon_disp=-ell_wrapped (ell increases LEFT)",
  "legend_summary": "two MAX + two MIN highlight bands, each 18\u00b0 end-to-end (\u00b19\u00b0 = 5% of 360\u00b0); four distinct colors",
  "colors": {
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    "MAX_185": "#e0107a",
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    "no_orange": true
  }
}

Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/loop_threepanel_matched_grad.png
Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/loop_threepanel_matched_grad.json
```


```
compare(FIGS / 'loop_threepanel_matched_grad.png', '/workspace/cosmology/papers/spreadsheet_model/loop_threepanel_matched_grad.png')
```

```
loop_threepanel_matched_grad.png: new (2262, 1027) vs ref (2262, 1027) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference papers/spreadsheet_model/loop_threepanel_matched_grad.png   |   right: notebook re-run
steps_v3/figs/loop_threepanel_matched_grad.png
```



Step 16c — Matched three-panel, gradient, 45° roll

What it shows. Step 16b with the loops rolled 45° about the CoE axis. This is the 45° roll version of the three-panel figure.

Source script: `/workspace/cosmology/figures/make_coe_rings_migkas_pin_axis_matched_grad_rot45.py` (last modified 2026-09-26 22:54 CT), run verbatim. *Output-path edit:* `/workspace/cosmology/papers/spreadsheet_model/loop_threepanel_matched_` → `/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/loop_threep`. *Reference image:* `papers/spreadsheet_model/loop_threepanel_matched_grad_rot45.png`.

```
%%runscript make_coe_rings_migkas_pin_axis_matched_grad_rot45.py
#!/usr/bin/env python3
"""
CoE rings + Migkas A(ell) extrema loops in the Galactic frame.

Concept:
  Pure CoE axisymmetry → constant A on small-circle rings about x = CoE↔anti-CoE.
  Scan A vs angle from +x (anti-CoE) around 360°. After-2ℓ A(ℓ) has TWO maxima
  and TWO minima (from best_fit_2ell_multiplier.npz A_after):
    MAX at θ = 30° and 185° from +x,
    MIN at θ = 115° and 265° from +x.

  Exactly FOUR loops — each ONE transparent highlight band, 18° end-to-end
  (= ±9° = 5% of 360°) about the center θ:
    MAX: 30±9 → [21°, 39°] and 185±9 → [176°, 194°]
    MIN: 115±9 → [106°, 124°] and 265±9 → [256°, 274°]
  Drawing: ONE smooth transparent filled ribbon between θ−9° and θ+9°
  + ONE solid mid-stroke at the exact center θ. No multi-stroke wash,
  no dense linspace hairball, no dual strong boundaries.
  Four distinct colors (two reds + two blues). No orange, no 283° companion.

  Galactic orientation: anti-CoE (+x) at ℓ=0°, b=+15°; CoE (−x) at ℓ=180°, b=−15°
  (axis tilted +15° so features near ℓ=0 move north on Mollweide; Migkas markers fixed).
  Sky markers (ℓ, b):
    Maxes: map extrema (17°, −9°) for ~30° max; strip (185°, 0°) for second max.
    Mins:  strip (115°, 0°) for ~115° min; map extrema (274°, −22°) for ~265° min.

Outputs:
  coe_rings_migkas_pin_axis.png
  coe_rings_migkas_pin_axis.json

# ... 946 more lines of the verbatim script: see coe_steps_v3.ipynb / .html / .md (or coe_steps_v3_fullcode.pdf)
```

```
PANEL_DIAM A=367.9px C=367.9x368.0px zoom=1.399
```

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    ]
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  "min_bands_deg": [
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      131.6
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      255.39999999999998,
      273.4
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  "band_intervals": {
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```

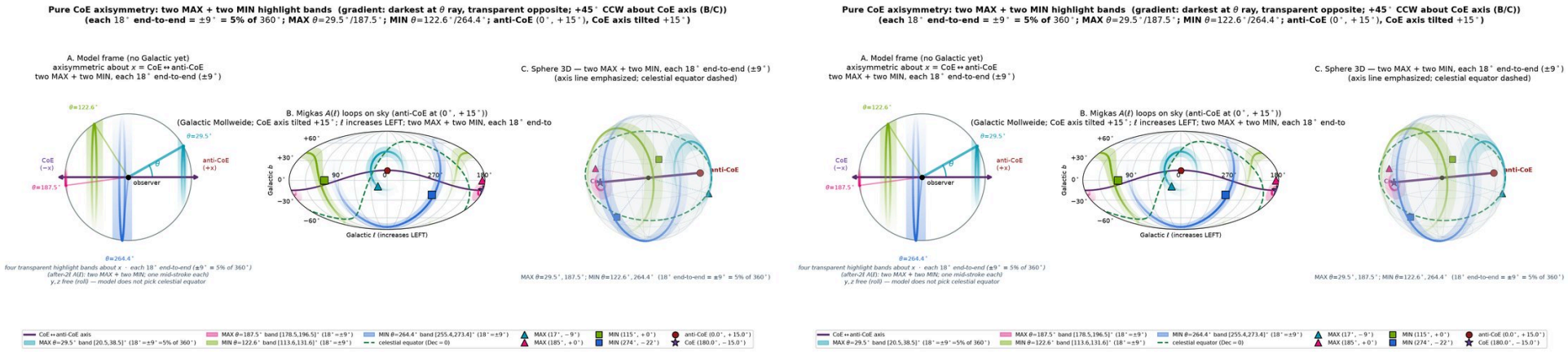
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    ],
    "MIN_265": [
      255.39999999999998,
      273.4
    ]
  },
  "scan_stop_source": "best_fit_2ell_multiplier.npz A_after extrema (two max, two min); each drawn as one transparent 18\u00b0 end-to-end highlight
(\u00b19\u00b0 = 5% of 360\u00b0) + one solid mid-stroke",
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  "anti_CoE_lb_deg": [
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    0.0,
```

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  ],
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  "orientation": "Galactic orientation: anti-CoE at (ell=0\u00b0, b=+15\u00b0), CoE at (ell=180\u00b0, b=\u221215\u00b0); +D = anti-CoE = +x; axis
tilted +15\u00b0 (Chris 2026-09-22 \u2014 markers fixed, rings rotate up 30\u00b0)",
  "drawing": "Exactly 4 features. Each = ONE smooth transparent highlight band between \u03b8\u22129\u00b0 and \u03b8+9\u00b0 (18\u00b0 end-to-end = 5%
of 360\u00b0) + ONE solid mid-stroke at the exact center \u03b8. No multi-stroke wash / hairball. Four distinct colors (two reds + two blues); no
orange; no 283\u00b0 companion. Panel A: clean vertical strip clipped to unit circle; theta rays from observer to each loop (emphasized at MAX
30\u00b0). Mollweide seam-split honestly. Sphere: single Poly3D strip (no n_wash stack).",
  "note": "Stops are after-2\u2113 A(\u2113) extrema from best_fit_2ell_multiplier.npz A_after: two MAX at \u03b8=29.5\u00b0/187.5\u00b0 and two MIN at
\u03b8=122.6\u00b0/264.4\u00b0, each drawn as one 18\u00b0 end-to-end highlight (\u00b19\u00b0 = 5% of 360\u00b0). Rings use \u03bc=cos(\u03b8); sky
markers are map extrema (17\u00b0,\u22129\u00b0) and (274\u00b0,\u221222\u00b0) plus strip stand-ins (185\u00b0,0\u00b0) and (115\u00b0,0\u00b0).",
  "plot_convention": "Migkas Galactic Mollweide: lon_disp=-ell_wrapped (ell increases LEFT)",
  "legend_summary": "two MAX + two MIN highlight bands, each 18\u00b0 end-to-end (\u00b19\u00b0 = 5% of 360\u00b0); four distinct colors",
  "colors": {
    "MAX_30": "#0097b2",
    "MAX_185": "#e0107a",
    "MIN_115": "#6fa800",
    "MIN_265": "#2166d4",
    "no_orange": true
  }
}

Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/loop_threepanel_matched_grad_rot45.png
Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/loop_threepanel_matched_grad_rot45.json
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compare(FIGS / 'loop_threepanel_matched_grad_rot45.png', '/workspace/cosmology/papers/spreadsheet_model/loop_threepanel_matched_grad_rot45.png')
```

```
loop_threepanel_matched_grad_rot45.png: new (2262, 1027) vs ref (2262, 1027) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference papers/spreadsheet_model/loop_threepanel_matched_grad_rot45.png   |   right: notebook re-run
steps_v3/figs/loop_threepanel_matched_grad_rot45.png
```



Step 16d — Matched three-panel, gradient, fitted 80.8° roll

What it shows. Step 16b with the roll set to the fitted 80.8°: the roll that brings the loops' θ-ray points closest to the Migkas markers.

Source script: `/workspace/cosmology/figures/make_coe_rings_migkas_pin_axis_matched_grad_rot_fit.py` (last modified 2026-09-26 22:54 CT), run verbatim. *Output-path edit:* `/workspace/cosmology/papers/spreadsheet_model/loop_threepanel_matched_` → `/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/loop_threep`. *Reference image:* `papers/spreadsheet_model/loop_threepanel_matched_grad_rotfit_rot81.png`.

```
%%runscript make_coe_rings_migkas_pin_axis_matched_grad_rot_fit.py
#!/usr/bin/env python3
"""
CoE rings + Migkas A(ell) extrema loops in the Galactic frame.

Concept:
  Pure CoE axisymmetry → constant A on small-circle rings about x = CoE↔anti-CoE.
  Scan A vs angle from +x (anti-CoE) around 360°. After-2ℓ A(ℓ) has TWO maxima
  and TWO minima (from best_fit_2ell_multiplier.npz A_after):
    MAX at θ = 30° and 185° from +x,
    MIN at θ = 115° and 265° from +x.

  Exactly FOUR loops — each ONE transparent highlight band, 18° end-to-end
  (= ±9° = 5% of 360°) about the center θ:
    MAX: 30±9 → [21°, 39°] and 185±9 → [176°, 194°]
    MIN: 115±9 → [106°, 124°] and 265±9 → [256°, 274°]
  Drawing: ONE smooth transparent filled ribbon between θ−9° and θ+9°
  + ONE solid mid-stroke at the exact center θ. No multi-stroke wash,
  no dense linspace hairball, no dual strong boundaries.
  Four distinct colors (two reds + two blues). No orange, no 283° companion.

  Galactic orientation: anti-CoE (+x) at ℓ=0°, b=+15°; CoE (−x) at ℓ=180°, b=−15°
  (axis tilted +15° so features near ℓ=0 move north on Mollweide; Migkas markers fixed).
  Sky markers (ℓ, b):
    Maxes: map extrema (17°, −9°) for ~30° max; strip (185°, 0°) for second max.
    Mins:  strip (115°, 0°) for ~115° min; map extrema (274°, −22°) for ~265° min.

Outputs:
  coe_rings_migkas_pin_axis.png
  coe_rings_migkas_pin_axis.json

# ... 946 more lines of the verbatim script: see coe_steps_v3.ipynb / .html / .md (or coe_steps_v3_fullcode.pdf)
```

```
PANEL_DIAM A=367.9px C=367.9x368.0px zoom=1.399
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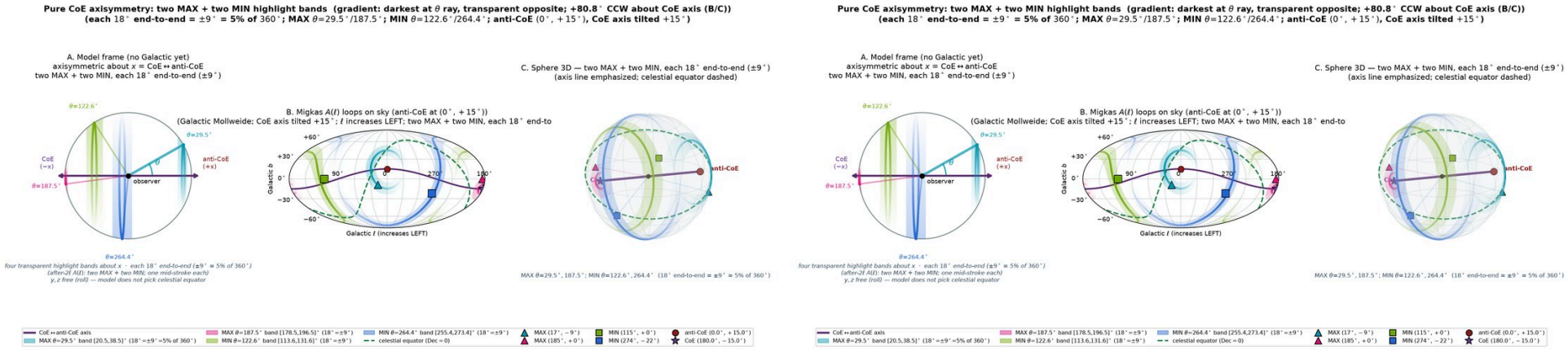
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  "scan_stop_source": "best_fit_2ell_multiplier.npz A_after extrema (two max, two min); each drawn as one transparent 18\u00b0 end-to-end highlight
(\u00b19\u00b0 = 5% of 360\u00b0) + one solid mid-stroke",
  "theta_max_from_plus_x_deg": [
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  "mu_min_bands": [
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    ],
    [
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  ],
  "CoE_lb_deg": [
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  "anti_CoE_lb_deg": [
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    0.0,
```

```
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  ],
  "angular_sep_migkas_max0_min1_deg": 98.476384,
  "axis_tilt_b_deg": 15,
  "axis_tilt_reason": "Chris 2026-09-22 \u2014 markers fixed, rings rotate up 30\u00b0",
  "orientation": "Galactic orientation: anti-CoE at (ell=0\u00b0, b=+15\u00b0), CoE at (ell=180\u00b0, b=\u221215\u00b0); +D = anti-CoE = +x; axis
tilted +15\u00b0 (Chris 2026-09-22 \u2014 markers fixed, rings rotate up 30\u00b0)",
  "drawing": "Exactly 4 features. Each = ONE smooth transparent highlight band between \u03b8\u22129\u00b0 and \u03b8+9\u00b0 (18\u00b0 end-to-end = 5%
of 360\u00b0) + ONE solid mid-stroke at the exact center \u03b8. No multi-stroke wash / hairball. Four distinct colors (two reds + two blues); no
orange; no 283\u00b0 companion. Panel A: clean vertical strip clipped to unit circle; theta rays from observer to each loop (emphasized at MAX
30\u00b0). Mollweide seam-split honestly. Sphere: single Poly3D strip (no n_wash stack).",
  "note": "Stops are after-2\u2113 A(\u2113) extrema from best_fit_2ell_multiplier.npz A_after: two MAX at \u03b8=29.5\u00b0/187.5\u00b0 and two MIN at
\u03b8=122.6\u00b0/264.4\u00b0, each drawn as one 18\u00b0 end-to-end highlight (\u00b19\u00b0 = 5% of 360\u00b0). Rings use \u03bc=cos(\u03b8); sky
markers are map extrema (17\u00b0,\u22129\u00b0) and (274\u00b0,\u221222\u00b0) plus strip stand-ins (185\u00b0,0\u00b0) and (115\u00b0,0\u00b0).",
  "plot_convention": "Migkas Galactic Mollweide: lon_disp=-ell_wrapped (ell increases LEFT)",
  "legend_summary": "two MAX + two MIN highlight bands, each 18\u00b0 end-to-end (\u00b19\u00b0 = 5% of 360\u00b0); four distinct colors",
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}

Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/loop_threepanel_matched_grad_rotfit_rot81.png
Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/loop_threepanel_matched_grad_rotfit_rot81.json
```


```
compare(FIGS / 'loop_threepanel_matched_grad_rotfit_rot81.png',
'/workspace/cosmology/papers/spreadsheet_model/loop_threepanel_matched_grad_rotfit_rot81.png')
```

```
loop_threepanel_matched_grad_rotfit_rot81.png: new (2262, 1027) vs ref (2262, 1027) | mean |diff| = 0.000/255, pixels differing >40: 0.000% ->
identical
left: approved reference papers/spreadsheet_model/loop_threepanel_matched_grad_rotfit_rot81.png   |   right: notebook re-run
steps_v3/figs/loop_threepanel_matched_grad_rotfit_rot81.png
```



Step 17 — Untilted, unrolled three-panel (anti-CoE $(0°, 0°)$, CoE $(180°, 0°)$)

What it shows. The matched three-panel with no tilt and no roll. The CoE axis lies in the Galactic plane, with anti-CoE at $(\ell, b) = (0^\circ, 0^\circ)$ and CoE at $(180^\circ, 0^\circ)$, so each loop is a small circle centred on the plane. Band centres (from the strip) are $\theta_{\max} = 29.5^\circ, 187.5^\circ$ and $\theta_{\min} = 122.6^\circ, 264.4^\circ$, each $\pm 9^\circ$. Panel C is at the same diameter as Panel A.

Source script: `/workspace/cosmology/figures/make_coe_rings_notilt.py` (last modified 2026-09-27 09:38 CT), run verbatim. *Output-path edit:* `/workspace/cosmology/papers/spreadsheet_model/loop_threepanel_notilt` → `/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/loop_threep`. *Reference image:* `papers/spreadsheet_model/loop_threepanel_notilt.png`.

```
%%runscript make_coe_rings_notilt.py
#!/usr/bin/env python3
"""
CoE rings + Migkas A(ell) extrema loops in the Galactic frame.

Concept:
  Pure CoE axisymmetry → constant A on small-circle rings about x = CoE↔anti-CoE.
  Scan A vs angle from +x (anti-CoE) around 360°. After-2ℓ A(ℓ) has TWO maxima
  and TWO minima (from best_fit_2ell_multiplier.npz A_after):
    MAX at θ = 30° and 185° from +x,
    MIN at θ = 115° and 265° from +x.

  Exactly FOUR loops — each ONE transparent highlight band, 18° end-to-end
  (= ±9° = 5% of 360°) about the center θ:
    MAX: 30±9 → [21°, 39°] and 185±9 → [176°, 194°]
    MIN: 115±9 → [106°, 124°] and 265±9 → [256°, 274°]
  Drawing: ONE smooth transparent filled ribbon between θ−9° and θ+9°
  + ONE solid mid-stroke at the exact center θ. No multi-stroke wash,
  no dense linspace hairball, no dual strong boundaries.
  Four distinct colors (two reds + two blues). No orange, no 283° companion.

  Galactic orientation: anti-CoE (+x) at ℓ=0°, b=+15°; CoE (−x) at ℓ=180°, b=−15°
  (axis tilted +15° so features near ℓ=0 move north on Mollweide; Migkas markers fixed).
  Sky markers (ℓ, b):
    Maxes: map extrema (17°, −9°) for ~30° max; strip (185°, 0°) for second max.
    Mins:  strip (115°, 0°) for ~115° min; map extrema (274°, −22°) for ~265° min.

Outputs:
  coe_rings_migkas_pin_axis.png
  coe_rings_migkas_pin_axis.json

# ... 837 more lines of the verbatim script: see coe_steps_v3.ipynb / .html / .md (or coe_steps_v3_fullcode.pdf)
```

```
PANEL_DIAM A=367.9px C=367.9x368.0px zoom=1.399
```

```
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    [
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  ],
  "migkas_min_lb_deg": [
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  ],
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  "strip_theta_min_deg": [
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  ],
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  ],
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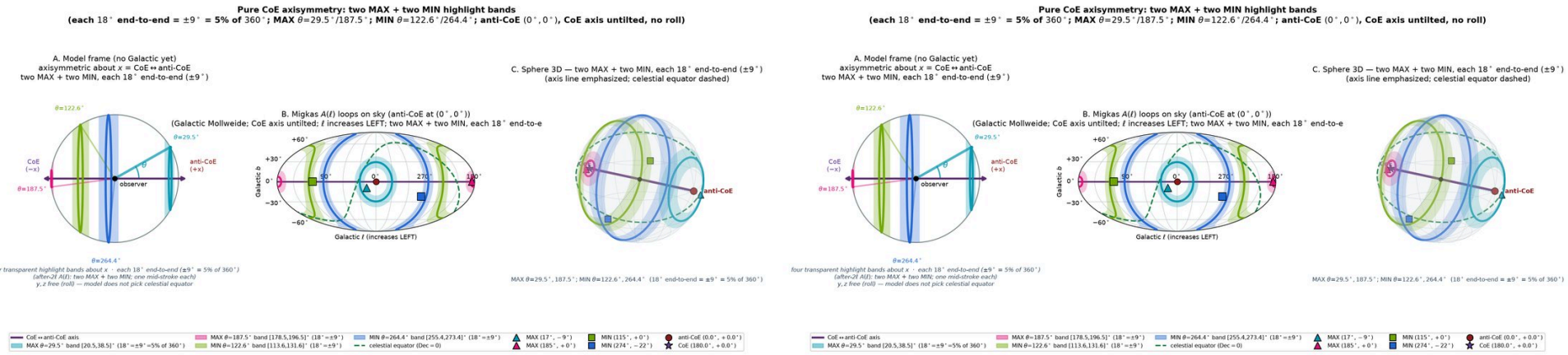
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(\u00b19\u00b0 = 5% of 360\u00b0) + one solid mid-stroke",
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  "mu_min_bands": [
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  "CoE_lb_deg": [
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```

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  "axis_tilt_reason": "Chris 2026-09-22 \u2014 markers fixed, rings rotate up 30\u00b0",
  "orientation": "Galactic orientation: anti-CoE at (ell=0\u00b0, b=+15\u00b0), CoE at (ell=180\u00b0, b=\u221215\u00b0); +D = anti-CoE = +x; axis
tilted +15\u00b0 (Chris 2026-09-22 \u2014 markers fixed, rings rotate up 30\u00b0)",
  "drawing": "Exactly 4 features. Each = ONE smooth transparent highlight band between \u03b8\u22129\u00b0 and \u03b8+9\u00b0 (18\u00b0 end-to-end = 5%
of 360\u00b0) + ONE solid mid-stroke at the exact center \u03b8. No multi-stroke wash / hairball. Four distinct colors (two reds + two blues); no
orange; no 283\u00b0 companion. Panel A: clean vertical strip clipped to unit circle; theta rays from observer to each loop (emphasized at MAX
30\u00b0). Mollweide seam-split honestly. Sphere: single Poly3D strip (no n_wash stack).",
  "note": "Stops are after-2\u2113 A(\u2113) extrema from best_fit_2ell_multiplier.npz A_after: two MAX at \u03b8=29.5\u00b0/187.5\u00b0 and two MIN at
\u03b8=122.6\u00b0/264.4\u00b0, each drawn as one 18\u00b0 end-to-end highlight (\u00b19\u00b0 = 5% of 360\u00b0). Rings use \u03bc=cos(\u03b8); sky
markers are map extrema (17\u00b0,\u22129\u00b0) and (274\u00b0,\u221222\u00b0) plus strip stand-ins (185\u00b0,0\u00b0) and (115\u00b0,0\u00b0).",
  "plot_convention": "Migkas Galactic Mollweide: lon_disp=-ell_wrapped (ell increases LEFT)",
  "legend_summary": "two MAX + two MIN highlight bands, each 18\u00b0 end-to-end (\u00b19\u00b0 = 5% of 360\u00b0); four distinct colors",
  "colors": {
    "MAX_30": "#0097b2",
    "MAX_185": "#e0107a",
    "MIN_115": "#6fa800",
    "MIN_265": "#2166d4",
    "no_orange": true
  }
}

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Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/loop_threepanel_notilt.json
```


```
compare(FIGS / 'loop_threepanel_notilt.png', '/workspace/cosmology/papers/spreadsheet_model/loop_threepanel_notilt.png')
```

```
loop_threepanel_notilt.png: new (2262, 1027) vs ref (2262, 1027) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference papers/spreadsheet_model/loop_threepanel_notilt.png   |   right: notebook re-run steps_v3/figs/loop_threepanel_notilt.png
```



Step 18 — The four-loop overlay on Migkas Fig. 8, recoloured for contrast against magma (4 versions)

What it shows. The matched loops of Step 16 drawn on the Migkas Fig. 8 $\theta = 75^\circ$ map, recoloured to stand out against the magma colour map. Four versions from one script: matched; with gradient; gradient with 45° roll; gradient with the fitted 80.8° roll.

Source script: `/workspace/cosmology/figures/make_migkas_fig8_overlay_matched.py` (last modified 2026-09-26 22:57 CT), run verbatim. *Output-path edit:* `SM_DIR = Path("/workspace/cosmology/papers/spreadsheet_model")` → `SM_DIR = Path("/workspace/cosmology/papers/A_equals_fourier/steps_v3/f`. *Reference image:* `papers/spreadsheet_model/fig8_overlay_matched.png`, `papers/spreadsheet_model/fig8_overlay_matched_grad.png`, `papers/spreadsheet_model/fig8_overlay_matched_grad_rot45.png`, `papers/spreadsheet_model/fig8_overlay_matched_grad_rot81.png`.

```
%%runscript make_migkas_fig8_overlay_matched.py
#!/usr/bin/env python3
"""
Superimpose CoE MAX/MIN highlight bands + Migkas markers onto the REAL Migkas
Fig. 8 theta=75 A/A_all Mollweide map.

Matches make_coe_rings_migkas_pin_axis.py (three-panel Panel B) geometry:
  Exactly FOUR features — each ONE transparent highlight band, 18° end-to-end
  (= ±9° = 5% of 360°) about the center θ + ONE solid mid-stroke:
    MAX 30°  → [21°, 39°]   color #00e5ff (bright cyan)
    MAX 185° → [176°, 194°] color #ff2d95 (magenta/hot pink)
    MIN 115° → [106°, 124°] color #b8f200 (lime)
    MIN 265° → [256°, 274°] color #4fc3f7 (sky blue)
  Drawing: clean PolyCollection fill + one mid-stroke (optional thin white
  underlay on mid only). NO black rim, NO thick white halo stacks, NO dual
  boundary-edge strokes. No 283° companion. No Dec=0 overlay.
  MAX triangles: (17°, −9°) and (185°, 0°)
  MIN squares:   (115°, 0°) and (274°, −22°)
  CoE axis tilted +30°: anti-CoE (0°,+30°), CoE (180°,−30°) (purple).

Registration (unchanged from working Fig. 8):
  GOOD_REG = (cx, cy, Rx, Ry) = (863.0, 763.25, 693.9927, 392.542425)
  Source: FULL_migkas_theta75.png
  CoE axis tilted +30°; clip/inset so overlays stay inside colored oval
  lon_disp = -wrap(ℓ)

Outputs:
  migkas_fig8_theta75_with_coe_rings.png
  migkas_fig8_theta75_with_coe_rings.json
"""

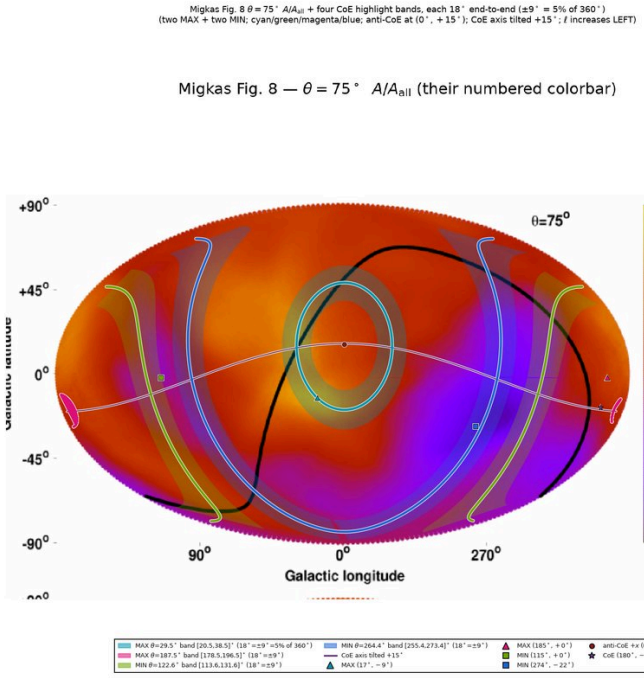
# ... 1164 more lines of the verbatim script: see coe_steps_v3.ipynb / .html / .md (or coe_steps_v3_fullcode.pdf)
```

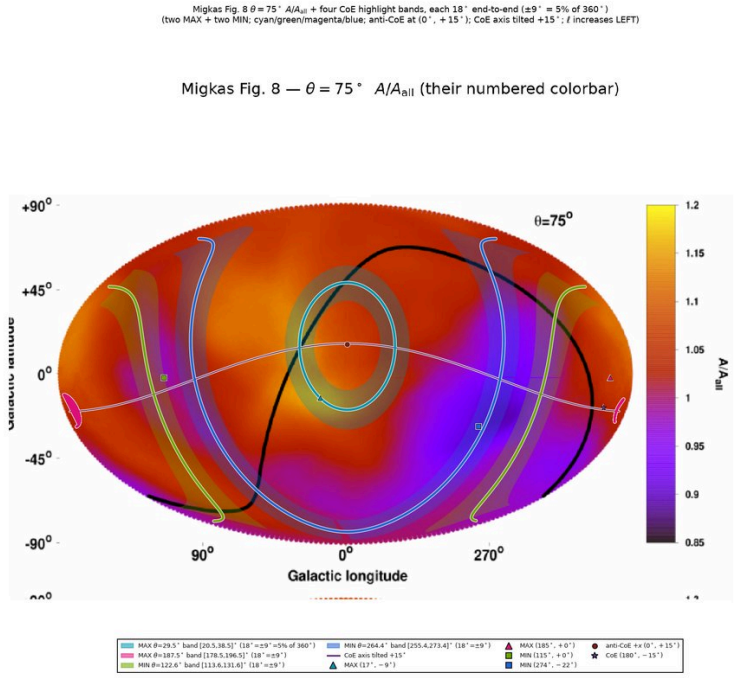
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restored reg: {'cx': 863.0, 'cy': 763.25, 'Rx': 693.9927, 'Ry': 392.542425, 'ceq_median_px': 1.0}
validation OK — max0 RGB [237.0, 166.33333333333334, 0.0] min1 RGB [96.61728395061728, 0.0, 205.39506172839506]
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```

```
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wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/fig8_overlay_matched_grad_rot81.png
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```

```
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compare(FIGS / 'fig8_overlay_matched_grad.png', '/workspace/cosmology/papers/spreadsheet_model/fig8_overlay_matched_grad.png')
compare(FIGS / 'fig8_overlay_matched_grad_rot45.png', '/workspace/cosmology/papers/spreadsheet_model/fig8_overlay_matched_grad_rot45.png')
compare(FIGS / 'fig8_overlay_matched_grad_rot81.png', '/workspace/cosmology/papers/spreadsheet_model/fig8_overlay_matched_grad_rot81.png')
```

```
fig8_overlay_matched.png: new (1975, 1515) vs ref (1975, 1515) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference papers/spreadsheet_model/fig8_overlay_matched.png   |   right: notebook re-run steps_v3/figs/fig8_overlay_matched.png
```

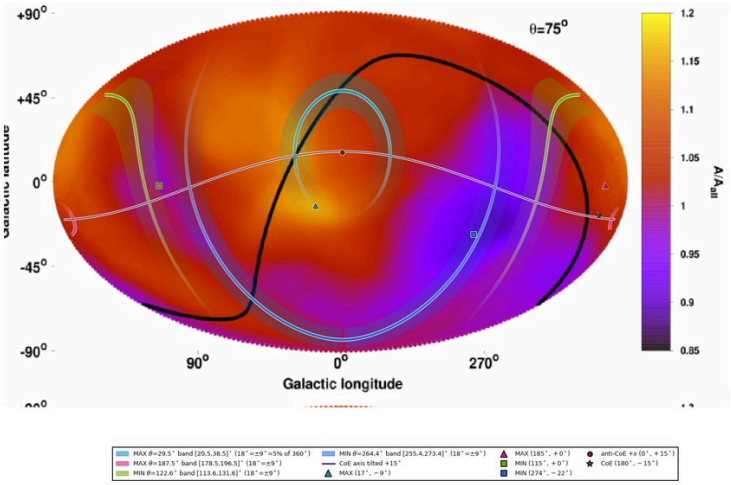





```
fig8_overlay_matched_grad.png: new (1975, 1515) vs ref (1975, 1515) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference papers/spreadsheet_model/fig8_overlay_matched_grad.png   |   right: notebook re-run
steps_v3/figs/fig8_overlay_matched_grad.png
```

Migkas Fig. 8 $\theta = 75^\circ$ A/A_{all} + four CoE highlight bands, each 18° end-to-end (±9° = 5% of 360°)
(two MAX + two MIN; cyan/green/magenta/blue; anti-CoE at (0°, +15°); CoE axis tilted +15°; l increases LEFT; gradient: darkest at θ ray, transparent opposite)

Migkas Fig. 8 — $\theta = 75^\circ$ A/A_{all} (their numbered colorbar)



Migkas Fig. 8 $\theta = 75^\circ$ A/A_{all} + four CoE highlight bands, each 18° end-to-end (±9° = 5% of 360°)
(two MAX + two MIN; cyan/green/magenta/blue; anti-CoE at (0°, +15°); CoE axis tilted +15°; l increases LEFT; gradient: darkest at θ ray, transparent opposite)

Migkas Fig. 8 — $\theta = 75^\circ$ A/A_{all} (their numbered colorbar)

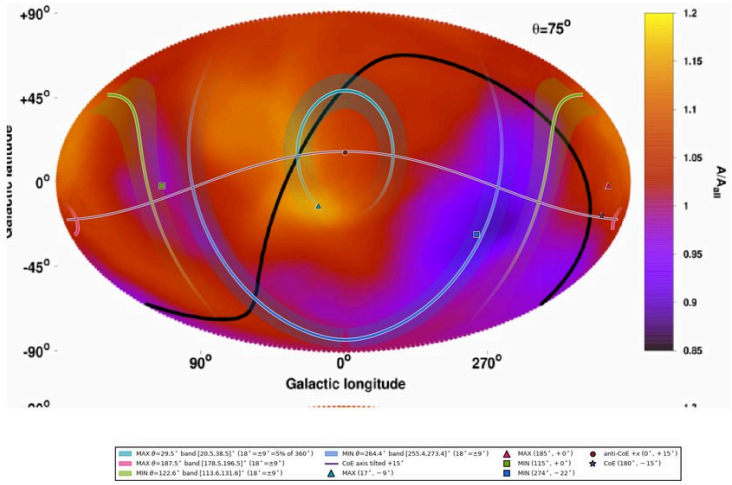
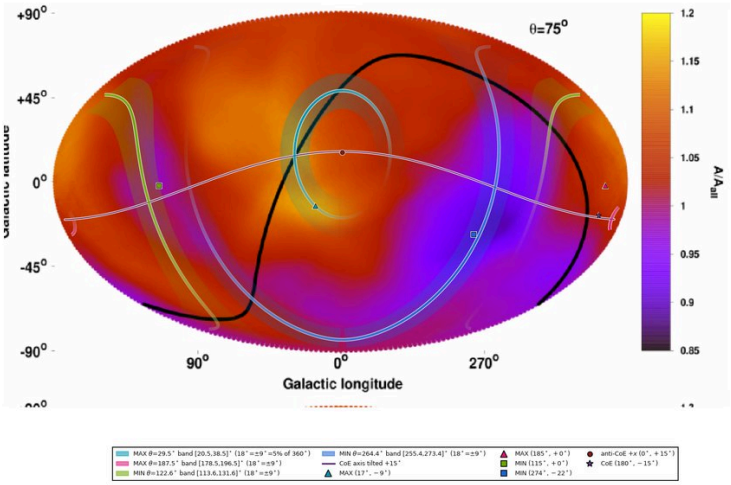


fig8_overlay_matched_grad_rot45.png: new (1975, 1515) vs ref (1975, 1515) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference papers/spreadsheet_model/fig8_overlay_matched_grad_rot45.png | right: notebook re-run steps_v3/figs/fig8_overlay_matched_grad_rot45.png

Migkas Fig. 8 $\theta = 75^\circ$ A/A_{all} + four CoE highlight bands, each 18° end-to-end (±9° = 5% of 360°)
(two MAX + two MIN; cyan/green/magenta/blue; anti-CoE at (0°, +15°); CoE axis tilted +15°; ℓ increases LEFT; gradient: darkest at θ ray, transparent opposite; +45° CCW about CoE axis)

Migkas Fig. 8 — $\theta = 75^\circ$ A/A_{all} (their numbered colorbar)



Migkas Fig. 8 $\theta = 75^\circ$ A/A_{all} + four CoE highlight bands, each 18° end-to-end (±9° = 5% of 360°)
(two MAX + two MIN; cyan/green/magenta/blue; anti-CoE at (0°, +15°); CoE axis tilted +15°; ℓ increases LEFT; gradient: darkest at θ ray, transparent opposite; +45° CCW about CoE axis)

Migkas Fig. 8 — $\theta = 75^\circ$ A/A_{all} (their numbered colorbar)

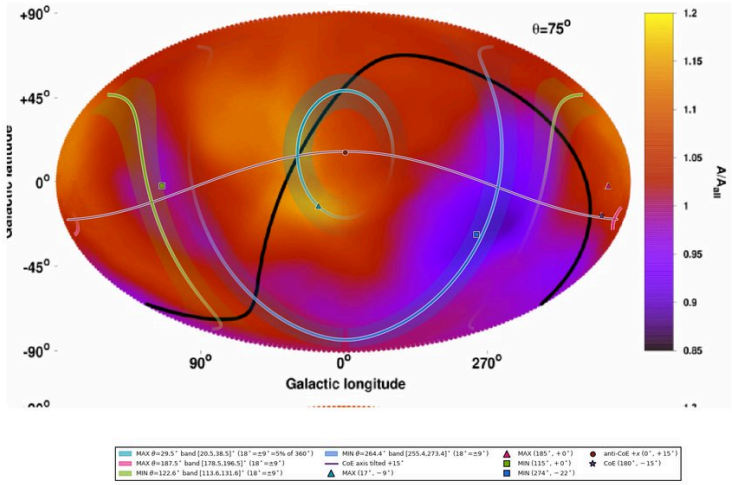
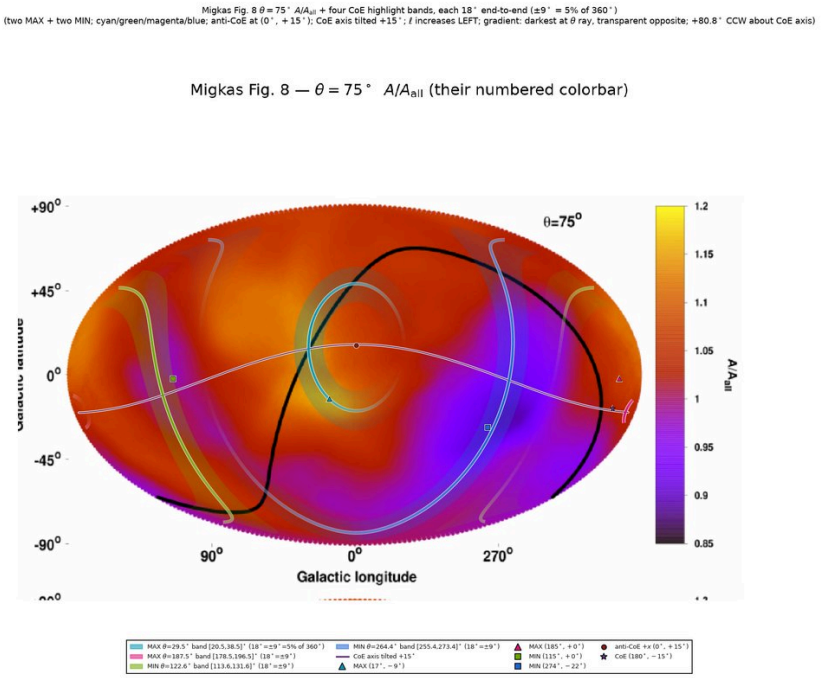
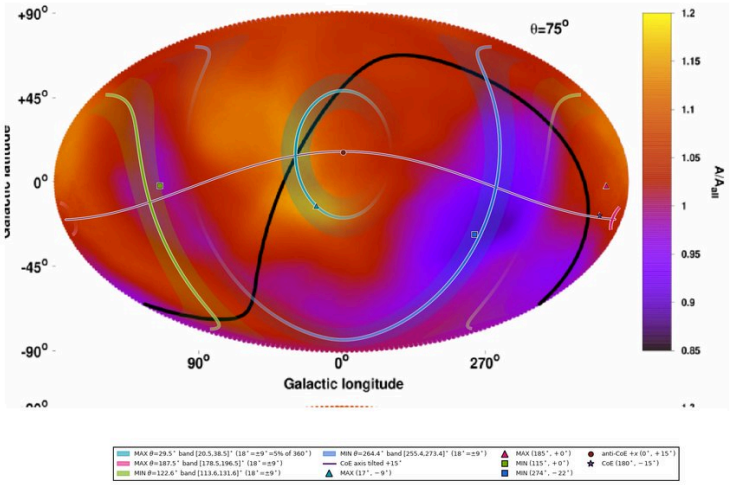


fig8_overlay_matched_grad_rot81.png: new (1975, 1515) vs ref (1975, 1515) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference papers/spreadsheet_model/fig8_overlay_matched_grad_rot81.png | right: notebook re-run steps_v3/figs/fig8_overlay_matched_grad_rot81.png



Migkas Fig. 8 θ = 75° A/A_all + four CoE highlight bands, each 18° end-to-end (±9° = 5% of 360°)
(two MAX + two MIN; cyan/green/magenta/blue; anti-CoE at (0°, +15°); CoE axis tilted +15°; ℓ increases LEFT; gradient: darkest at θ ray, transparent opposite; +80.8° CCW about CoE axis)
Migkas Fig. 8 — θ = 75° A/A_all (their numbered colorbar)



Step 19 — Loop-flow page (the four transformation steps, earlier 26 Sep version)

What it shows. The step-by-step loop-flow figure: the $A(\ell)$ graph with its extrema bands, then the loops untilted, tilted, and with the gradient and roll, ending on the Migkas Fig. 8 overlay. The explanatory text comes from `stack_text.py`. Run from the `loops/` folder, whose `loop_fit.json` holds the fitted tilt and roll.

Source script: `/workspace/cosmology/papers/spreadsheet_model/loops/loop_flow2.py` (last modified 2026-09-26 20:56 CT), run verbatim. *Output-path edit:* `../loop_` → `/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/loop_`. *Reference image:* `papers/spreadsheet_model/loop_flow_thinshell.png`.

```
%%runscript loop_flow2.py --cwd /workspace/cosmology/papers/spreadsheet_model/loops
import json, sys, os, numpy as np, matplotlib
matplotlib.use("Agg"); import matplotlib.pyplot as plt
from matplotlib.collections import PolyCollection, LineCollection
from matplotlib.patches import Ellipse, FancyArrowPatch, Patch
from matplotlib.lines import Line2D
from PIL import Image
sys.path.insert(0, "/workspace/cosmology/figures"); sys.path.insert(0, "../bubble")
from coe_half_loop_geometry import lb_to_n, n_to_lb, ring_basis, MIGKAS_MAXES, MIGKAS_MINS, lon_disp_from_ell, great_circle_through_poles
_here = os.getcwd(); os.chdir("../bubble")
```

```
import analysis_curve as AC
from bubble_core import TH1, ML, MA, MLO, MHI, AFfour
os.chdir(_here)
F = json.load(open("loop_fit.json")); d = json.load(open("../bubble/thin_b_free_hug.json"))
A = AC.curve("ptshell", d); l0 = F["l0"]
FIT = F["fits"]["b0_only"]; UNT = F["fits"]["untilted"]
COLS = {"MAX1": "#0097b2", "MAX2": "#e0107a", "MIN1": "#6fa800", "MIN2": "#2166d4"}
pairs = F["pairs"]
keys = []
cm = {"MAX": 0, "MIN": 0}
for p in pairs: cm[p["kind"]] += 1; keys.append("%s%d" % (p["kind"], cm[p["kind"]]))
BH = 9.0
def moll(l, b):
    lon = np.radians(lon_disp_from_ell(np.asarray(l, float))); lat = np.radians(np.asarray(b, float))
    th = lat.copy()
    for _ in range(40):
        f = 2 * th + np.sin(2 * th) - np.pi * np.sin(lat); th = th - f / (2 + 2 * np.cos(2 * th) + 1e-12)
    return 2 * np.sqrt(2) / np.pi * lon * np.cos(th), np.sqrt(2) * np.sin(th)
def pts_lb(P):
    b = np.degrees(np.arcsin(np.clip(P[:, 2], -1, 1))); l = np.degrees(np.arctan2(P[:, 1], P[:, 0])) % 360; return l, b

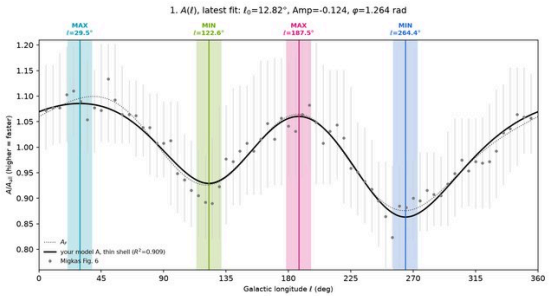
# ... 229 more lines of the verbatim script: see coe_steps_v3.ipynb / .html / .md (or coe_steps_v3_fullcode.pdf)
```

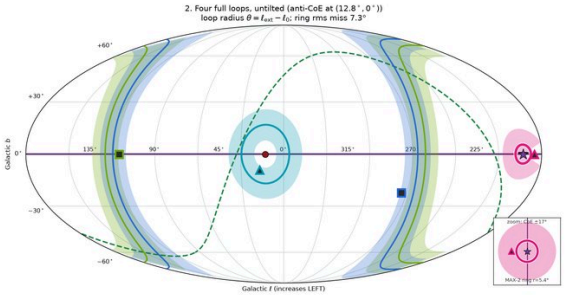
```
page 1 (2550, 2040) (1700, 1360)
page 2 (2550, 2970) (1460, 1700)
ok (865, 1700)
```

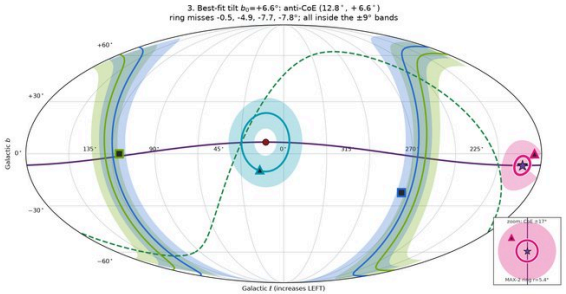
```
compare(FIGS / 'loop_flow_thinshell.png', '/workspace/cosmology/papers/spreadsheet_model/loop_flow_thinshell.png')
```

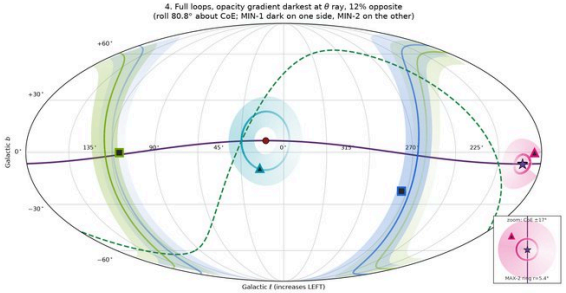
```
loop_flow_thinshell.png: new (2550, 5010) vs ref (2550, 5010) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference papers/spreadsheet_model/loop_flow_thinshell.png   |   right: notebook re-run steps_v3/figs/loop_flow_thinshell.png
```

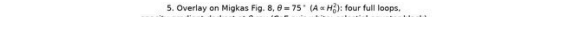
Stacked transformations: from your model A(ℓ) to loops on the sky and the Migkas Fig. 8 overlay
(thin-shell best fit; Galactic Mollweide, ℓ increases LEFT)











Step 1. Gravity breaks homology: your model A

A pure homologous blast (V = H R) would look isotropic from anywhere, so the CoE would drop out. Gravity decelerates matter at different positions relative to the CoE by different amounts, so the expansion rate seen from Earth depends on direction. Your model A (thin-shell best fit) turns this into $A(\ell) \propto H0^2$ along the Galactic plane, fitted to Migkas Fig. 6 with the parameters in Table 1a. Its two maxima and two minima are read off the curve (Table 1b); each coloured band is ±9° about an extremum, and the same colours are used for the loops on every map below.

Table 1a. Model A fit to Migkas Fig. 6

Parameter	Value
ℓ_0 (anti-CoE longitude)	12.82°
Amp	-0.124
φ	1.264 rad
R^2	0.909

Table 1b. Extrema of model A

Loop	ℓ_{ext} (°)	A/A_all
MAX-1	29.5	1.085
MIN-1	122.6	0.929
MAX-2	187.5	1.060
MIN-2	264.4	0.863

Step 2. Each extremum becomes a loop (untilted)

Model A is axisymmetric about the CoE axis, so a value read at longitude ℓ_{ext} in the plane holds on the whole cone of half-angle $\theta = \ell_{ext} - \ell_0$ about the anti-CoE direction. Drawing each cone on the sky turns the four extrema into four full loops, here with the axis kept in the Galactic plane (b0 = 0). The ring miss is the signed angular distance from each Migkas extremum to its loop (Table 2). MAX-2 is a small ring 5.4° round the CoE (see inset), and the two MIN loops have almost equal radii, so they are drawn 2° apart for display.

Table 2. Loop radii and ring misses, untilted

Loop	θ (°)	cone radius (°)	Migkas (ℓ, b)	miss (°)
MAX-1	16.7	16.7	(17°, -9°)	-6.8
MIN-1	109.8	109.8	(115°, +0°)	-7.6
MAX-2	174.6	5.4 (CoE)	(185°, +0°)	-2.5
MIN-2	251.6	108.4	(274°, -22°)	-10.2
RMS				7.3

Step 3. Tilt the CoE axis out of the plane

The anti-CoE longitude is fixed by ℓ_0, but the axis may point out of the Galactic plane. Tilting it to latitude b0 moves all four loops rigidly on the sphere; b0 is fitted by least squares on the ring misses, giving b0 = +6.6° and anti-CoE (12.8°, +6.6°). All four Migkas extrema then fall inside the ±9° bands (Table 3). Freeing ℓ_0 as well only reaches 5.8° RMS, so the remaining ~6° comes from the offset between the model and Migkas extremum longitudes, not from the orientation.

Table 3. Ring misses before and after the tilt (°)

Loop	b0 = 0	b0 = +6.6°
MAX-1	-6.8	-0.5
MIN-1	-7.6	-7.7
MAX-2	-2.5	-4.9
MIN-2	-10.2	-7.8
RMS	7.3	6.0

Step 4. Roll about the CoE axis: the gradient

The tilt fixes where the loops lie but not their orientation about the axis. A roll about the CoE axis sets each loop's θ-ray point, the point where the loop meets the great circle through the axis that plays the role of step 1's slice; there the loop value is the one actually read from A(ℓ). Each loop is drawn darkest at its θ-ray point and fades to 12% opacity opposite, so the dark end marks the measured direction and the faded part is the axisymmetric extrapolation. Because sin θ has opposite signs for MIN-1 and MIN-2, their dark ends lie on opposite sides. The roll, 80.8°, minimises the distance from the θ-ray points to the Migkas extrema (Table 4).

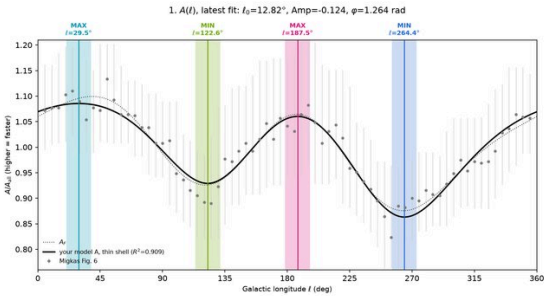
Table 4. θ-ray miss after roll 80.8° (°)

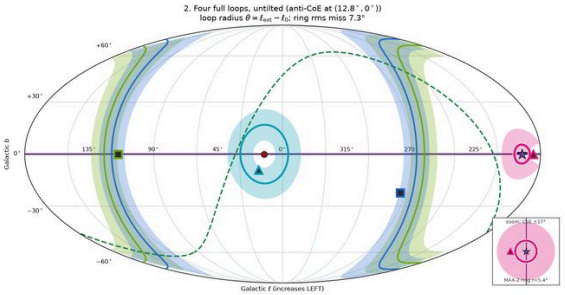
Loop	θ-ray miss
MAX-1	21.9
MIN-1	10.7
MAX-2	6.3
MIN-2	13.9
RMS	14.4

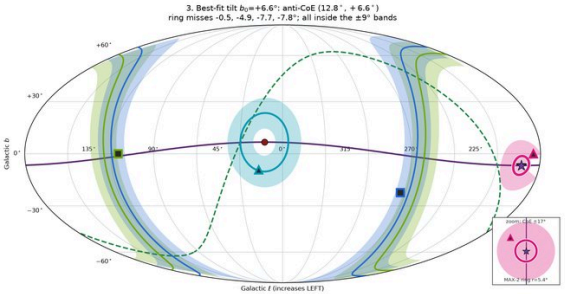
Step 5. Compare with Migkas Fig. 8

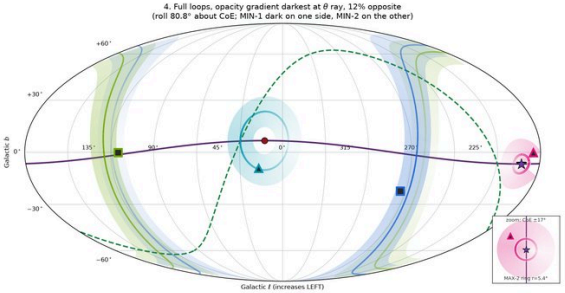
Finally the tilted and rolled loops are laid on Migkas et al.'s

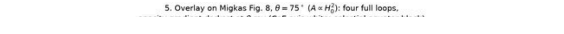
Stacked transformations: from your model A(ℓ) to loops on the sky and the Migkas Fig. 8 overlay
(thin-shell best fit; Galactic Mollweide, ℓ increases LEFT)











Step 1. Gravity breaks homology: your model A

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Step 2. Each extremum becomes a loop (untilted)

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Table 2. Loop radii and ring misses, untilted

Loop	θ (°)	cone radius (°)	Migkas (ℓ, b)	miss (°)
MAX-1	16.7	16.7	(17°, -9°)	-6.8
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MIN-2	251.6	108.4	(274°, -22°)	-10.2
RMS				7.3

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The anti-CoE longitude is fixed by ℓ_0, but the axis may point out of the Galactic plane. Tilting it to latitude b0 moves all four loops rigidly on the sphere; b0 is fitted by least squares on the ring misses, giving b0 = +6.6° and anti-CoE (12.8°, +6.6°). All four Migkas extrema then fall inside the ±9° bands (Table 3). Freeing ℓ_0 as well only reaches 5.8° RMS, so the remaining ~6° comes from the offset between the model and Migkas extremum longitudes, not from the orientation.

Table 3. Ring misses before and after the tilt (°)

Loop	b0 = 0	b0 = +6.6°
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MIN-1	-7.6	-7.7
MAX-2	-2.5	-4.9
MIN-2	-10.2	-7.8
RMS	7.3	6.0

Step 4. Roll about the CoE axis: the gradient

The tilt fixes where the loops lie but not their orientation about the axis. A roll about the CoE axis sets each loop's θ-ray point, the point where the loop meets the great circle through the axis that plays the role of step 1's slice; there the loop value is the one actually read from A(ℓ). Each loop is drawn darkest at its θ-ray point and fades to 12% opacity opposite, so the dark end marks the measured direction and the faded part is the axisymmetric extrapolation. Because sin θ has opposite signs for MIN-1 and MIN-2, their dark ends lie on opposite sides. The roll, 80.8°, minimises the distance from the θ-ray points to the Migkas extrema (Table 4).

Table 4. θ-ray miss after roll 80.8° (°)

Loop	θ-ray miss
MAX-1	21.9
MIN-1	10.7
MAX-2	6.3
MIN-2	13.9
RMS	14.4

Step 5. Compare with Migkas Fig. 8

Finally the tilted and rolled loops are laid on Migkas et al.'s

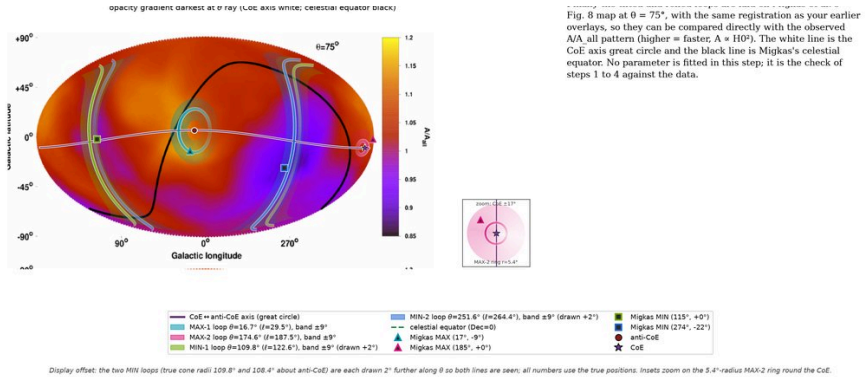


Fig. 8 map at θ = 75°, with the same registration as your earlier overlays, so they can be compared directly with the observed A/A_all pattern (higher = faster, A ∝ H0²). The white line is the CoE axis great circle and the black line is Migkas's celestial equator. No parameter is fitted in this step; it is the check of steps 1 to 4 against the data.

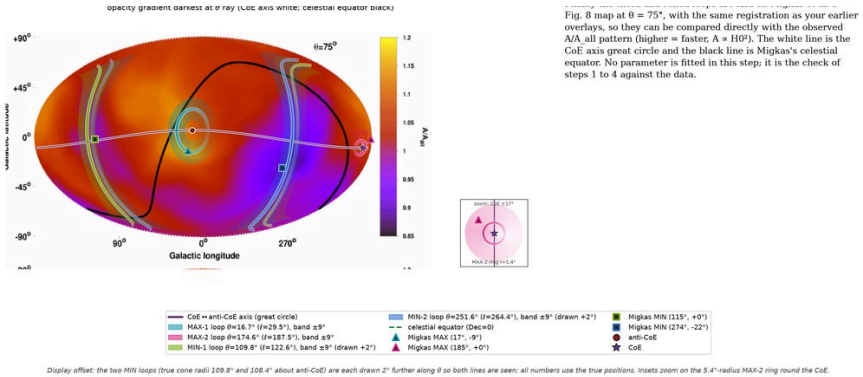


Fig. 8 map at θ = 75°, with the same registration as your earlier overlays, so they can be compared directly with the observed A/A_all pattern (higher = faster, A ∝ H0²). The white line is the CoE axis great circle and the black line is Migkas's celestial equator. No parameter is fitted in this step; it is the check of steps 1 to 4 against the data.

Step 20 — Journal page of the loop transformations (26 Sep)

What it shows. The journal-style page: the A graph (row 0), then Transformation 1 (each extremum becomes a full loop of radius $\theta = \ell_{\rm ext} - \ell_0$ about anti-CoE, untilted), Transformation 2 (tilt), Transformation 3 (roll, with the gradient darkest at the θ ray), and the Fig. 8 overlay. Each caption gives the RMS miss.

Source script: `/workspace/cosmology/papers/spreadsheet_model/loops/loop_journal.py` (last modified 2026-09-26 21:21 CT), run verbatim. *Output-path edit:* `../loop_` → `/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/loop_`; `Path("journal_rows")` → `Path("/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/journ`. *Reference image:* `papers/spreadsheet_model/loop_journal_page.png`.

```
%%runscript loop_journal.py --cwd /workspace/cosmology/papers/spreadsheet_model/loops
"""Journal page: A(l) + Chris's own 3-panel figure (make_coe_rings_gradient_loops.main) for three transformations
+ his Fig. 8 overlay (make_migkas_fig8_gradient_loops_overlay.main). Only colours, theta angles and axis orientation
(and the numbers in titles/legends) are changed; figures, layout and styling are his, unchanged."""
import json, sys, os, re, numpy as np, matplotlib
matplotlib.use("Agg"); import matplotlib.pyplot as plt
from matplotlib.figure import Figure
from pathlib import Path
from PIL import Image
sys.path.insert(0, "/workspace/cosmology/figures"); sys.path.insert(0, "../bubble")
import coe_gradient_loop_geometry as CG
import make_coe_rings_gradient_loops as RB
import make_migkas_fig8_gradient_loops_overlay as OV
from coe_half_loop_geometry import lb_to_n, n_to_lb, MIGKAS_MAXES, MIGKAS_MINS, band_edges, mu_pair_for_band, BAND_HALF
_here = os.getcwd(); os.chdir("../bubble")
import analysis_curve as AC
from bubble_core import TH1, ML, MA, MLO, MHI
os.chdir(_here)
OUT = Path("/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/journal_rows").resolve()
F = json.load(open("loop_fit.json")); d = json.load(open("../bubble/thin_b_free_hug.json"))
A = AC.curve("ptshell", d); l0 = F["l0"]; FIT = F["fits"]["b0_only"]; UNT = F["fits"]["untilted"]
```

```
P = {(p["kind"], round(p["migkas"][0])): p for p in F["pairs"]}
pMAX1, pMAX2, pMIN1, pMIN2 = P[("MAX", 17)], P[("MAX", 185)], P[("MIN", 115)], P[("MIN", 274)]
C = {"MAX1": "#0097b2", "MIN1": "#6fa800", "MAX2": "#e0107a", "MIN2": "#2166d4"}
THMAX = [pMAX1["theta"], pMAX2["theta"]]; THMIN = [pMIN1["theta"], pMIN2["theta"]]
# ---- colours and angles (only changes) ----
for M in (RB, OV):
    M.C_MAX_30, M.C_MAX_185, M.C_MIN_115, M.C_MIN_265 = C["MAX1"], C["MAX2"], C["MIN1"], C["MIN2"]
    M.COLORS_MAX = [C["MAX1"], C["MAX2"]]; M.COLORS_MIN = [C["MIN1"], C["MIN2"]]
OV.THETA_MAX, OV.THETA_MIN = list(THMAX), list(THMIN); OV.MAX_BANDS, OV.MIN_BANDS = band_edges(THMAX), band_edges(THMIN)

# ... 123 more lines of the verbatim script: see coe_steps_v3.ipynb / .html / .md (or coe_steps_v3_fullcode.pdf)
```

```
{
  "png": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/journal_rows/coe_rings_gradient_loops_tilt0.png",
  "pdf": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/journal_rows/coe_rings_gradient_loops_tilt0.pdf",
  "script": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/journal_rows/make_coe_rings_gradient_loops.py",
  "geometry_module": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/journal_rows/coe_half_loop_geometry.py",
  "gradient_module": "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/journal_rows/coe_gradient_loop_geometry.py",
  "FULL_GRADIENT_LOOPS": true,
  "gradient_rule": "full loops; alpha *= (1+cos psi)/2, psi = ring azimuth from the \u03b8-ray point (ring point most along
sign(sin\u03b8_mid)\u00b7\u0176, \u0176 rolled for B/C; A unrolled); equivalently alpha_scale = (1 + sign(sin\u03b8_mid)\u00b7sin(phi))/2",
  "legacy_half_loop_rule_equiv_alpha_scale_ge_half": "\u03b8-ray side: keep (n\u00b7\u0176)*sin(\u03b8_mid)>=0 (Panel A: ey*sin(\u03b8_mid)>=0; TOP if
sin\u03b8>0, BOTTOM if sin\u03b8<0)",
  "panel_A_roll_deg": 0.0,
  "roll_applies_to_panels": [
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    "C"
  ],
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    "fill_B": 0.26,
    "fill_C": 0.25,
    "mid_line": 0.95
  },
  "axis_tilt_b_deg": 0.0,
  "roll_about_coe_deg": 0.0,
  "migkas_max_lb_deg": [
    [
      17.0,
      -9.0
    ],
    [
      185.0,
      0.0
    ]
  ],
  "migkas_min_lb_deg": [
    [
      115.0,
      0.0
    ],
    [
      274.0,
      -22.0
    ]
  ],
  "strip_theta_max_deg": [
    16.702461873818333,
    174.63115386586279
  ],
  "strip_theta_min_deg": [
    109.78243150454531,
    251.59236170210514
  ],
  "band_half_width_deg": 9.0,
  "band_full_width_deg": 18.0,
  "max_bands_deg": [
    [
      7.702461873818333,
```

```
      25.702461873818333
    ],
    [
      165.63115386586279,
      183.63115386586279
    ]
  ],
  "min_bands_deg": [
    [
      100.78243150454531,
      118.78243150454531
    ],
    [
      242.59236170210514,
      260.5923617021051
    ]
  ],
  "CoE_lb_deg": [
    192.8215,
    -0.0
  ],
  "anti_CoE_lb_deg": [
    12.8215,
    0.0
  ],
  "D_hat_galactic_xyz": [
    0.9750662140684075,
    0.2219141234403582,
    0.0
  ],
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opacity opposite; roll about CoE = 0\u00b0 CCW (panels B/C only; panel A unrolled)",
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```

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opacity opposite; roll about CoE = 0\u00b0 CCW (panels B/C only; panel A unrolled)",
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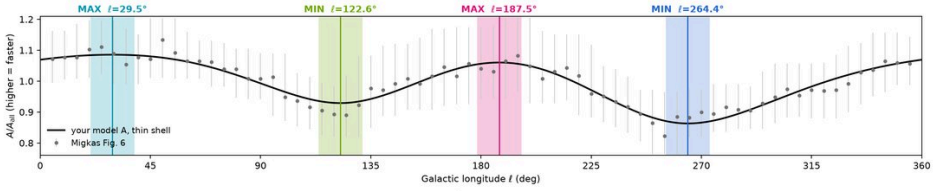
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  "legacy_half_loop_rule_equiv_alpha_scale_ge_half": "\u03b8-ray side: keep (n\u00b7\u0176)*sin(\u03b8_mid)>=0 (Panel A: ey*sin(\u03b8_mid)>=0; TOP if
sin\u03b8>0, BOTTOM if sin\u03b8<0)",
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    "fill_C": 0.25,
    "mid_line": 0.95
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    174.63115386586279
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    251.59236170210514
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```

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    0.968547882712532,
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opacity opposite; roll about CoE = 81\u00b0 CCW (panels B/C only; panel A unrolled)",
  "plot_convention": "Migkas Galactic Mollweide: lon_disp=-ell_wrapped (ell increases LEFT)",
  "colors": {
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    "MIN_265": "#2166d4"
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A vs C diameters (px at 160 dpi), zoom: [(418.1, 418.5, np.float64(1.4)), (418.1, 418.5, np.float64(1.4)), (418.1, 418.5, np.float64(1.4)), (418.1,
418.5, np.float64(1.4)), (418.1, 418.5, np.float64(1.4)), (418.1, 418.5, np.float64(1.4))]
page (2278, 5713) web (678, 1700) [(2278, 528), (2278, 99), (2278, 1027), (2278, 99), (2278, 1027), (2278, 99), (2278, 988), (2278, 99), (2278, 1747)]
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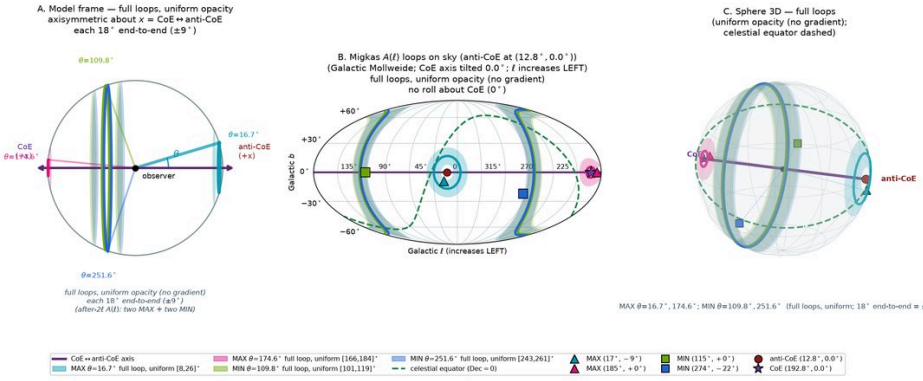

```
compare(FIGS / 'loop_journal_page.png', '/workspace/cosmology/papers/spreadsheet_model/loop_journal_page.png')
```

```
loop_journal_page.png: new (2278, 5713) vs ref (2278, 5713) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference papers/spreadsheet_model/loop_journal_page.png   |   right: notebook re-run steps_v3/figs/loop_journal_page.png
```



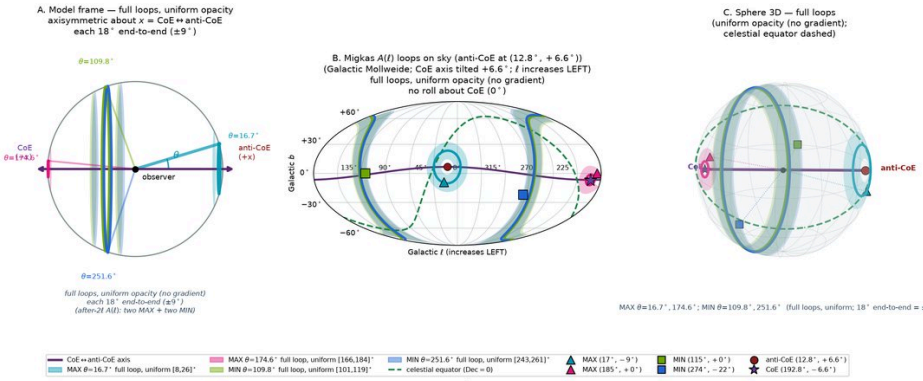
Transf. 1 *Transformation 1: each extremum becomes a full loop of radius $\theta = \ell_{ext} - \ell_0$ about anti-CoE (12.8°, 0°), untilted; ring RMS miss 7.3°*

Pure CoE axisymmetry: full loops, uniform opacity (no gradient)
(each 18° end-to-end = ±9°; MAX θ=16.7°/174.6°; MIN θ=109.8°/251.6°; anti-CoE (12.8°, 0.0°), CoE axis tilted 0.0°)



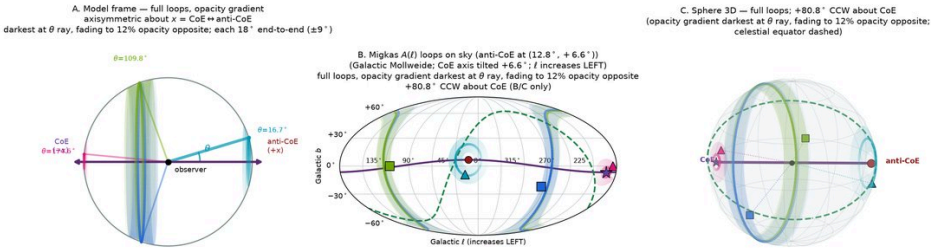
Transf. 2 *Transformation 2: tilt CoE axis b_0 = +6.6° (anti-CoE (12.8°, +6.6°)); ring RMS miss 7.3° to 6.0°*

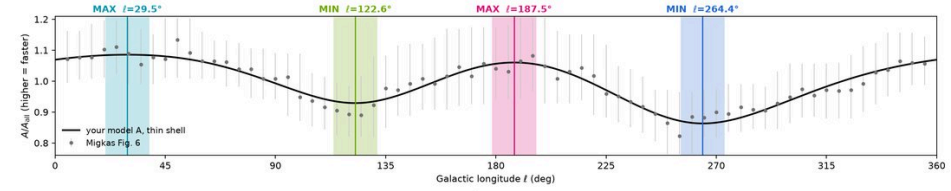
Pure CoE axisymmetry: full loops, uniform opacity (no gradient)
(each 18° end-to-end = ±9°; MAX θ=16.7°/174.6°; MIN θ=109.8°/251.6°; anti-CoE (12.8°, +6.6°), CoE axis tilted +6.6°)



Transf. 3 *Transformation 3: roll 80.8° about the CoE axis; loop gradient darkest at the θ ray, fading to 12% opposite; θ-ray RMS miss 14.4°*

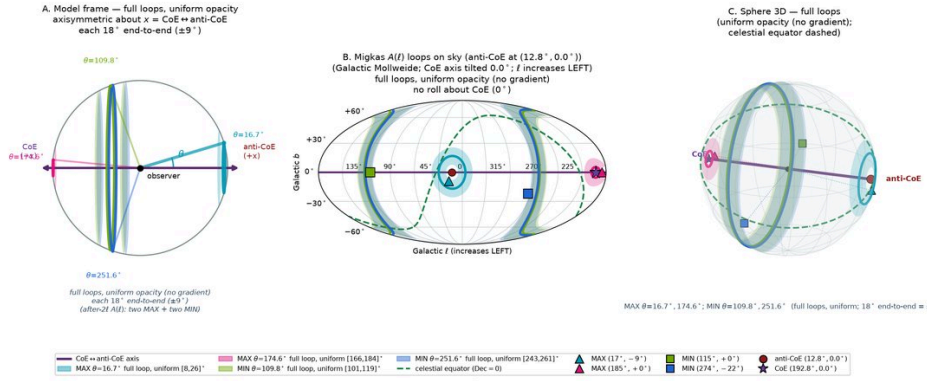
Pure CoE axisymmetry: full loops, opacity gradient darkest at θ ray, fading to 12% opacity opposite ($\alpha \propto 1 - 0.88|\psi|/\pi$)
(each 18° end-to-end = ±9°; MAX θ=16.7°/174.6°; MIN θ=109.8°/251.6°; anti-CoE (12.8°, +6.6°), CoE axis tilted +6.6°; +80.8° CCW about CoE axis)





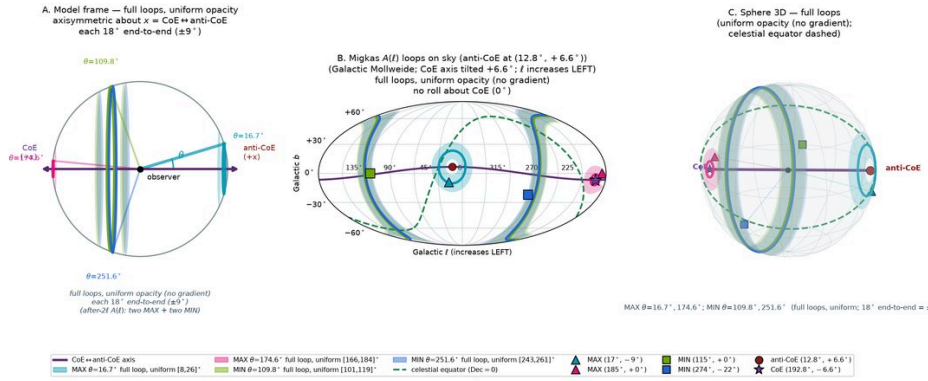
Transf. 1 *Transformation 1: each extremum becomes a full loop of radius $\theta = \ell_{ext} - \ell_0$ about anti-CoE (12.8°, 0°), untilted; ring RMS miss 7.3°*

Pure CoE axisymmetry: full loops, uniform opacity (no gradient)
(each 18° end-to-end = ±9°; MAX θ=16.7°/174.6°; MIN θ=109.8°/251.6°; anti-CoE (12.8°, 0.0°), CoE axis tilted 0.0°)



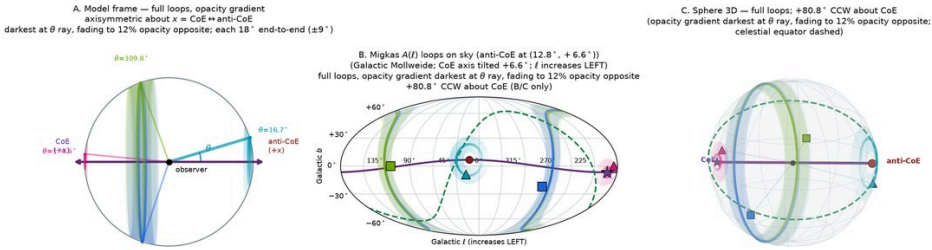
Transf. 2 *Transformation 2: tilt CoE axis b_0 = +6.6° (anti-CoE (12.8°, +6.6°)); ring RMS miss 7.3° to 6.0°*

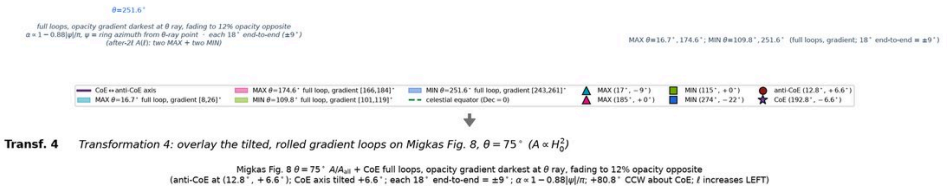
Pure CoE axisymmetry: full loops, uniform opacity (no gradient)
(each 18° end-to-end = ±9°; MAX θ=16.7°/174.6°; MIN θ=109.8°/251.6°; anti-CoE (12.8°, +6.6°), CoE axis tilted +6.6°)



Transf. 3 *Transformation 3: roll 80.8° about the CoE axis; loop gradient darkest at the θ ray, fading to 12% opposite; θ-ray RMS miss 14.4°*

Pure CoE axisymmetry: full loops, opacity gradient darkest at θ ray, fading to 12% opacity opposite ($\alpha \propto 1 - 0.88|\psi|/\pi$)
(each 18° end-to-end = ±9°; MAX θ=16.7°/174.6°; MIN θ=109.8°/251.6°; anti-CoE (12.8°, +6.6°), CoE axis tilted +6.6°; +80.8° CCW about CoE axis)

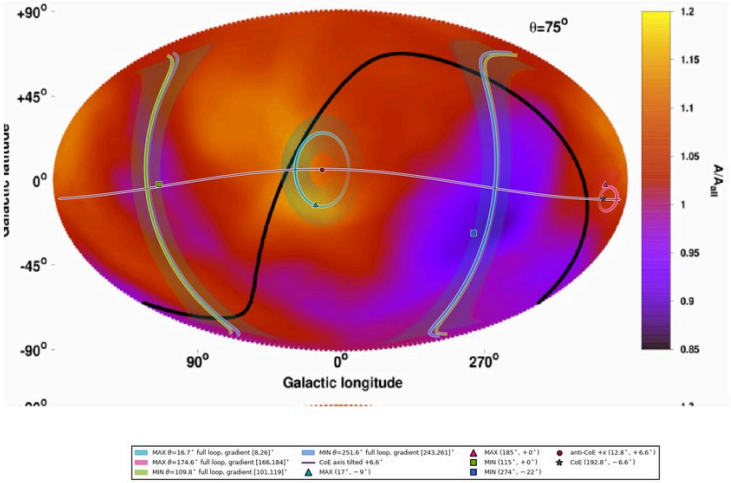




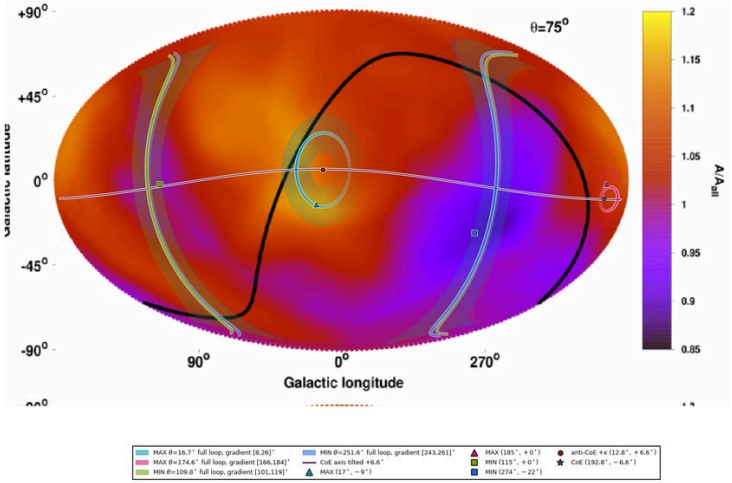
Transf. 4 *Transformation 4: overlay the tilted, rolled gradient loops on Migkas Fig. 8, $\theta = 75^\circ$ ($A \propto H_0^2$)*

Migkas Fig. 8 $\theta = 75^\circ$ $A/A_{\rm all}$ + CoE full loops, opacity gradient darkest at θ ray, fading to 12% opacity opposite (anti-CoE at (12.8°, +6.6°); CoE axis tilted +6.6°; each 18° end-to-end = ±9°; $\alpha \propto 1 - 0.88|\psi|/\pi$; +80.8° CCW about CoE; ℓ increases LEFT)

Migkas Fig. 8 — $\theta = 75^\circ$ $A/A_{\rm all}$ (their numbered colorbar)



Migkas Fig. 8 — $\theta = 75^\circ$ $A/A_{\rm all}$ (their numbered colorbar)



Step 21 — Loop stacks for the 45° and 80.8° rolls (26 Sep)

What it shows. For each roll (45° and the fitted 80.8°): the A graph, the three-panel loops in the Galactic-ℓ frame, and the Fig. 8 overlay, stacked on one page. This script loads the gradient module version it was written with (`coe_gradient_loop_geometry.cos_bak.py`).

Source script: `/workspace/cosmology/papers/spreadsheet_model/loops/loop_stack.py` (last modified 2026-09-26 22:28 CT), run verbatim. *Output-path edit:* `../loop_` → `/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/loop_` ; `../%s.png` → `/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/%s.png` ; `../%s_web.jpg` → `/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/%s_web.jpg` ; `Path("stack_rows")` → `Path("/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/stack` . *Reference image:* `papers/spreadsheet_model/loop_stack_roll45.png` , `papers/spreadsheet_model/loop_stack_roll80p8.png` .

```
%%runscript loop_stack.py --cwd /workspace/cosmology/papers/spreadsheet_model/loops
"""Journal page: A(l) + Chris's own 3-panel figure (make_coe_rings_gradient_loops.main) for three transformations
+ his Fig. 8 overlay (make_migkas_fig8_gradient_loops_overlay.main). Only colours, theta angles and axis orientation
(and the numbers in titles/legends) are changed; figures, layout and styling are his, unchanged."""
import json, sys, os, re, numpy as np, matplotlib
matplotlib.use("Agg"); import matplotlib.pyplot as plt
from matplotlib.figure import Figure
from pathlib import Path
from PIL import Image
sys.path.insert(0, "/workspace/cosmology/figures"); sys.path.insert(0, "../bubble")
import importlib.util
_sp = importlib.util.spec_from_file_location("coe_gradient_loop_geometry", "/workspace/cosmology/figures/coe_gradient_loop_geometry.cos_bak.py")
CG = importlib.util.module_from_spec(_sp); sys.modules["coe_gradient_loop_geometry"] = CG; _sp.loader.exec_module(CG)
import make_coe_rings_gradient_loops as RB
import make_migkas_fig8_gradient_loops_overlay as OV
from coe_half_loop_geometry import lb_to_n, n_to_lb, MIGKAS_MAXES, MIGKAS_MINS, band_edges, mu_pair_for_band, BAND_HALF
_here = os.getcwd(); os.chdir("../bubble")
import analysis_curve as AC
from bubble_core import TH1, ML, MA, MLO, MHI, AFfour
os.chdir(_here)
OUT = Path("/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/stack_rows").resolve(); OUT.mkdir(exist_ok=True)
F = json.load(open("loop_fit.json")); d = json.load(open("../bubble/thin_b_free_hug.json"))
A = AC.curve("ptshell", d); l0 = F["l0"]; FIT = F["fits"]["b0_only"]; UNT = F["fits"]["untilted"]
P = {(p["kind"], round(p["migkas"][0])): p for p in F["pairs"]}
pMAX1, pMAX2, pMIN1, pMIN2 = P[("MAX", 17)], P[("MAX", 185)], P[("MIN", 115)], P[("MIN", 274)]
C = {"MAX1": "#0097b2", "MIN1": "#6fa800", "MAX2": "#e0107a", "MIN2": "#2166d4"}
THMAX = [pMAX1["l_model"], pMAX2["l_model"]]; THMIN = [pMIN1["l_model"], pMIN2["l_model"]]   # model-A extremum angles ARE the angles from the anti-CoE axis
# ---- colours and angles (only changes) ----
for M in (RB, OV):
    M.C_MAX_30, M.C_MAX_185, M.C_MIN_115, M.C_MIN_265 = C["MAX1"], C["MAX2"], C["MIN1"], C["MIN2"]

# ... 155 more lines of the verbatim script: see coe_steps_v3.ipynb / .html / .md (or coe_steps_v3_fullcode.pdf)
```

```
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sign(sin\u03b8_mid)\u00b7\u0176, \u0176 rolled for B/C; A unrolled); equivalently alpha_scale = (1 + sign(sin\u03b8_mid)\u00b7sin(phi))/2",
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    "C"
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  ],
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  ],
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    264.41384529244493
  ],
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```

```
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opacity opposite; roll about CoE = 45\u00b0 CCW (panels B/C only; panel A unrolled)",
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    "MAX_185": "#e0107a",
    "MIN_115": "#6fa800",
    "MIN_265": "#2166d4"
  }
}

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```

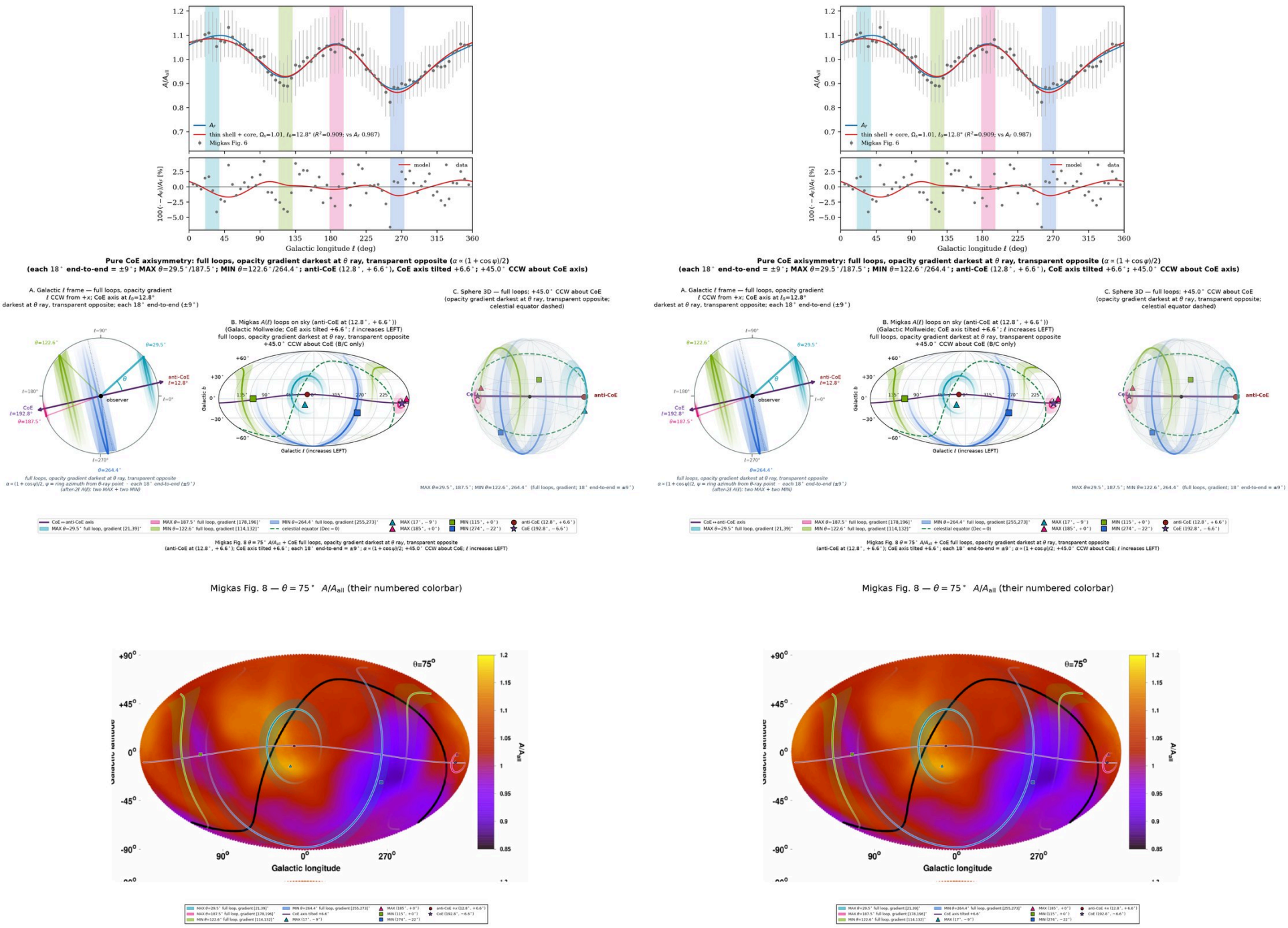
```
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  "gradient_rule": "full loops; alpha *= (1+cos psi)/2, psi = ring azimuth from the \u03b8-ray point (ring point most along
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  "legacy_half_loop_rule_equiv_alpha_scale_ge_half": "\u03b8-ray side: keep (n\u00b7\u0176)*sin(\u03b8_mid)>=0 (Panel A: ey*sin(\u03b8_mid)>=0; TOP if
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    "C"
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    "mid_line": 0.95
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  ],
  "migkas_min_lb_deg": [
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      0.0
    ],
    [
      274.0,
      -22.0
    ]
  ],
  "strip_theta_max_deg": [
    29.523945464158125,
    187.45263745620258
  ],
  "strip_theta_min_deg": [
    122.60391509488511,
    264.41384529244493
  ],
  "band_half_width_deg": 9.0,
  "band_full_width_deg": 18.0,
  "max_bands_deg": [
    [
      20.523945464158125,
```

```
      38.523945464158125
    ],
    [
      178.45263745620258,
      196.45263745620258
    ]
  ],
  "min_bands_deg": [
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      113.60391509488511,
      131.6039150948851
    ],
    [
      255.41384529244493,
      273.41384529244493
    ]
  ],
  "CoE_lb_deg": [
    192.8215,
    -6.6287
  ],
  "anti_CoE_lb_deg": [
    12.8215,
    6.6287
  ],
  "D_hat_galactic_xyz": [
    0.968547882712532,
    0.2204306244038185,
    0.1154354309474087
  ],
  "angular_sep_migkas_max0_min1_deg": 98.476384,
  "orientation": "anti-CoE at (0\u00b0, +7\u00b0), CoE at (180\u00b0, \u22127\u00b0); full loops, opacity gradient darkest at \u03b8 ray, fading to 12%
opacity opposite; roll about CoE = 81\u00b0 CCW (panels B/C only; panel A unrolled)",
  "plot_convention": "Migkas Galactic Mollweide: lon_disp=-ell_wrapped (ell increases LEFT)",
  "colors": {
    "MAX_30": "#0097b2",
    "MAX_185": "#e0107a",
    "MIN_115": "#6fa800",
    "MIN_265": "#2166d4"
  }
}

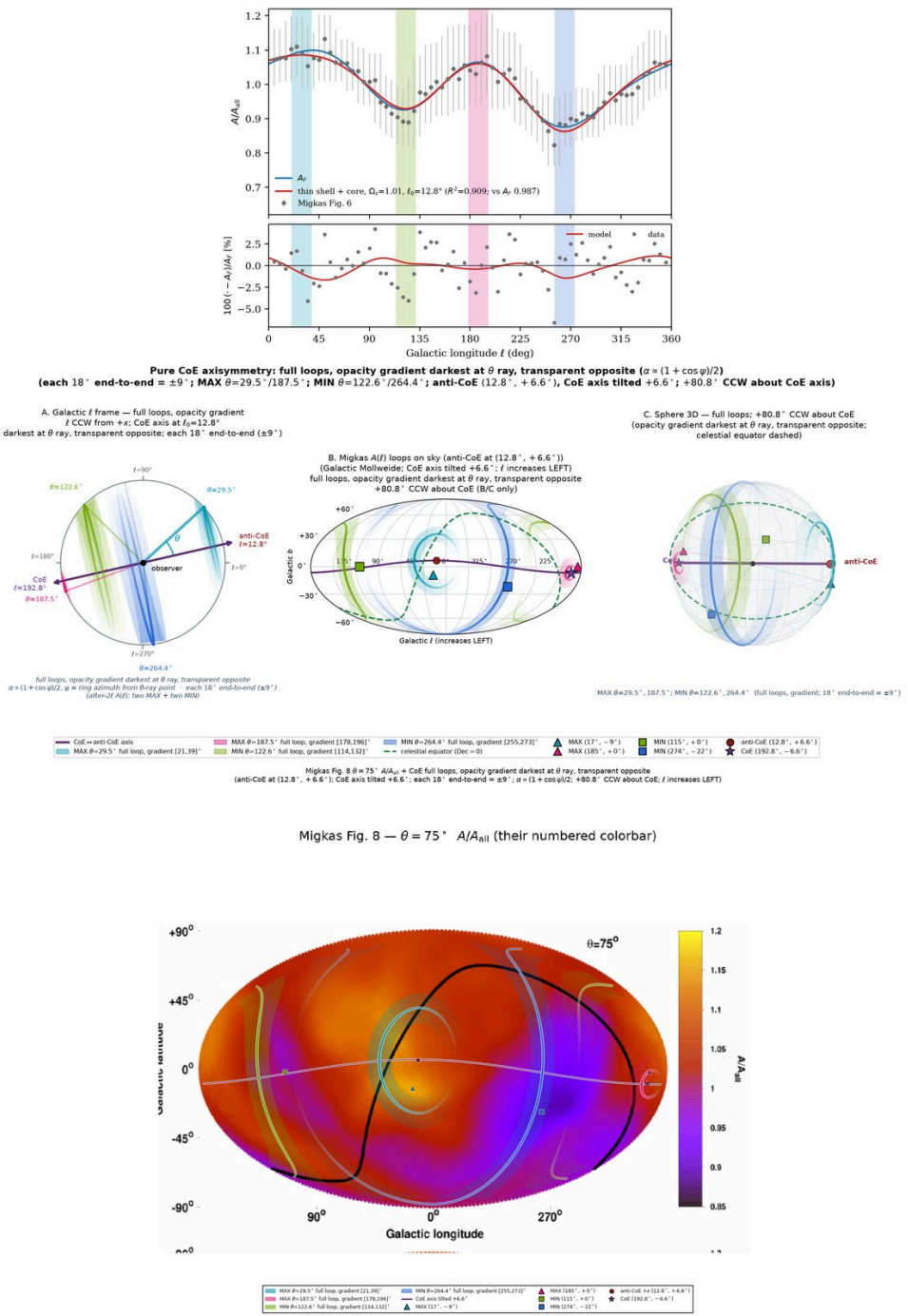
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Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/stack_rows/coe_rings_gradient_loops_tilt7_rot81ccw.pdf
Wrote /workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/stack_rows/coe_rings_gradient_loops_tilt7_rot81ccw.json
roll45 [(1176, 905), (2302, 1018), (1516, 1410)] (2302, 3333)
roll80p8 [(1176, 905), (2302, 1018), (1516, 1410)] (2302, 3333)
sizes [(418.1, 418.5, np.float64(1.4)), (418.1, 418.5, np.float64(1.4)), (418.1, 418.5, np.float64(1.4)), (418.1, 418.5, np.float64(1.4))]
```


```
compare(FIGS / 'loop_stack_roll45.png', '/workspace/cosmology/papers/spreadsheet_model/loop_stack_roll45.png')
compare(FIGS / 'loop_stack_roll80p8.png', '/workspace/cosmology/papers/spreadsheet_model/loop_stack_roll80p8.png')
```

```
loop_stack_roll45.png: new (2302, 3333) vs ref (2302, 3333) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference papers/spreadsheet_model/loop_stack_roll45.png   |   right: notebook re-run steps_v3/figs/loop_stack_roll45.png
```

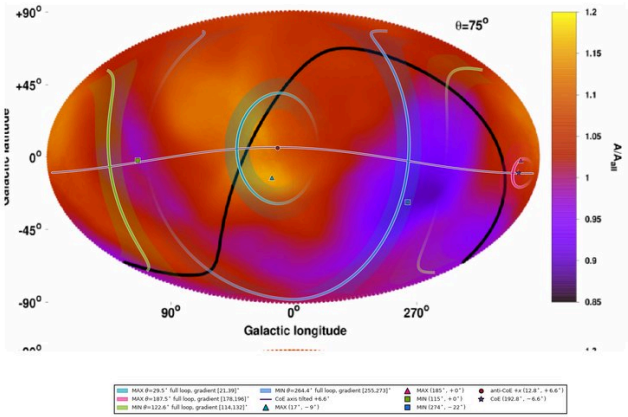


loop_stack_roll80p8.png: new (2302, 3333) vs ref (2302, 3333) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference papers/spreadsheet_model/loop_stack_roll80p8.png | right: notebook re-run steps_v3/figs/loop_stack_roll80p8.png



Migkas Fig. 8 — $\theta = 75^\circ$ $A/A_{\rm all}$ (their numbered colorbar)

Migkas Fig. 8 — $\theta = 75^\circ$ $A/A_{\rm all}$ (their numbered colorbar)



Step 22 — Stacked figures: 3-column and 4-column (stack_4col) versions

What it shows. The summary stack: the model $A(\ell)$ graph (Newtonian $\Omega_s = 0.049$ plus the relativistic curve) at the top, then four loop rows (untilted/no roll; $+15^\circ$ tilt; $+15^\circ$ tilt with gradient; $+15^\circ$ tilt with gradient and $+80.8^\circ$ roll), plus the Migkas Fig. 8 overlays (matched, gradient, 80.8° roll). The 3-column version puts the overlays in a row at the bottom; the 4-column version puts each overlay beside its loop row at the original scale. This script composes existing images: the approved row and overlay images on disk, including three row images and two overlay images kept in the chat attachments folder.

Source script: `/workspace/cosmology/figures/make_stack_clean.py` (last modified 2026-09-27 12:28 CT), run verbatim. *Output-path edit:* `P = "/workspace/cosmology/papers/spreadsheet_model/` → `P = "/workspace/cosmology/papers/spreadsheet_model/"` `OUTP = "/workspac`; `im.save(P + name + ".jpg` → `im.save(OUTP + name + ".jpg`; `.save(P + name + "_web.jpg` → `.save(OUTP + name + "_web.jpg`. *Reference image:* `papers/spreadsheet_model/stack_3col_clean.jpg`, `papers/spreadsheet_model/stack_4col_clean.jpg`.

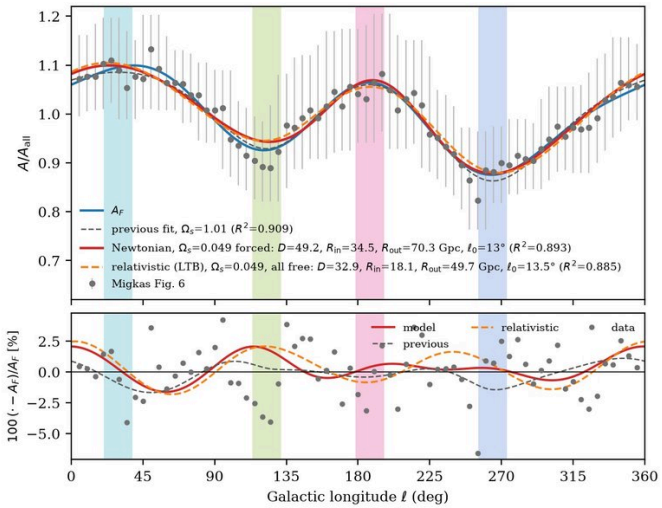
```
%%runscript make_stack_clean.py
from PIL import Image, ImageDraw, ImageFont
import numpy as np
A = "/home/box/agent-data/agents/196ab413-0708-4f81-99fc-58a32c5366c5/attachments/"
P = "/workspace/cosmology/papers/spreadsheet_model/"
OUTP = "/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/"   # notebook: outputs redirected
graph = Image.open(P + "bubble/newt_omb_rel_Agraph.png").convert("RGB")   # was attachment f480c67c... (newt_omb_Agraph); + relativistic curve
rows_src = [Image.open(f).convert("RGB") for f in (P + "loop_threepanel_notilt_web.jpg",
            A + "e2fab4ef4115016f7bc638a6ac1075749b6acbab6e4c2ca6b2f20e29f8735139.jpg",
            A + "fe905a57d5aaf4dede004a4cbdc50883bbf92c984372f9f260251bbfc057b757.jpg",
            A + "d6982109d332a918353df473def180356c133fb81196c90b6e0dea313238ef05.jpg")]
ovs = [Image.open(f).convert("RGB") for f in (A + "3b64499f8c49ab6537b8a73b1fd80965e8b5f8ae82d93d6c99b61dc4f5b8344d.jpg",
       A + "83d1ae71406713257f0623d2b732446ef1ee220de6167592d4a45c5cac55637c.jpg", P + "fig8_overlay_matched_grad_rot81_web.jpg")]
LAB = ["Untilted, no roll", "+15° tilt", "+15° tilt, gradient", "+15° tilt, gradient, +80.8° roll"]
OLAB = ["+15° tilt", "gradient", "+80.8° roll"]
FB = ImageFont.truetype("/usr/share/fonts/truetype/dejavu/DejaVuSans-Bold.ttf", 22)
FH = ImageFont.truetype("/usr/share/fonts/truetype/dejavu/DejaVuSans-Bold.ttf", 18)
FT = ImageFont.truetype("/usr/share/fonts/truetype/dejavu/DejaVuSans.ttf", 17)
W = 1700; LH = 40
def row_content(i):
    im = rows_src[i]
    if i == 0:
        return im.crop((0, 110, W, 628))            # keep A./B./C. headings + footnotes (first row only)
    c = im.crop((0, 222, W, 572)).copy()          # drops suptitle, A/C headings, footnotes, legend
    ImageDraw.Draw(c).rectangle((385, 234 - 222, 1212, 279 - 222), fill="white")   # B heading
    return c
legend = rows_src[3].crop((0, 712, W, 768))        # one legend strip (last row's)
def ov_crop(k, cbar):
    return ovs[k].crop((140, 355, 1445 if cbar else 1276, 1060))
def label(dr, x, y, t, f=FB, anchor="mm"):

# ... 50 more lines of the verbatim script: see coe_steps_v3.ipynb / .html / .md (or coe_steps_v3_fullcode.pdf)
```

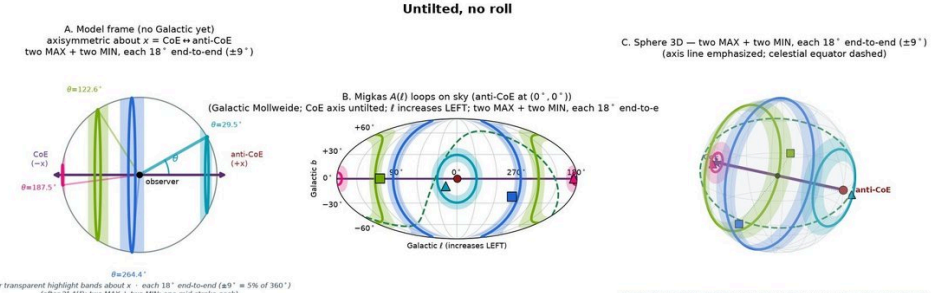
```
stack_3col_clean (1700, 3158)
stack_4col_clean (2493, 2876)
```

```
compare(FIGS / 'stack_3col_clean.jpg', '/workspace/cosmology/papers/spreadsheet_model/stack_3col_clean.jpg')
compare(FIGS / 'stack_4col_clean.jpg', '/workspace/cosmology/papers/spreadsheet_model/stack_4col_clean.jpg')
```

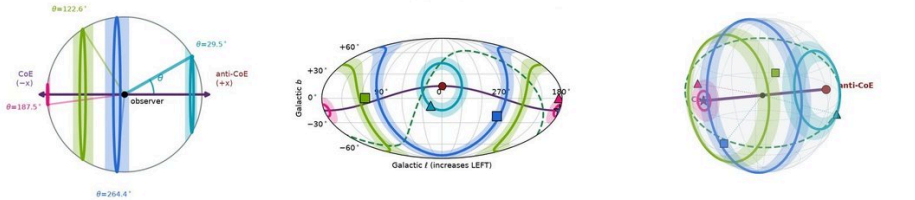
```
stack_3col_clean.jpg: new (1700, 3158) vs ref (1700, 3158) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference papers/spreadsheet_model/stack_3col_clean.jpg   |   right: notebook re-run steps_v3/figs/stack_3col_clean.jpg
```



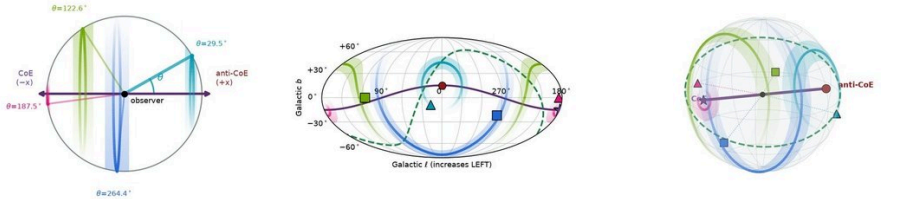
Untilted, no roll



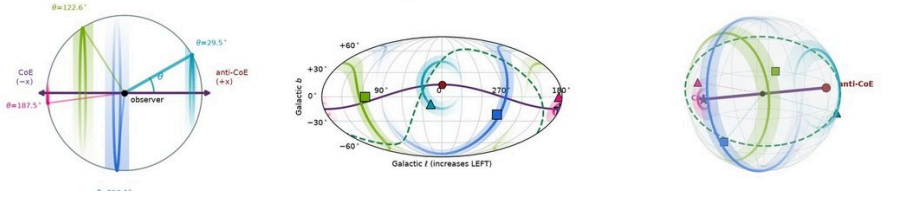
+15° tilt

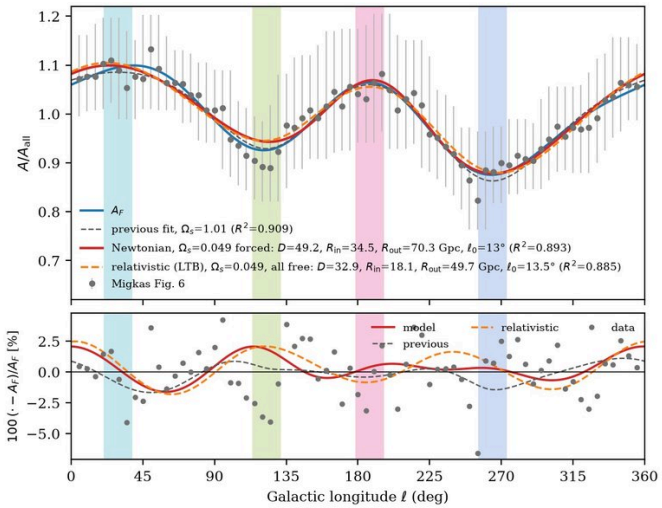


+15° tilt, gradient

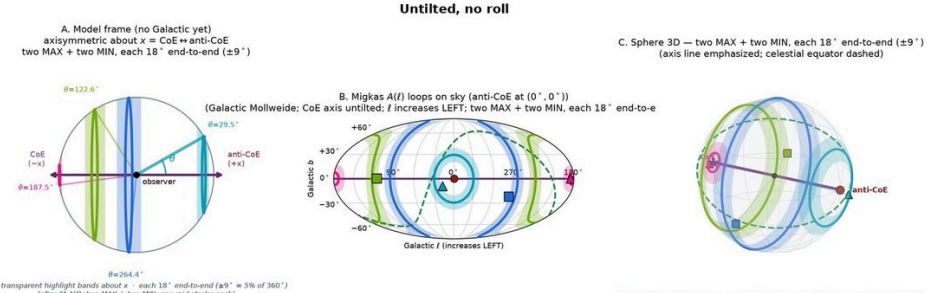


+15° tilt, gradient, +80.8° roll

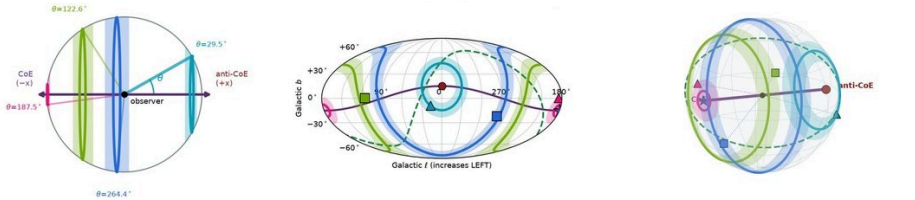




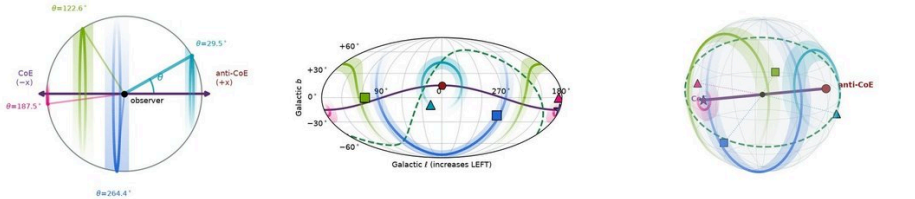
Untilted, no roll



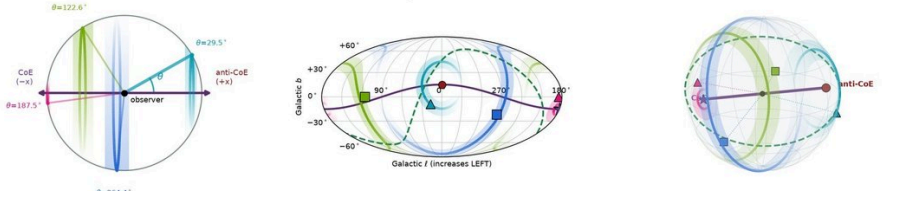
+15° tilt

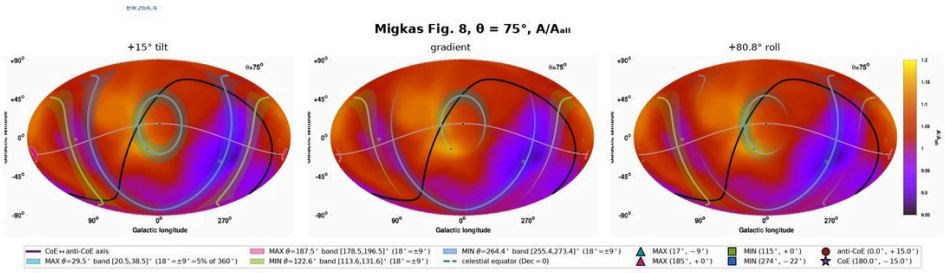


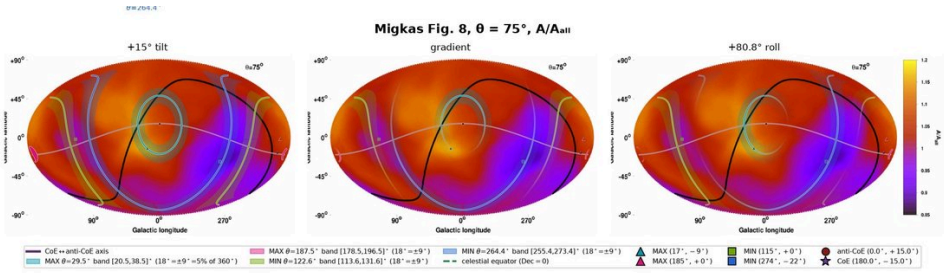
+15° tilt, gradient



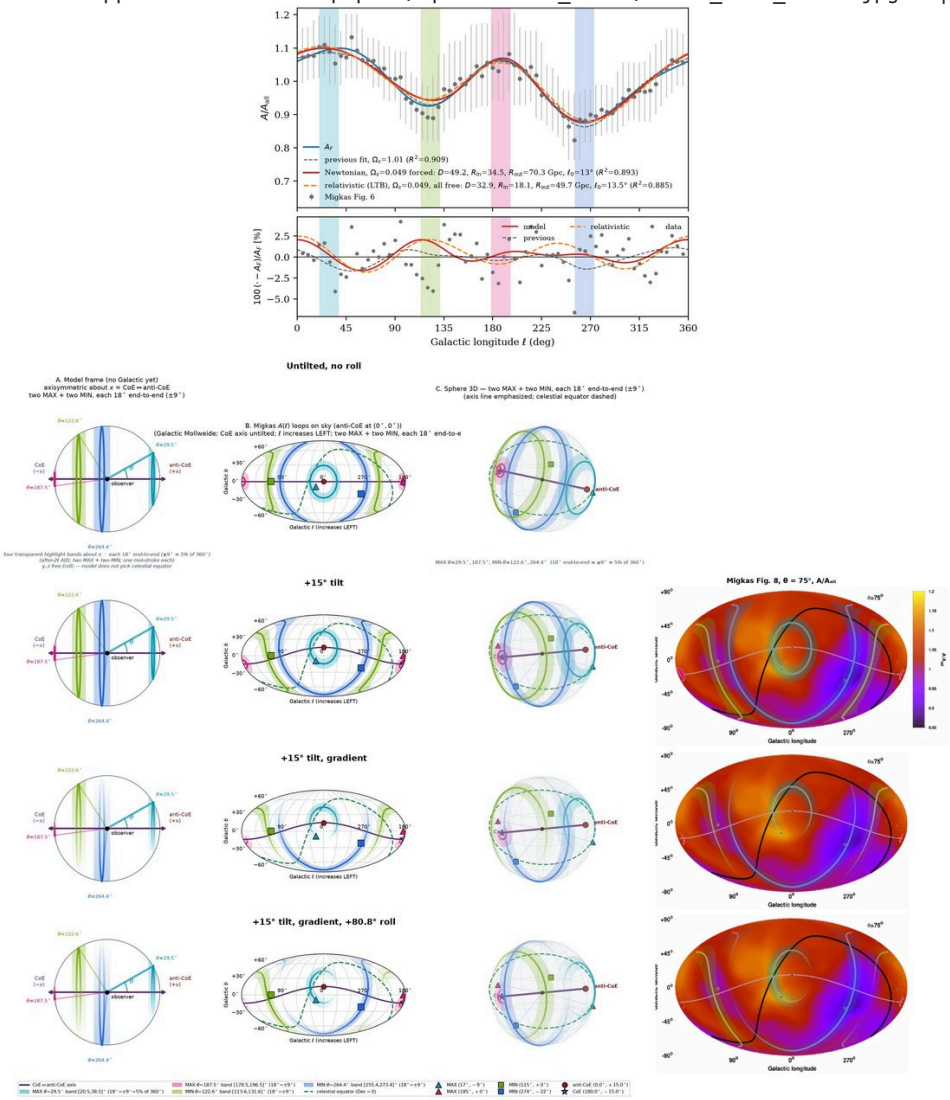
+15° tilt, gradient, +80.8° roll

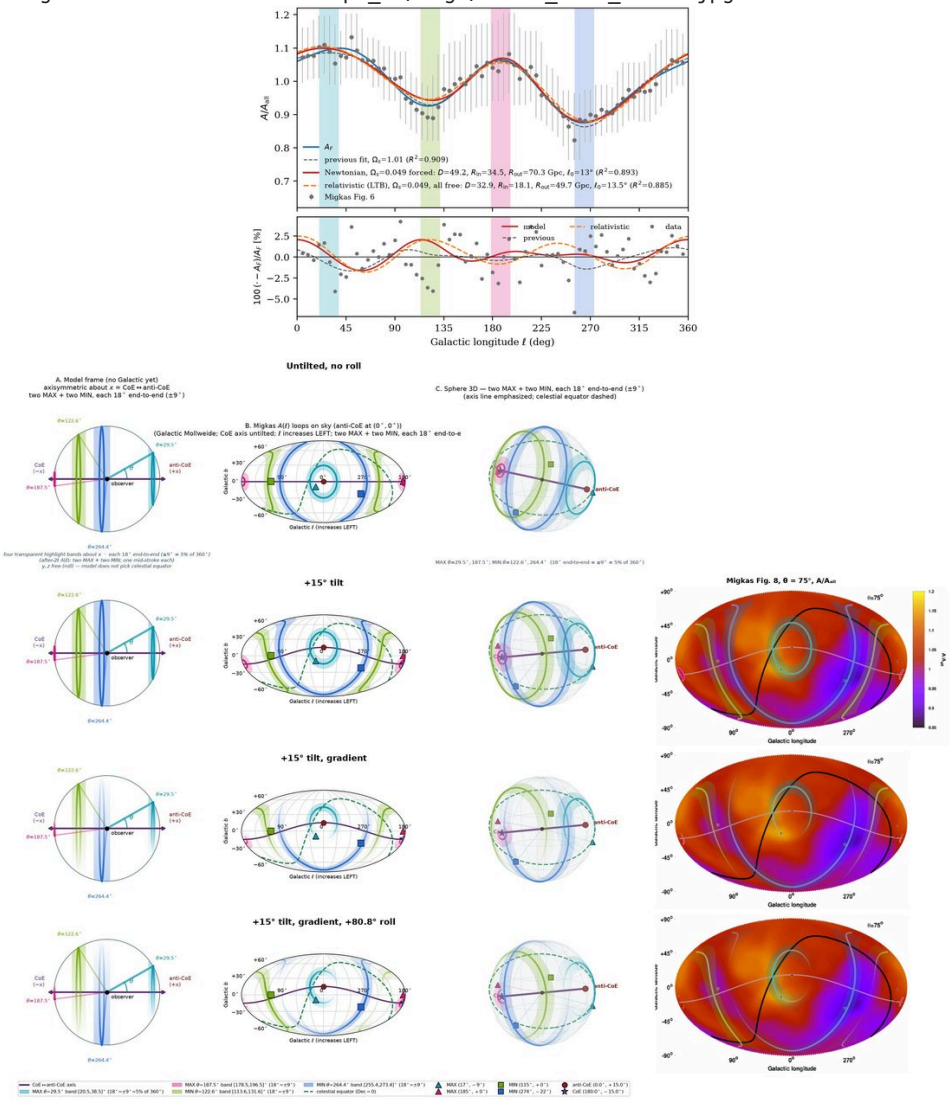






stack_4col_clean.jpg: new (2493, 2876) vs ref (2493, 2876) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference papers/spreadsheet_model/stack_4col_clean.jpg | right: notebook re-run steps_v3/figs/stack_4col_clean.jpg





Step 23 — Thin-shell article, Figure 1 (model panels) and Figure 2 (A(angle) fits incl. relativistic 0.885)

What it shows. The two figures of `universe_bestfit_thinshell.pdf`, from its `.tex` source (`\includegraphics{universe_thin_drawing.png}` and `{universe_thin_Achart.png}`). *Figure 1*: three scales at the relativistic best fit. (a) Hubble velocity vectors: in a Sedov–Taylor blast wave (Taylor 1950; Sedov 1959) $\vec{V} = k\vec{r}$, so $\vec{V}_P - \vec{V}_O = k\vec{r}$ is Hubble's law and the CoE drops out. (b) The Hubble bubble to scale in Gpc: ordinary-matter shell ($\Omega_s = 0.049$) from $R_{\rm in} = 18.1$ to $R_{\rm out} = 49.7$ Gpc around an empty interior and the CoE core, the $D = 32.9$ Gpc circle through Earth, and the $R = 14.3$ Gpc vision sphere. (c) The cosmic shell with the core's Schwarzschild radius $r_s = 2GM_c/c^2$. *Figure 2*: Migkas points with error bars, A_F and the three model curves, normalised to the sky mean: Newtonian thin shell ($R^2 = 0.909$), Newtonian $\Omega_s = 0.049$ (0.893), and relativistic LTB (0.885). The lower panel shows percent differences from A_F. The script also rewrites the table file `universe_thin.json`; that copy is redirected to `steps_v3/figs/` too, so the data file on disk is untouched. Run from the `bubble/` folder, which holds its fit inputs.

Source script: `/workspace/cosmology/papers/spreadsheet_model/bubble/universe_thin.py` (last modified 2026-09-27 12:03 CT), run verbatim. *Output-path edit:* `open("universe_thin.json", "w")` → `open("/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/unive`; `../universe_thin_` → `/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/universe_th`; `open("../" + f + ".png")` → `open("/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/" + f`; `im.save("../" + f + "_web.jpg` → `im.save("/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/"`. *Reference image:* `papers/spreadsheet_model/universe_thin_drawing.png`, `papers/spreadsheet_model/universe_thin_Achart.png`.

```
%%runscript universe_thin.py --cwd /workspace/cosmology/papers/spreadsheet_model/bubble
import json, os, numpy as np, matplotlib
matplotlib.use("Agg"); import matplotlib.pyplot as plt
from matplotlib.patches import Circle, Wedge
from scipy.optimize import brentq
from PIL import Image
from bubble_core import AFfour, ML, MA, MLO, MHI, TH1, H0, GSI, RKM, GPC, IMIG, r2
import analysis_curve as AC
plt.rcParams.update({"font.size": 9, "font.family": "serif"})
c = 2.99792458e8; Msun = 1.989e30; Gm = GPC * 1e3
def phys(k):
    d = json.load(open("thin_%s.json" % k.replace("|", "_")))
    Dm = d["D_Gpc"] * Gm; rhoc = 3 * H0**2 / (8 * np.pi * GSI); rhos = d["Os"] * rhoc; Mcrit = 4 * np.pi / 3 * rhoc * Dm**3
    Mc = d["mcc"] * Mcrit; rin = d["rin"] * Dm; rout = d["rout"] * Dm
    Msh = lambda r: 4 * np.pi / 3 * rhos * (np.clip(r, rin, rout)**3 - rin**3)
    def Phi(r):
        if r < rin: ps = -GSI * 2 * np.pi * rhos * (rout**2 - rin**2)
        elif r < rout: ps = -GSI * (4 * np.pi / 3 * rhos * (r**3 - rin**3) / r + 2 * np.pi * rhos * (rout**2 - r**2))
        else: ps = -GSI * Msh(rout) / r
```

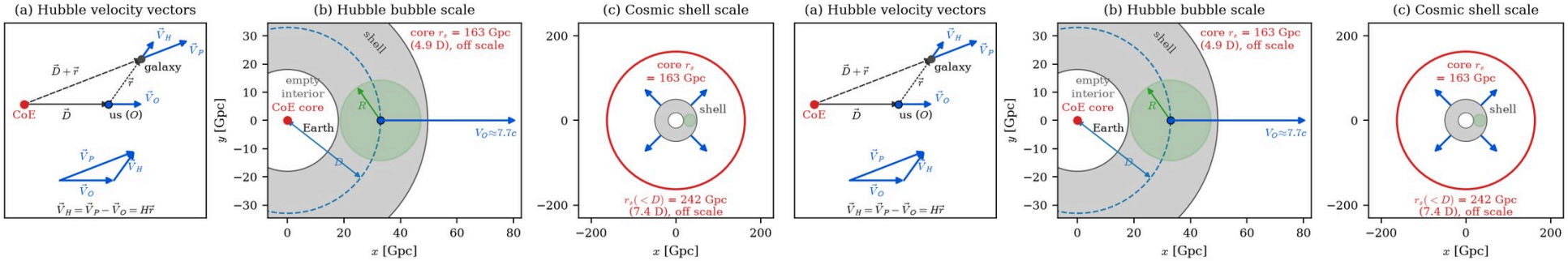
```
        return -GSI * Mc / r + ps
    v2 = d["vesc2"] * 2 * abs(Phi(Dm)); E = v2 / 2 + Phi(Dm)
    rt = brentq(lambda r: Phi(r) - E, Dm, 1e6 * Dm) if E < 0 else np.inf
    rs = lambda M: 2 * GSI * M / c**2 / Gm
    Mt = Mc + Msh(min(rt, 1e300))
    return dict(d=d, D=d["D_Gpc"], R=RKM / GPC, DR=d["DR"], Os=d["Os"], rhos=rhos, Mc=Mc, Mcrit=Mcrit, mcc=d["mcc"], Msh=Msh(Dm), Mtot=Mc + Msh(Dm),
                Mhole=4 * np.pi / 3 * rhos * rin**3, Rin=rin / Gm, Rout=rout / Gm, rs_c=rs(Mc), rs_D=rs(Mc + Msh(Dm)), rt=rt / Gm, rs_t=rs(Mt),
                vE=np.sqrt(v2) / c, vesc2=d["vesc2"], age=d["age"], t_turn=d["t_turn"], amp=d["amp"], phi=d["phi"], l0=d["l0"], R2=d["R2_mig"],
R2F=d["R2_four"],
                Lc=d["Lc"] * d["D_Gpc"], Dkm=d["DR"] * RKM, Lckm=d["Lc"] * d["DR"] * RKM)
P1, P2 = phys("b|free|hug"), phys("b|0.05|hug")
json.dump({k: {kk: vv for kk, vv in p.items() if kk != "d"} for k, p in (("free", P1), ("baryons", P2))},
open("/workspace/cosmology/papers/A_equals_fourier/steps_v3/figs/universe_thin.json", "w"), indent=1, default=float)

# ... 131 more lines of the verbatim script: see coe_steps_v3.ipynb / .html / .md (or coe_steps_v3_fullcode.pdf)
```

```
{'D': 93.7201, 'R': 14.3005, 'DR': 6.5536, 'Os': 1.0124, 'rhos': 0.0, 'Mc': 2.472258139581755e+54, 'Mcrit': 9.324647886726921e+56, 'mcc': 0.0027,
'Msh': np.float64(3.695681787866017e+56), 'Mtot': np.float64(3.7204043692618345e+56), 'Mhole': 5.744951904153747e+56, 'Rin': 79.4197, 'Rout': 108.0206,
'rs_c': 118.9912, 'rs_D': np.float64(17906.5219), 'rt': 990.7581, 'rs_t': np.float64(42041.6443), 'vE': np.float64(19.685), 'vesc2': 0.9013, 'age':
13.8309, 't_turn': 54.5691, 'amp': -0.124, 'phi': 1.2644, 'l0': 12.8215, 'R2': 0.9088, 'R2F': 0.9867, 'Lc': 0.4686, 'Dkm': 2.8919225210400707e+24,
'Lckm': 1.4459612605200357e+22}
{'D': 49.685, 'R': 14.3005, 'DR': 3.4744, 'Os': 0.05, 'rhos': 0.0, 'Mc': 2.383318897221138e+53, 'Mcrit': 1.3893452409706232e+56, 'mcc': 0.0017, 'Msh':
np.float64(4.437473459712923e+54), 'Mtot': np.float64(4.675805349435037e+54), 'Mhole': 2.509252745140193e+54, 'Rin': 35.3845, 'Rout': 63.9855, 'rs_c':
11.4711, 'rs_D': np.float64(225.0492), 'rt': inf, 'rs_t': np.float64(604.8166), 'vE': np.float64(10.8627), 'vesc2': 10.5579, 'age': 13.8872, 't_turn':
inf, 'amp': -0.0622, 'phi': 0.9331, 'l0': 13.2482, 'R2': 0.8931, 'R2F': 0.9769, 'Lc': 0.2484, 'Dkm': 1.5331308837249658e+24, 'Lckm':
7.665654418624828e+21}
REL {'D': 32.9401, 'R': 14.3005, 'DR': 2.3034, 'Os': 0.049, 'rhos': 0.0, 'Mc': 3.3795154244053916e+54, 'Mcrit': 4.0486197275839253e+55, 'mcc': 0.0835,
'Msh': 1.65666926404884e+54, 'Mtot': 5.0361846884542316e+54, 'Mshtot': 6.480355778284555e+54, 'Mhole': 3.2704699040605176e+53, 'Mvis':
1.6232363858405436e+53, 'Rin': 18.0618, 'Rout': 49.6841, 'rs_c': 162.658, 'rs_D': 242.3945, 'vE': 7.6913, 'twoE': 45.1086, 'age': 12.9769, 'amp':
-0.0558, 'phi': 0.8475, 'l0': 13.5, 'R2': 0.8848, 'R2ns': 0.5696, 'R2F': np.float64(0.9632), 'H_r': 78.153, 'H_perp': 65.9235, 'z': 0.3, 'Lc': 0.1647,
'Dkm': 1.0164316238833838e+24, 'Lckm': 5.082158119416919e+21}
EDGE 'Earth' 18.1 [ 6.2 -4.9 16.7 -1. ]
EDGE 'empty\ninterior' 18.1 [-1.6  7.4 12.6 16. ]
EDGE '' 32.9 [25.1  0.  32.9 11.2]
EDGE '' 18.1 [  0.  -20.   25.5   0. ]
EDGE '' 32.9 [  0.  -20.   25.5   0. ]
EDGE 'core $r_s$ = 163 Gpc' 49.7 [43.5 24.4 79.1 33. ]
EDGE '' 49.7 [32.9 -0.9 80.2  0.9]
EDGE '' 32.9 [32.9 -0.9 80.2  0.9]
EDGE 'core $r_s$\n= 163 Gpc' 162.7 [-64.9  84.3  64.9 141.5]
EDGE 'shell' 49.7 [ 55.7  11.9 113.1  38.1]
EDGE '$r_s(<D)$ = 242 Gpc\n' 162.7 [-122.4 -229.1  122.4 -177.1]
EDGE '' 49.7 [36.9 36.9 71.5 71.5]
EDGE '' 49.7 [-71.5  36.9 -36.9  71.5]
EDGE '' 49.7 [-71.5 -71.5 -36.9 -36.9]
EDGE '' 49.7 [ 36.9 -71.5  71.5 -36.9]
axes heights (in) [np.float64(1.842), np.float64(1.842), np.float64(1.842)] H 1.842 tight W 7.26
```

```
compare(FIGS / 'universe_thin_drawing.png', '/workspace/cosmology/papers/spreadsheet_model/universe_thin_drawing.png')
compare(FIGS / 'universe_thin_Achart.png', '/workspace/cosmology/papers/spreadsheet_model/universe_thin_Achart.png')
```

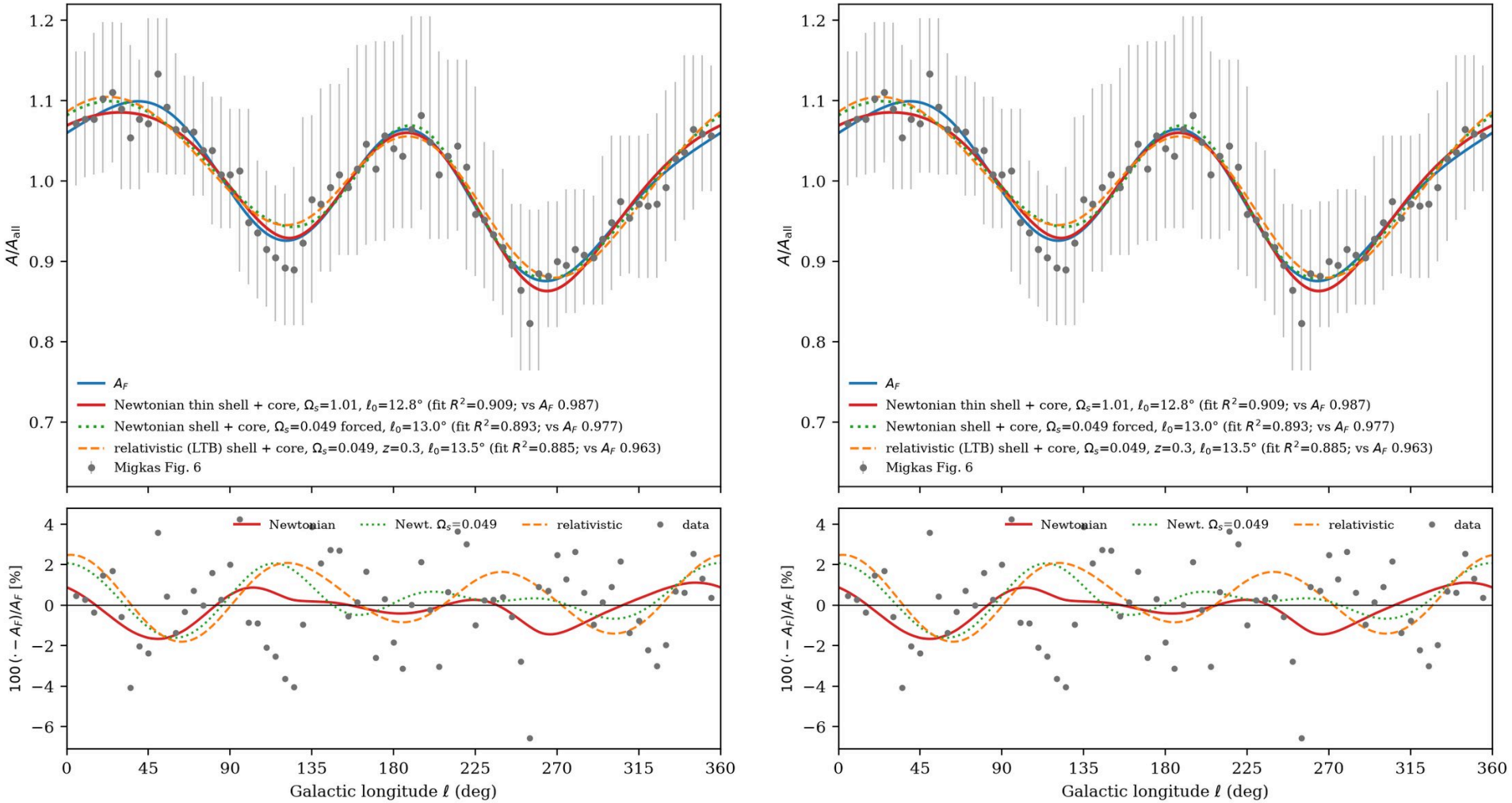
```
universe_thin_drawing.png: new (1460, 488) vs ref (1460, 488) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical
left: approved reference papers/spreadsheet_model/universe_thin_drawing.png   |   right: notebook re-run steps_v3/figs/universe_thin_drawing.png
```



(a) Hubble velocity vectors
(b) Hubble bubble scale
(c) Cosmic shell scale
galaxy
CoE
us (O)
shell
empty interior
CoE core
Earth
core r_s = 163 Gpc
(4.9 D), off scale
$r_s(<D)$ = 242 Gpc
(7.4 D), off scale
$V_O \approx 7.7c$
$\vec{V}_H = \vec{V}_P - \vec{V}_O = H\vec{r}$
x [Gpc]
y [Gpc]

universe_thin_Achart.png: new (1182, 1287) vs ref (1182, 1287) | mean |diff| = 0.000/255, pixels differing >40: 0.000% -> identical

left: approved reference papers/spreadsheet_model/universe_thin_Achart.png | right: notebook re-run steps_v3/figs/universe_thin_Achart.png



A/A_{all}
A_F
Newtonian thin shell + core, Ω_s=1.01, ℓ_0=12.8° (fit R^2=0.909; vs A_F 0.987)
Newtonian shell + core, Ω_s=0.049 forced, ℓ_0=13.0° (fit R^2=0.893; vs A_F 0.977)
relativistic (LTB) shell + core, Ω_s=0.049, z=0.3, ℓ_0=13.5° (fit R^2=0.885; vs A_F 0.963)
Migkas Fig. 6
Newtonian
Newt. Ω_s=0.049
relativistic
data
$100\,(\cdot - A_F)/A_F$ [%]
Galactic longitude ℓ (deg)

```
# Part B summary - match check for every figure
from IPython.display import Markdown
t = "| figure | reference | size ref | size new | mean abs diff (/255) | pixels differing >40 | verdict |\n|---|---|---|---|---|---|---|\n"
t += "\n".join(f"| {m['figure']} | {m['reference'].replace('/workspace/cosmology/', '')} | {m['size_ref'][0]}x{m['size_ref'][1]} | {m['size_new']
[0]}x{m['size_new'][1]} | {m['mean_abs_diff']:.3f} | {100*m['frac_px_diff']:.3f}% | {m['verdict']} |" for m in MATCH)
display(Markdown(t))
```

figure	reference	size ref	size new	mean abs diff (/255)	pixels differing >40	verdict
migkas_A_extrema_table2_mollweide.png	figures/migkas_A_extrema_table2_mollweide.png	2079x2612	2079x2612	0.000	0.000%	identical
migkas_fig8_theta75_A_extrema_only.png	figures/migkas_fig8_theta75_A_extrema_only.png	1975x1486	1975x1486	0.000	0.000%	identical
coe_rings_migkas_pin_axis.png	figures/coe_rings_migkas_pin_axis.png	2283x1027	2283x1027	0.000	0.000%	identical
migkas_fig8_theta75_with_coe_rings.png	figures/migkas_fig8_theta75_with_coe_rings.png	1975x1515	1975x1515	0.000	0.000%	identical
coe_rings_half_loops_tilt15.png	figures/coe_rings_half_loops_tilt15.png	2272x1027	2272x1027	0.000	0.000%	identical
coe_rings_half_loops_tilt15_rot45ccw.png	figures/coe_rings_half_loops_tilt15_rot45ccw.png	2272x1027	2272x1027	0.000	0.000%	identical
migkas_fig8_half_loops_tilt15.png	figures/migkas_fig8_half_loops_tilt15.png	1975x1515	1975x1515	0.000	0.000%	identical
migkas_fig8_half_loops_tilt15_rot45ccw.png	figures/migkas_fig8_half_loops_tilt15_rot45ccw.png	1975x1515	1975x1515	0.000	0.000%	identical
coe_rings_gradient_loops_tilt15.png	figures/coe_rings_gradient_loops_tilt15.png	2334x1027	2334x1027	0.000	0.000%	identical
coe_rings_gradient_loops_tilt15_rot45ccw.png	figures/coe_rings_gradient_loops_tilt15_rot45ccw.png	2334x1027	2334x1027	0.000	0.000%	identical
migkas_fig8_gradient_loops_tilt15.png	figures/migkas_fig8_gradient_loops_tilt15.png	1975x1515	1975x1515	0.000	0.000%	identical
migkas_fig8_gradient_loops_tilt15_rot45ccw.png	figures/migkas_fig8_gradient_loops_tilt15_rot45ccw.png	1975x1515	1975x1515	0.000	0.000%	identical
loop_threepanel_matched.png	papers/spreadsheet_model/loop_threepanel_matched.png	2262x1027	2262x1027	0.000	0.000%	identical
loop_threepanel_matched_grad.png	papers/spreadsheet_model/loop_threepanel_matched_grad.png	2262x1027	2262x1027	0.000	0.000%	identical
loop_threepanel_matched_grad_rot45.png	papers/spreadsheet_model/loop_threepanel_matched_grad_rot45.png	2262x1027	2262x1027	0.000	0.000%	identical
loop_threepanel_matched_grad_rotfit_rot81.png	papers/spreadsheet_model/loop_threepanel_matched_grad_rotfit_rot81.png	2262x1027	2262x1027	0.000	0.000%	identical
loop_threepanel_notilt.png	papers/spreadsheet_model/loop_threepanel_notilt.png	2262x1027	2262x1027	0.000	0.000%	identical
fig8_overlay_matched.png	papers/spreadsheet_model/fig8_overlay_matched.png	1975x1515	1975x1515	0.000	0.000%	identical
fig8_overlay_matched_grad.png	papers/spreadsheet_model/fig8_overlay_matched_grad.png	1975x1515	1975x1515	0.000	0.000%	identical
fig8_overlay_matched_grad_rot45.png	papers/spreadsheet_model/fig8_overlay_matched_grad_rot45.png	1975x1515	1975x1515	0.000	0.000%	identical
fig8_overlay_matched_grad_rot81.png	papers/spreadsheet_model/fig8_overlay_matched_grad_rot81.png	1975x1515	1975x1515	0.000	0.000%	identical
loop_flow_thinshell.png	papers/spreadsheet_model/loop_flow_thinshell.png	2550x5010	2550x5010	0.000	0.000%	identical
loop_journal_page.png	papers/spreadsheet_model/loop_journal_page.png	2278x5713	2278x5713	0.000	0.000%	identical
loop_stack_roll45.png	papers/spreadsheet_model/loop_stack_roll45.png	2302x3333	2302x3333	0.000	0.000%	identical
loop_stack_roll80p8.png	papers/spreadsheet_model/loop_stack_roll80p8.png	2302x3333	2302x3333	0.000	0.000%	identical
stack_3col_clean.jpg	papers/spreadsheet_model/stack_3col_clean.jpg	1700x3158	1700x3158	0.000	0.000%	identical

figure	reference	size ref	size new	mean abs diff (/255)	pixels differing >40	verdict
stack_4col_clean.jpg	papers/spreadsheet_model/stack_4col_clean.jpg	2493x2876	2493x2876	0.000	0.000%	identical
universe_thin_drawing.png	papers/spreadsheet_model/universe_thin_drawing.png	1460x488	1460x488	0.000	0.000%	identical
universe_thin_Achart.png	papers/spreadsheet_model/universe_thin_Achart.png	1182x1287	1182x1287	0.000	0.000%	identical